

ENHANCING E-GOVERNANCE IN LAW ENFORCEMENT: DEVELOPING
AN AUTOMATED DOMESTIC POLICE CLEARANCE CERTIFICATE
SYSTEM FOR A MORE EFFICIENT AND TRANSPARENT
APPLICATION PROCESS

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NSBM Green University
in partial fulfillment of the requirements for the degree of
Bachelor of Science (Special) Degree in Management Information Systems

by

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DECLARATION

I, the undersigned R.G.I.S. Senarathna hereby declare that I am the sole author of this dissertation. To the best of my knowledge this dissertation contains no material previously published by any other person except where due acknowledgement has been made. This thesis contains no material which has been accepted as part of the requirements of any other academic degree or non-degree program, in English or in any other language. This is a true copy of the dissertation, including final revisions.

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DEDICATION

This thesis is dedicated to my research supervisors, Dr. Mohamed Shafraz and Mr. Diluka Wijesinghe, whose understanding, support, and invaluable guidance have been pivotal throughout this journey. Their encouragement to pursue further achievements, including the submission of a paper derived from this research, has been a constant source of inspiration. I also dedicate this work to my family, whose endless love, encouragement, and unwavering support gave me the strength and determination to accomplish my goals. Finally, I extend my heartfelt appreciation to the Officer-in-Charge (OIC) of Raddolugama Police Station, the OIC of Seeduwa Police Station, and the Police Clearance Unit, for their invaluable cooperation and support, which played a crucial role in shaping the foundation of this research.

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ABSTRACT

This research explores the development of an automated Police Clearance Certificate (PCC) system for domestic applications, situated within the broader domain of e-governance and public service digital transformation. The study addresses the problem of inefficiencies, delays, and risks of document loss associated with manual PCC processes. It seeks to answer how a digitally driven approach can enhance transparency, efficiency, and accountability in law enforcement while improving citizen trust in government institutions.

The research adopts a systems development approach to design a standardized digital platform. Applicants are enabled to submit requests online, upload required documents, track the progress of applications in real time, and securely access issued certificates. Police officers utilize role-based portals for verification and decision-making, while administrators manage oversight functions through monitoring and reporting tools.

The findings demonstrate that the automated PCC system significantly reduces manual inefficiencies, accelerates processing times, strengthens data accuracy, and supports transparent service delivery. The system's design incorporates data security, real-time tracking, and administrative accountability, which collectively enhance operational effectiveness.

The study contributes to the growing body of e-governance literature by showing how digital solutions can modernize critical law enforcement services in developing countries. Its implications highlight improved public trust, citizen-centric service delivery, and potential adaptability to other areas of governance. Further research may expand the model to cross-border applications or integration with national digital identity frameworks.

Keywords: Administrative Efficiency, Automated Police Clearance Certificate, Digital Transformation, Domestic PCC, E-Governance, Public Trust, Transparency

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CHAPTER 01: INTRODUCTION

1.1 Chapter Overview

This chapter sets the foundation for the study by examining the influence of e-governance on the effectiveness of law enforcement, with particular reference to the Police Clearance Certificate (PCC) regime. It begins with a review of the challenges associated with the traditional PCC issuing process and investigates how the processes are changing in light of the digital revolution.

The discussion then moves to key aspects of e-governance, highlighting its current limitations and potential benefits. General issues in public administration, as well as specific shortcomings in the domestic PCC issuance process, are discussed in the problem statement. The research question, justification, and objectives are then given, outlining the focus of the study and anticipated contributions.

A conceptual overview of the proposed solution is provided, illustrating how an integrated e-governance platform can enhance efficiency, transparency, and user experience. The chapter also identifies essential resources, including hardware and software requirements, and outlines the project scope to establish clear research boundaries.

Through such vital elements, this chapter forms a platform for a proper analysis of the role of e-governance in the area of law enforcement. It provides the necessary background required to understand the research problem and its significance, thereby guiding the direction of the inquiry.

1.2 Problem Background

E-governance in Sri Lanka has evolved slowly with the objective of incorporating information technology into diverse administration areas to enhance the efficiency of operations, transparency, and delivery of services. The government has introduced various initiatives to modernize public services as part of this digitalization process. However, some issues like poor public access to some of the services, dispersed infrastructure facilities, and low system compatibility have hindered e-governance from achieving its optimal level. These issues are felt acutely in the area of law enforcement, where smooth processes and seamless exchange of information are essential for service delivery and public confidence.

The Police Clearance Certificate (PCC) is an important certificate in Sri Lanka, required for employment, migration, higher studies, and professional licensing, among other reasons. The certificate serves as a formal guarantee that the applicant has no criminal history. The process of getting a PCC in Sri Lanka can be broadly divided into foreign PCC and local PCC, both of which have different processes and limitations.

For international PCCs, there is an online system to address overseas employment and migration needs. It allows applicants to apply online, check the status of the application, and obtain certificates electronically. The system offers convenience, minimizes mistakes, and enhances transparency. [1] It is, however, restricted to those individuals who are applying to work or migrate outside the country, especially to nations with stringent foreign employment policies.

Apart from that, the domestic PCC process is mainly paper-based and manual. Applicants are required to go to their Grama Niladhari (GN) division and neighboring police stations to file documents, check residence, and fill out forms. Most of the tasks, including obtaining GN certificates and entering information into logbooks, are performed manually. Moreover, the internal centralized police system provides officers with the ability to utilize an individual's National Identity Card (NIC) number to access criminal history, yet it is not connected to other administrative organizations and is solely for internal consumption.

This manual approach causes several issues:

- Slow processing, typically 7 to 14 days
- Repeated trips and expensive travel for applicants
- Human errors in data entry
- Greater risk of fraud and corruption
- Poor traceability, transparency, and accountability

Current studies of the available solutions reflect significant dissatisfaction among users, highlighting constraints related to delays, limited access, and a lack of transparency. Statistics from the Sri Lanka Bureau of Foreign Employment.[2] indicate that over 200,000 applicants annually require Police Clearance Certificates (PCCs) for foreign employment, reflecting the high demand for efficient clearance services. While foreign PCC applications enjoy some level of digital support, those applying within the country continue to face delayed and cumbersome processes

Relative to global benchmarks, like India's e-PCC or the integration of Singapore's SingPass, Sri Lanka is behind in terms of digitalizing local PCCs. The absence of automation impacts the productivity of law enforcement services as well as public confidence. This research suggests creating a unified digital platform to computerize the domestic PCC issuing process. This platform will network police stations, Grama Niladhari offices, and divisional secretariats under one system at the divisional level. Computerization of the process will minimize delays, enhance accuracy, enhance transparency, and offer citizens a more convenient and reliable service. Ultimately, this system is expected to eradicate existing inefficiencies and facilitate the ongoing development of e-governance in Sri Lanka's policing sector through contemporary information systems.

1.2.1 As-Is Workflow (Manual PCC Process)

In the existing manual process, applicants are required to physically visit multiple institutions, including the Grama Niladhari (GN) office and local police stations. The process is heavily dependent on paperwork and manual verification, which results in delays, redundancy, and inefficiency.

Key Characteristics of the As-Is Process:

- Applicants must travel to police stations with required documentations.
- Paper-based submission and verification of documents.
- Manual communication between police stations, and higher authorities.
- Delays caused on by a high workload, mismanaged paperwork, and a lack of consolidated data access.
- No real time tracking; applicant must make repeated visits for updates

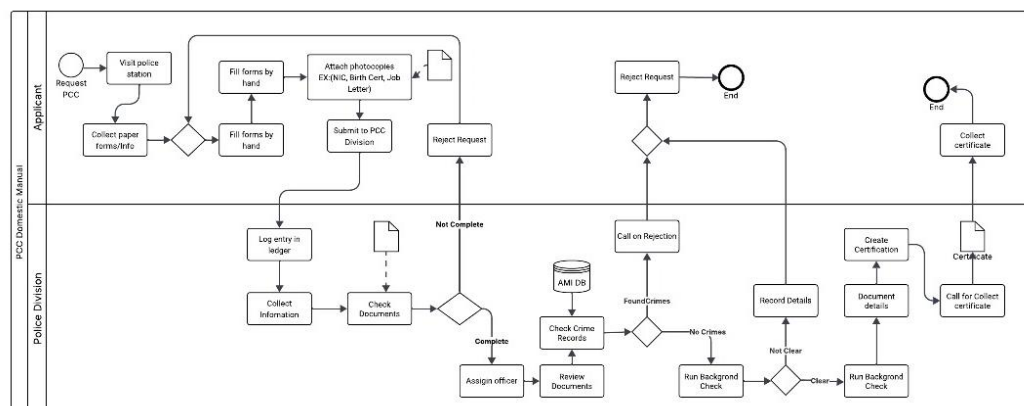


Figure 1: As-Is Process Diagram

Source: https://drive.google.com/file/d/18HGcWlqABHFtDpkkiQitT5Id--Gzx_7m/view?usp=drive_link

1.3 Problem Statement

The traditional process of issuing Police Clearance Certificates (PCC) in Sri Lanka relies on manual verification methods that are inefficient, time-consuming, and prone to delays. The requirement for applicants to physically obtain a Grama Sevaka Certificate to verify their residence further complicates the process. Citizens face multiple visits, uncertain waiting times, and lack of transparency in application status. This directly impacts employment opportunities, visa processing, and business operations requiring background verification. The below diagram represents the structured components that were identified in this manual process.

Applicant physically visits GN division → submits documents (NIC , Birth certification) → GN certification (only for period of residence receive)→ forward to police station → criminal and background record check → police officer approval (after manual background and information checks) → clearance issued (manual, paper-based, takes 7-14 days)

This outdated approach slows down law enforcement operations and creates frustration among applicants due to long waiting times and procedural inefficiencies. The integration of e-governance solutions presents an opportunity to streamline the PCC issuance process, improve transparency, and enhance overall law enforcement efficiency. However, limited research exists on how digital transformation in law enforcement, particularly in PCC systems, can address these inefficiencies.

1.3.1. General Problem (High-Level Implications)

The inefficiencies in the current domestic PCC process create challenges for both citizens and police stations. Delays in obtaining PCCs hinder employment opportunities, visa applications, and legal procedures. Police stations face operational difficulties due to manual record-keeping and verification, and the government incurs additional costs to maintain these processes. Without a standardized procedure, these inefficiencies reduce transparency, increase processing times, and raise the risk of document fraud, negatively affecting public trust. A survey conducted with experienced users revealed widespread dissatisfaction, emphasizing the need for digital solutions to enhance efficiency, accessibility, and client satisfaction.

1.3.2. Specific Problem (Domain-Specific Shortcomings & Research Gap)

The existing local PCC process does not tackle sectoral issues in law enforcement, specifically the automation requirement for the issuance of Police Clearance Certificates. A look at previous research shows that there has been no study of a fully automated PCC issuing system for the local process since 2015 for Sri Lanka [3]. Because of this, there is no end-to-end digital solution, and delays, errors, and inefficiencies in operations result.

This study seeks to bridge that gap by putting forward a completely automated system that utilizes contemporary information systems in order to render the local PCC process more speedy, accurate, and accessible. It is relevant and essential to investigate how the PCC system might be modernized to improve its effectiveness, transparency, and end-user satisfaction given the growing need for reliable and timely public services.

1.4 Research Question

Before formulating the research question, it is worth mentioning that this research builds upon lessons from previous surveys, secondary data sources (e.g., Sri Lanka Bureau of Foreign Employment, 2023), and comparative international best practices (e.g., India's e-PCC, Philippines NPCPS). Since the research will not be gathering sensitive applicant records directly from the Sri Lanka Police, the research question is posed to test the effectiveness, relevance, and applicability of a proposed e-governance system. In this manner, the study remains focused on addressing systemic inefficiencies, identifying critical gaps, and analyzing how digital transformation can streamline the domestic Police Clearance Certificate (PCC) application process in Sri Lanka.

Main Research Question:

- **“How can a digital e-Governance system automate and streamline the domestic police clearance process in Sri Lanka to improve efficiency, transparency, and user satisfaction?”**

Sub Questions:

1. To what extent does integrating digital identity verification (e.g., NIC database and OTP login) improve accuracy and reduce fraudulent applications compared to the current manual process?

2. How can automated integration with the People's Registration Department (PRD) and Automated Management of Information (AMI) systems enhance verification accuracy and reduce fraudulent PCC applications?
3. Which workflow components (e.g., applicant registration, GN verification, police criminal record check, certificate issuance) contribute most significantly to delays in the manual PCC process, and how can automation address these bottlenecks?
4. What are the primary technical, organizational, and legal challenges in implementing a fully automated PCC system in Sri Lanka, and what strategies can mitigate these barriers?
5. How effective is the proposed automated PCC system in reducing processing time and applicant travel costs, as measured by performance metrics (e.g., average completion time, error rate, user satisfaction)?
6. What best practices from global e-governance initiatives can be adapted to ensure scalability and long-term sustainability of the Sri Lankan PCC system?

1.5 Research Motivation

Public service efficiency is a critical concern in Sri Lanka, with citizens facing significant challenges in accessing essential government services. The inefficiencies in the domestic Police Clearance Certificate (PCC) issuance process have contributed to this issue, as the manual, paper-based system continues to create barriers for citizens seeking employment, education opportunities, and other life-changing prospects. The consequences of these inefficiencies have been far-reaching, resulting in missed employment opportunities, delayed visa processing, frustrated citizens, and erosion of public trust in law enforcement services. The urgency of this problem is underscored by the staggering statistics, with over 200,000 applicants annually requiring Police Clearance Certificates for various purposes, according to the Sri Lanka Bureau of Foreign Employment (SLBFE, 2023) [2], yet domestic applicants face processing delays of 7-14 days and multiple office visits that could be easily avoided with proper digitalization.

This main consequence has deeply motivated me to address this problem, and after thorough analysis, I realized that I cannot completely eliminate all administrative inefficiencies or transform the entire government service delivery overnight. However, I can significantly reduce the processing delays and citizen frustration by ensuring that the domestic PCC system operates with

the same efficiency and transparency as the existing foreign PCC digital platform. If so, we can minimize the service delivery gaps that occur due to outdated manual processes and improve overall citizen satisfaction with law enforcement services. Also, we cannot simply create just an information display system or basic digitization because that will not be sufficient to address the systematic issues, and it wouldn't be an effective method to minimize on processing errors and human intervention. So, I have been motivated to develop a comprehensive, automated e-governance solution that can deliver PCC services without excessive human involvement, multiple office visits, and lengthy processing delays.

1.6 Research Aim

The aim of this research is to develop and implement a comprehensive automated e-governance system integrated with digital verification technologies to enhance the domestic Police Clearance Certificate (PCC) issuance process and improve public service delivery in Sri Lanka. The proposed system seeks to leverage existing government databases and digital identity verification mechanisms to make accurate, data-driven decisions about applicant eligibility and criminal record verification. By automating the assessment and approval process, the system aims to reduce human intervention and potential inefficiencies, ensuring that only legitimate applications are processed while providing citizens with a transparent, efficient, Faster and reliable service delivery experience.

1.7 Research Objectives

1.7.1 To Identify

1. Identify key bottlenecks and inefficiencies in the current manual domestic PCC issuance process through workflow analysis and stakeholder consultation
2. Major technical, organizational, and legal requirements for the introduction of an automated PCC system in the Sri Lankan environment
3. Interfaces between current government databases (PRD, AMI systems) and the suggested digital PCC platform
4. Security threats and fraud prevention mechanisms required for digital identity verification in law enforcement systems

1.7.2 For Analysis

1. The efficiency of existing manual verification procedures over suggested digital counterparts when it comes to accuracy, speed, and fraud detection
2. International best practices in automated police clearance systems and their applicability to Sri Lankan governmental structures
3. Cost-benefit considerations of a shift from manual to electronic PCC processing for citizens and government agencies
4. Stakeholder expectations and requirements of various user groups (applicants, police officers, system administrators)

1.7.3 To Design

1. An exhaustive digital e-governance solution for automating the domestic PCC issuance procedure with incorporated workflow administration
2. Easy-to-use interfaces for various stakeholder groups for web application (mobile responsive) for applicants and administrative dashboards for officials
3. Secure e-governance system that integrates applicant, police officer, and admin functionalities into a one centralized platform for domestic PCC issuance
4. Integration protocols for linking the suggested system to government databases and authentication systems that already exist

1.7.4 To Evaluate

1. Effectiveness of the proposed system through surveys, mock testing, and user feedback, focusing on performance measures such as processing time, accuracy, transparency, and user satisfaction.
2. Scalability and sustainability of the system to handle increasing numbers of applications and potential expansion to other law enforcement services
3. Feasibility of implementation based on technical infrastructure, organizational preparedness, and resource needs in the Sri Lankan situation

1.7.5 Deliverables

1. Research Paper:

A scholarly paper summarizing the research methodology, findings, and implications of the study, suitable for publication in academic journals or presentations at conferences.

2. System Prototype:

A working prototype of the automated PCC system, demonstrating the integration of digital verification mechanisms, real-time tracking capabilities, and showcasing its operational functionalities within the pilot divisions.

1.8 Rich Picture of the Proposed Solution

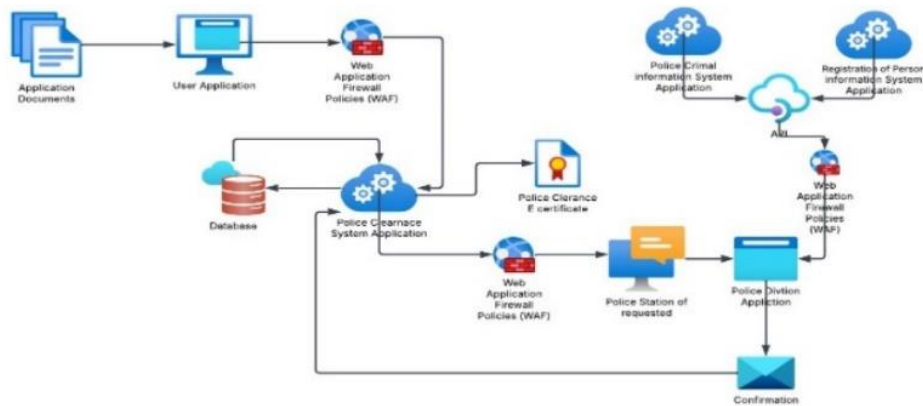


Figure 2: Rich Picture

The solution proposed here is a complete automated domestic PCC system with three major portals:

1. Applicant Portal → Allows citizens to apply using NIC and OTP verification, upload required documents (NIC copy, birth certificate, employment letter), and monitor their application status via a personalized dashboard.
2. Police Officer Portal → Enables officers to confirm applicant details, cross-check records against the People's Registration Department (PRD) and the Automated Management of Information (AMI) criminal database, and accept or refuse applications.
3. Admin Portal → Offers administrative oversight to manage officers, track application workflows, generate reports, and maintain accountability.

Normal workflow of solution: **Applicant Submission** → **Digital Verification (PRD/AMI)** → **Review by GN/Police Officer** → **Admin Oversight** → **Generation of Certificate** → **Applicant Downloads Digital PCC**

1.9 Resource Requirement

1.9.1 Hardware

1. Development Laptops/PCs (8GB RAM, i5/i7 processors, 256GB+ SSD).
2. Server hosting (AWS Free Tier / Azure / Heroku).
3. Optional: Test smartphones for OTP-based login testing.

1.9.2 Software

1. Frontend: React.js (for applicant and officer portals)).
2. Backend: Node.js with Express (server-side logic).
3. Database: MongoDB Atlas (cloud database) / SQL / Firbase.
4. Authentication: JWT + Twilio/Email API for OTP verification.
5. Design Tools: Figma, Draw.io, Lucidchart (for UI and diagram design).
6. Version Control: GitHub for code collaboration.
7. Testing Tools: Selenium, JMeter (for system evaluation).

1.10 Project scope

Table 1: Project scope and out of scope

In Scope	Out Scope
Development of automated domestic PCC request processing system with complete workflow management.	International PCC requests or modifications to existing foreign PCC systems and processes.
Focus on applicants within Seeduwa & Raddolugama divisional areas as pilot implementation zones for testing and validation.	Real-world deployment beyond pilot testing phases or full-scale production implementation across multiple divisions.

Integration with sample/mock PRD and AMI databases to simulate government system connectivity and data exchange.	Direct integration with live government databases or access to actual production-level government system infrastructure.
Implement of an extensive digital applicant dashboard with real-time tracking and status monitoring features for application processing.	Full national rollout implementation in initial development phase or country-wide system deployment.
SMS/Email notifications for status updates	Biometric verification (fingerprint/iris)
OTP verification mechanisms via mobile phone and email for secure user authentication and identity confirmation	Blockchain-based certificate storage solutions, artificial intelligence prediction models, or advanced cryptographic implementations.

1.11 Chapter Summary

This chapter has initiated the research by setting the background of e-governance in Sri Lanka with specific reference to inefficiencies in the local Police Clearance Certificate (PCC) procedure. The problem background delineated the pitfalls of the existing manual system, such as [3], several office visits, no transparency, and risks of fraud. The problem statement articulated general implications as well as the specific research niche where this research is filling the gap.

The research question was developed to examine how an electronic e-governance system could digitize and automate the domestic PCC process to improve efficiency and user satisfaction. The research motivation emphasized the dire need for modernization of the manual system that impacts more than 200,000 applicants each year [2], followed by specific research aims and objectives for creating an automated e-governance system with integration of digital verification technologies.

A comprehensive diagram depicted the workflow of the proposed solution, linking all relevant stakeholders via a centralized digital platform. This section delineated critical hardware and software prerequisites, defined the project parameters with an emphasis on the Seeduwa and Raddolugama divisions as pilot regions and by addressing these aspects, this chapter provides the framework for investigating the function of digital transformation in law enforcement and provides guidance for the following chapters of the study.

CHAPTER 02: LITERATURE REVIEW

2.1 Chapter Overview

This chapter provides a detailed review of the current literature on automating the local Police Clearance Certificate (PCC) issuing process under the broader framework of e-governance with specific emphasis on pilot implementation strategies appropriate for divisional-level testing. Although Sri Lanka has partly embraced digital services for PCCs related to foreign employment, the local process is still remains largely manual, fragmented, and paper-intensive, as seen in divisional regions such as Seeduwa and Raddolugama where this study will be piloted.

The study examines the root causes of persistent inefficiencies in local implementations, contrasts the practices in Sri Lanka with those of pilot models internationally, and outlines the specific design principles required for the effective automation of divisional-level systems. The core aim of this research is to evaluate the deficiencies encountered within the Sri Lankan domestic model at the divisional level, clarify the causes of these inadequacies, and analyze how lessons learned from pilot programs in other countries can inform the creation of a system that successfully balances policy imperatives and technical pragmatism for localized implementation.

Particular attention is given to three critical considerations for pilot implementation: the effectiveness of divisional workflow automation versus maintaining existing manual processes, the gap between the availability of sample database integration and the lack of automated citizen services at the local level, and the ongoing challenge of balancing policy compliance with modern technological opportunities within a limited pilot scope.

Through an exploration of both international best practices in piloting as well as the weaknesses of Sri Lanka's existing domestic PCC process within individual divisions, this chapter lays the foundation of knowledge necessary to inform the solution proposed. The discussion identifies localized automation, simulated integration functionality, and workflow standardization as essential principles that need to underlie any effective digital transformation pilot of the PCC system.

2.2 Conceptual Map of the literature review

The conceptual map positions the literature around the problem of Sri Lanka's domestic PCC system at the divisional level and extracts design implications for the proposed pilot solution. At the core lies the current fragmentation of the process within specific divisions: while Sri Lanka already possesses internal police databases, these are disconnected from the citizen-facing PCC process at the local level. Applicants in divisions like Seeduwa and Raddolugama still rely on physical Grama Niladhari (GN) letters and manual police station verifications, producing delays, opacity, and inconsistency.

In contrast, comparative studies highlight pilot automated systems in selected regions of countries such as the Philippines and India that demonstrate reduced processing times, minimized corruption, and enhanced transparency within specific operational areas. These localized implementations show that integrating mock operational databases with standardized citizen workflows, underpinned by digital verification and audit trails, can be achieved effectively at the pilot level.

From the Sri Lankan pilot perspective, the conceptual map emphasizes three key requirements: divisional workflow orchestration to eliminate station-level disparities within the pilot area, integration of sample data sources with external citizen-facing services for testing purposes, and embedding GN and local verification requirements into digital templates and processes rather than bypassing them. Collectively, these themes guide the strategic design of the proposed pilot PCC automation system, ensuring it is technically feasible for divisional implementation, policy-compliant, and citizen-friendly.

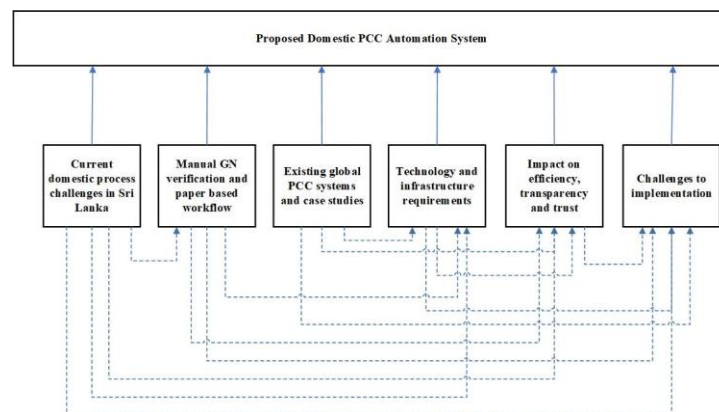


Figure 3: Conceptual Map

2.3 Thematic Review

2.3.1 The context e governance and digital transformation

With increased efficiency, transparency, and citizen-oriented service delivery, e-governance and digitalization have been identified as key drivers of public sector modernization globally. Integration of information and communication technology is especially crucial for law and order and administrative services like the Police Clearance Certificate (PCC) issuing process due to the fact that it simplifies procedures, reduces delays and enhances public satisfaction at the divisional level.

The potential of digital platforms to solve longtime public sector issues can be seen through pilot deployments and effective regional rollouts within nations that have emerging e-governance frameworks. These pilot studies emphasize the value of automation, inter-agency interoperability, and system centralization in targeted areas of operation. For instance, regional rollouts that have piloted automated background checks and localized law enforcement database access have shown reduced processing times, less human error, and enhanced public trust in their pilot units [4].

The current PCC issuing process in Sri Lanka's divisional areas like Seeduwa and Raddolugama, which remains dependent on manual processes and disjointed institutional coordination, can be transformed with the help of such pilot experience. This study illustrates how the inefficiencies of Sri Lanka's law enforcement departments at the divisional level can be addressed by localized technology interventions through a best-practice analysis of pilots. The lessons from this can enable the establishment of a more consistent and easily accessed divisional PCC system as a platform for replication on a broader scale.

2.3.1.1 E-governance and Digital Transformation in Sri Lankan Divisions

E-governance implementation in Sri Lankan divisions remains at a relatively low level when compared to pilot programs in more developed regions, where digital government services have become more localized, citizen-centric, and accessible within specific operational areas. Although Sri Lanka has taken important steps towards adopting e-governance, particularly through initiatives aimed at digitalizing selected public services, the overall implementation at the divisional level is still limited in scope, integration, and efficiency. [5]

Several divisional public sector processes, including those related to law enforcement and administrative services like the Police Clearance Certificate (PCC) issuance in areas such as Seeduwa and Raddolugama, continue to rely on manual and paper-based systems and according to findings all around the country used the same manual approach in different (no standardization). This has resulted in inefficiencies such as delays, lack of transparency, and fragmented communication between local departments. In contrast, pilot implementations in other regions have adopted centralized, automated systems that allow real-time data sharing within specific divisions, cross-institutional collaboration at the local level, and online service delivery, significantly improving divisional responsiveness and citizen experience [6].

While Sri Lanka has made progress in developing digital infrastructure and launching platforms for selected services, the lack of system interoperability at the divisional level, limited access in specific areas, and varying digital literacy levels remain barriers to effective pilot implementation. Moreover, the absence of a unified digital governance strategy across divisional government institutions weakens the overall impact of localized e-governance efforts.

Therefore, there is a growing need to strengthen Sri Lanka's divisional digital framework by learning from international pilot practices and tailoring those approaches to local divisional needs. Enhancing digital transformation, particularly in critical sectors like law enforcement at the divisional level, can not only improve operational efficiency but also build public trust and accountability. This research aims to contribute toward that transformation by focusing on the development of an automated and pilot PCC issuing system as a key step in modernizing Sri Lanka's divisional e-governance landscape [7].

2.3.2 Current Status of PCC System in Pilot Divisions: Public Trust, Transparency and Accuracy Rates

The current Police Clearance Certificate (PCC) issuance system in Sri Lankan divisions like Seeduwa and Raddolugama, though functional, remains largely outdated and inefficient when measured against pilot digital service delivery standards. It operates through a partially digital framework where only applications for foreign employment are facilitated through an online platform. However, this service remains limited in scope for domestic applications within these divisions, excluding local employment needs and other categories of applications.

The manual nature of the process, especially for domestic PCC applications within the pilot divisions, has led to a number of issues that directly affect public trust and satisfaction at the local level. Research findings and feedback from regular users within these divisional areas indicate widespread dissatisfaction due to delays, lack of transparency, and the need for repetitive documentation, such as obtaining a Grama Niladhari certificate confirms to the resident's status inside the designated division.

Record keeping within these divisions is still performed using physical logbooks and paper files, which increases the risk of human error, misplacement, and delays in data retrieval, ultimately affecting the accuracy and timelines of issued certificates. A survey conducted as part of this research among users from the Seeduwa and Raddolugama divisions highlighted significant shortcomings in the current divisional system [8].

Respondents from these pilot areas reported challenges including long waiting periods, limited tracking of application status, inconsistent service levels across local police stations, and poor communication between applicants and the divisional authorities. The survey results also suggested that the manual process lacks transparency at the divisional level, as applicants are not clearly informed about the status or progress of their requests within these specific areas, which undermines public trust in the local service.

Moreover, the absence of an integrated system across police stations and relevant administrative divisions within the Seeduwa and Raddolugama areas contributes to inefficiencies and fragmented communication. While internal systems exist for police officers to access criminal history records,

these are not linked to an end-to-end automated PCC issuance platform at the divisional level, limiting their utility for streamlining the full clearance process within these pilot areas.

These findings from the pilot divisions reflect the urgent need for a more centralized, transparent, and accurate divisional system that can restore public confidence and improve service quality at the local level. By adopting a pilot automated PCC platform that enables real-time data sharing and application tracking within these divisions, Sri Lanka can test and refine law enforcement services alignment with modern standards, enhance accountability, and meet the evolving expectations of citizens for faster, more reliable public services [9].

2.4 Case Studies from Pilot Implementations: Digital Transformation in Divisional Governance Services

To improve divisional public sector delivery, especially in the context of issuing Police Clearance Certificates (PCCs), several regions have embraced digital transformation through pilot e-governance initiatives. These case studies from selected divisional implementations demonstrate how the adoption of localized systems, sample databases, and verification mechanisms have significantly improved transparency, accuracy, and public satisfaction in law enforcement services at the regional level.

India - Regional CCTNS Implementation India implemented pilot phases of the Crime and Criminal Tracking Network and Systems (CCTNS) in selected districts to enhance policing through digital transformation. These regional pilots connected specific police stations to centralized databases, facilitating real-time access to criminal records and complaints tracking within designated areas. Through divisional Digital Police Portals, services like online police clearance certificates (PCC) and criminal background checks were made available to the public in pilot regions. These divisional implementations enhanced transparency, reduced paperwork, and accelerated the verification process within specific operational areas [10].

Philippines - Divisional NPCP Pilots, National Police Clearance System (NPCP) in the Philippines conducted pilot implementations in selected regions to test and refine the process of obtaining police clearance. Through biometric verification and integration with regional databases, applicants in pilot areas could request and receive PCCs digitally. These regional transformations reduced manual errors, improved processing speeds within specific divisions, and supported real-

time validation. The divisional NPCS pilots reflect a strong Government to Citizen (G2C) e-governance model by enhancing efficiency and reducing corruption within the pilot service delivery areas [11].

Kenya - Regional eCitizen Platform Testing, Kenya's eCitizen platform conducted pilot testing of digitalized public services in selected regions, including police clearance certificates issued by regional Directorate of Criminal Investigations (DCI) offices. The pilot platform enabled applicants in test areas to request PCCs online, pay via mobile money, and receive digital notifications. Kenya's pilot model, driven by accessibility and mobile-first design, reduced delays within test regions, minimized corruption, and brought services closer to specific communities, making it an excellent example of regional digital public sector reform [12].

Singapore - Pilot COC Implementation Singapore conducted pilot testing of its digitalized Certificates of Clearance (COC) application process in selected administrative areas, integrating it with regional access to Sing-pass, the national digital ID system. Citizens and residents in pilot areas could apply for PCCs online, upload required documents, and receive digital certificates without visiting police stations. These pilot implementations demonstrated how trusted digital identity infrastructure enhances public trust and operational efficiency at the regional level [13].

Ghana, is progressing toward an e Policing model, integrated through the GCNet platform, to enhance crime management and public service delivery. The digitization of police clearance services enables quicker processing, reduces human interaction, and increases transparency. Ghana's approach, focusing on building inter agency digital infrastructure, is a critical step in its broader public sector reform and highlights the role of technology in modernizing law enforcement in African contexts [14].

Case studies from these countries demonstrates the transformative potential of digital governance in improving law enforcement services, especially in delivering efficient and transparent police clearance systems. Countries like India, the Philippines and Kenya showcase how digital platforms can reduce delays, minimize corruption, and enhance public access to critical services. Moreover, examples from the UAE [15] and Singapore highlight the role of advanced technologies and national digital identity systems in delivering seamless and citizen centric services. By studying these diverse implementations, Sri Lanka can draw valuable lessons to design and implement its own data driven, secure and efficient digital police clearance system. Such a system would not

only improve administrative efficiency but also strengthen public trust, enhance national security, and contribute to broader e governance reforms.

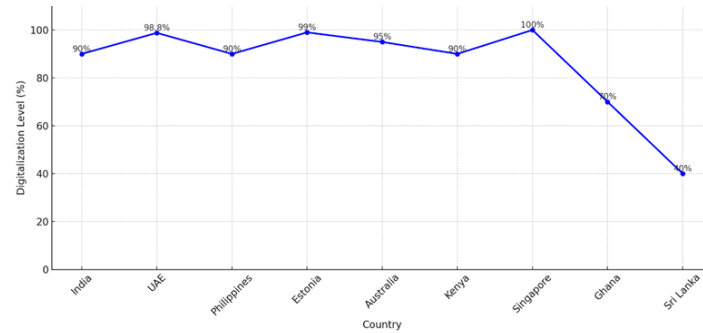


Figure 4: User Satisfaction Rates

In conclusion, the case studies from these regional and divisional implementations demonstrate the transformative potential of pilot digital governance programs in improving law enforcement services, specifically in delivering efficient and transparent police clearance systems within specific operational areas. implementations in countries like India, the Philippines, and Kenya showcase how regional digital platforms can reduce delays, minimize corruption, and enhance public access to critical services within designated test areas.

By studying these diverse pilot implementations, Sri Lanka can draw valuable lessons to design and implement its own divisional, data-driven, secure and efficient digital police clearance system within the Seeduwa and Raddolugama pilot areas. Such a pilot system would not only test administrative efficiency improvements but also help build public trust, enhance local security processes, and contribute to broader divisional e-governance testing and refinement [16].

2.4.1 Information Systems Implementation in Law Enforcement for Divisional E-governance

The integration of information systems into law enforcement within divisional e-governance frameworks has become a critical aspect of modern public administration testing and implementation. In the context of Sri Lanka's Police Clearance Certificate (PCC) issuing process within pilot divisions like Seeduwa and Raddolugama, the implementation of a digitalized information system can significantly improve the efficiency, reliability, and accessibility of the service at the local level.

Currently, the PCC process in these Sri Lankan divisions is largely manual, fragmented, and often delayed, causing frustration among the local public and leading to issues in transparency, data accuracy and administrative burden within these specific areas. By introducing an information system-based digital solution for pilot testing, the entire PCC process (from application submission and identity verification to record checking and certificate issuance) can be streamlined and tested within these designated divisions [17].

Such a pilot system can be integrated with sample national identity databases, mock criminal record systems, and local public service platforms, enabling real-time data validation and automated decision-making testing within the pilot scope. The advantages of implementing an information system-based solution for the pilot PCC issuance process include improved efficiency within the test area, enhanced transparency and accountability at the divisional level, increased public trust in local services, and improved data security and accuracy for the pilot implementation.

Therefore, implementing such a pilot system in Sri Lanka's designated divisions, inspired by successful regional digital models, would contribute greatly to testing the digital transformation of public services and divisional e-governance goals. It would also help align with regional digital strategies and provide a foundation for broader implementation, ensuring that law enforcement becomes more citizen-friendly, efficient, and future-ready at the divisional level [18].

2.4.2 Pilot Standardization, Public Trust, Transparency and Efficiency

The current Police Clearance Certificate (PCC) issuing process varies across different divisions in Sri Lanka, leading to inconsistencies in service quality, delays, and lack of public confidence at the local level. Through pilot digitalization within designated divisions like Seeduwa and Raddolugama, it is possible to establish and test standardized and unified processes, ensuring that citizens within these pilot areas receive consistent service quality.

Pilot standardization involves implementing a single digital platform that designated police stations within the pilot divisions must use to process PCC applications. This unified system would test common guidelines, data formats, verification protocols, and certificate issuance procedures, ensuring that the entire process within the pilot area is consistent and free from regional variations or individual discretion. The pilot system can be monitored and maintained by designated project teams during the testing phase [19].

To make this pilot transformation successful, focused training programs should be conducted for staff within the pilot divisions. Officers and clerical staff in Seeduwa and Raddolugama areas must be trained to use the pilot digital system effectively, including how to handle digital identity verification, update or query sample records, manage applicant information securely, and resolve technical issues. These training sessions should be designed in phases: initial onboarding for pilot staff, hands-on practice within the test environment, and continuous support during the pilot period.

The benefits of this pilot digitalized standardization include increased efficiency within the test divisions, enhanced transparency at the local level, boosted public trust in pilot services, better decision-making capabilities during testing, reduced corruption potential, and inclusive access testing within the designated areas [20].

In conclusion, pilot digital standardization not only tests operational efficiency and transparency improvements but also plays a vital role in building public trust in law enforcement at the divisional level. A centralized and well-trained police force within the pilot areas, supported by robust digital infrastructure testing, is essential for delivering equitable and accountable services in the modern e-governance era.

2.5 Technological Analysis for Divisional PCC System Implementation

The technological infrastructure required for implementing automated PCC systems in divisional areas requires careful consideration of system architecture, database integration, and security frameworks. Modern e-governance systems rely on service-oriented architectures (SOA) that enable seamless integration between multiple government databases and citizen-facing applications [21]. For Sri Lankan divisional implementations, cloud-based solutions offer scalability advantages, particularly for pilot testing in areas like Seeduwa and Raddolugama, where infrastructure limitations may constrain traditional on-premises deployments.

Database integration represents a critical technological challenge for PCC automation. Research demonstrates that successful implementations require Application Programming Interface (API) frameworks that can securely connect disparate systems such as national identity databases, criminal records management systems, and local administrative platforms. The use of RESTful APIs and microservices architecture has proven effective in similar pilot implementations, enabling modular development and testing while maintaining system [22].

Authentication and digital identity verification constitute essential technological components for automated PCC systems. Multi-factor authentication mechanisms, including One-Time Password (OTP) verification and digital signature integration, have shown effectiveness in preventing fraudulent applications while maintaining user accessibility [23]. Additionally, blockchain-based certificate validation systems are emerging as potential solutions for ensuring document authenticity and preventing tampering in law enforcement applications (out scope).

Security considerations encompass data encryption, secure transmission protocols, and audit trail mechanisms that are fundamental for maintaining public trust in divisional e-governance systems. Implementation of ISO 27001 security standards and regular vulnerability assessments are recommended for pilot systems handling sensitive law enforcement data [24].

2.6 Research Gap

Although several countries have taken steps to digitalize public services in law enforcement through pilot programs, there is a clear lack of research focused specifically on automating the Police Clearance Certificate (PCC) process at the divisional level, particularly in Sri Lankan contexts like Seeduwa and Raddolugama divisions. Although many countries have implemented regional digital verification systems, Sri Lanka still relies on outdated manual or partially digital methods that are slow and inefficient at the divisional level.

A review of past studies shows that no significant research has been done on a fully automated divisional PCC system since 2015 for local PCC in Sri Lanka. This has resulted in continued operational challenges within specific divisions, including delays, inconsistent procedures across local police departments, lack of integration with other relevant local government bodies, and limited accessibility for the public within these areas.

Important local aspects such as collaboration with divisional Grama Niladharis and relevant police officers, standardization of processes within specific areas, and public digital awareness at the local level have also been overlooked in existing literature. Therefore, there is a noticeable gap in research and implementation when it comes to a complete, information system-driven PCC solution tailored to Sri Lankan divisional needs, specifically for pilot testing within designated areas like Seeduwa and Raddolugama.

This study addresses that gap by proposing a modern, fully automated pilot system that improves speed, accuracy, transparency, and accessibility within these designated divisions. It aims to support Sri Lanka's broader e-governance goals while enhancing public trust and service delivery in the law enforcement sector at the divisional level through focused pilot implementation and testing [25].

CHAPTER 03: METHODOLOGY

3.1 Research Paradigm

This study is grounded in the pragmatism research paradigm, which emphasizes practical problem-solving through a balanced application of both positivist and interpretivist perspectives. Pragmatism is particularly relevant for research projects that aim not only to generate knowledge but also to produce tangible solutions to real-world challenges. Since the inefficiency of the Police Clearance Certificate (PCC) issuance system in Sri Lanka is both a practical issue (delays, errors, lack of transparency) and a social problem (the dissatisfaction of citizens, the workload of officers), pragmatism offers the perfect philosophical basis.

Unlike purely positivist or purely interpretivist paradigms, pragmatism allows to combine quantitative evidence with qualitative insights to design, implement, and evaluate an innovative IT-based artifact. This dual lens ensures that the research goes beyond measuring variables and also considers the lived experiences of stakeholders, thereby creating a system that is both effective and user-friendly.

Justification for Pragmatism

The justification for adopting pragmatism lies in its ability to integrate both positivist and interpretivist perspectives in a balanced and solution-oriented manner. From a positivist standpoint, the study employs quantitative analysis to measure the efficiency of the proposed PCC automation system, using indicators such as average processing times, user satisfaction levels, and adoption willingness across different demographic groups. The efficacy of the artifact is empirically demonstrated by these measurable outcomes. At the same time, an interpretivist perspective is necessary to capture the lived experiences and contextual insights of stakeholders, including applicants, police officers, and administrators. Through interviews and feedback sessions, the research identifies usability concerns, trust issues, and barriers such as digital literacy, ensuring that the system is not only technically robust but also socially acceptable. The pragmatic stance allows the integration of these complementary approaches, avoiding the rigidity of a single paradigm and focusing instead on practical outcomes. This integration is operationalized through the Design Science Research (DSR) methodology, where quantitative evaluation validates the artifact while qualitative insights guide iterative improvements. By combining empirical evidence

with stakeholder perspectives, pragmatism ensures that the research outcome is both effective in solving the inefficiencies of the current PCC process and adaptable to the real-world needs of its users.

3.2 Research Approach

This research employs a hybrid deductive inductive methodology, integrating the advantages of both reasoning techniques to guarantee that the creation of the PCC automation system is both theoretically supported and empirically verified. [26]

From a deductive perspective, the study begins by examining existing e-governance frameworks and international PCC automation models, such as those implemented in India, Singapore, and the Philippines. These countries have demonstrated improvements in efficiency, transparency, and user trust through digital transformation in law enforcement services. By drawing upon these established theories and models, the study formulates initial assumptions and hypotheses, such as:

- Automating the PCC process will significantly reduce processing time compared to the manual system.
- A digital platform will enhance transparency, thereby improving citizen trust in law enforcement services.
- Streamlined workflows will minimize human error and resource inefficiencies.

These deductively derived assumptions provide a structured foundation for the design and development of the proposed IT-based artifact.

At the same time, an inductive reasoning approach is employed to ensure that the solution is contextually relevant and adaptable to the unique socio-technical environment of Sri Lanka. Survey data from applicants and law enforcement officers in Seeduwa and Raddolugama police divisions are analyzed to uncover patterns and insights regarding existing challenges. These include citizens' dissatisfaction with long waiting periods, difficulties in document submission, and officers' concerns over manual workload. Furthermore, qualitative feedback gathered during prototype testing is used to identify usability issues, trust concerns, and potential barriers to adoption, such as varying levels of digital literacy across age groups.

By integrating deductive and inductive reasoning, the research follows an iterative cycle: theory-driven assumptions are tested against real-world data, and the insights gained from field evidence refine both the system design and the underlying hypotheses. For example, while the deductive assumption may be that “automation improves efficiency,” inductive findings from pilot testing highlight practical concerns such as system navigation issues or network accessibility in certain areas, which require iterative adjustments.

Thus, the hybrid approach ensures that the research is not limited to theoretical validation alone but also produces a practical and user-centered solution. It aligns with the broader Design Science Research (DSR) methodology, where knowledge flows in both directions: from established theory to practice (deduction) and from observed practice back to refined theory (induction).

3.3 Research Strategy

The overall research strategy for this study is grounded in the Design Science Research (DSRM) methodology, which is particularly well-suited for projects aiming to create and evaluate IT artifacts that address real-world problems. In this research, the artifact is the Police Clearance Certificate (PCC) automation system prototype, designed to improve efficiency, transparency, and citizen satisfaction compared to the existing manual process. DSRM provides a strong emphasis on two fundamental components: creating artifacts and evaluating them. The development phase of this study focuses on designing the prototype through iterative cycles of requirement gathering, system design, and implementation, while the evaluation phase assesses the artifact by comparing its performance against the current manual PCC process using both quantitative and qualitative measures. [27]

To complement the core DSRM strategy, two additional research strategies are integrated: survey research and a case study approach.

- **Survey Research:**

A structured survey was conducted among applicants and police officers to collect both quantitative data (e.g., average processing time, satisfaction levels, adoption willingness) and qualitative insights (e.g., challenges faced, suggestions for improvement). This approach enables the study to capture a broad understanding of stakeholder perspectives and validate assumptions

derived from existing e-governance literature. Survey results also provide the statistical basis for evaluating the effectiveness of the proposed system.

- **Case Study Approach:**

To ensure contextual relevance and practical applicability, a pilot case study was carried out in the Seeduwa and Raddolugama police divisions. These locations were chosen as representative study areas to examine how the system performs in real-world operational environments. The case study approach enables in-depth analysis of local workflows, citizen interactions, and administrative challenges, offering insights that go beyond generalized survey data. By observing and testing the system in these divisions, the study ensures that the artifact is refined based on actual field conditions, usability concerns, and stakeholder feedback.

Together, these strategies form a triangulated approach: DSR provides the methodological backbone for artifact creation and evaluation; survey research ensures broad stakeholder representation; and the case study approach grounds the findings in real-world practice. This triangulation strengthens the research's validity and reliability, ensuring that the final solution is both theoretically justified and practically implementable in Sri Lanka's policing context.

3.4 Fact Collection Mechanisms

The study employed a combination of primary and secondary data collection mechanisms to ensure both empirical accuracy and contextual depth. Additionally, using many sources of evidence made data triangulation easier, which improves the conclusions' validity and dependability. [28]

3.4.1 Primary Data Collection

1. Surveys

A structured survey was administered to 184 respondents, comprising 152 applicants who had previously applied for a PCC and 32 police officers directly involved in the clearance process.

- For applicants, the survey captured experiences with the manual PCC process, including average waiting time, satisfaction levels, and perceived challenges (e.g., delays, document handling, lack of transparency). It also assessed applicants' willingness to adopt the proposed online system, as well as differences across age groups.

- For officers, the survey measured administrative workload, error rates, communication delays, and overall readiness to adopt a digital solution.

Responses were analyzed using descriptive and inferential statistics to identify trends and gaps, providing a quantitative basis for evaluating system performance.

2. Interviews

Semi-structured interviews were conducted with Grama Niladhari (GN) officers and police administrators to collect qualitative insights that could not be captured through surveys.

- Topics included workflow bottlenecks, resource limitations, trust in digital systems, and suggestions for process improvements.
- The semi-structured format allowed flexibility to probe deeper into context-specific challenges, thereby enriching the research with practical perspectives from system stakeholders.

3. Prototype Testing

A demo version of the PCC automation system was developed and tested by a selected group of applicants and police officers.

- Applicants assessed usability factors such as ease of navigation, responsiveness, and system transparency (e.g., status tracking).
- Officers tested the verification workflow, document access, and approval process within the system.

Feedback gathered from prototype testing was used to refine system functionalities, ensuring that the artifact was both technically efficient and socially acceptable.

3.4.2 Secondary Data Collection

1. Literature Review on Global PCC Models

Research included case studies of countries such as India, Singapore, and the Philippines, which have successfully implemented digital PCC systems. These models provided valuable insights into best practices, challenges in adoption, and technological frameworks that informed the design of the Sri Lankan prototype.

2. Official Reports and Statistics

Data from the Sri Lanka Police, the Sri Lanka Bureau of Foreign Employment (SLBFE), and related government institutions were reviewed to understand:

- Application volumes processed annually.
- Average turnaround times for manual PCCs.
- Resource allocation and staffing for PCC processing units.

These secondary sources provided a baseline for comparative analysis, allowing the study to quantify improvements introduced by the automated system.

By integrating survey data, interview insights, prototype testing feedback, and official records, the study ensured that the fact collection process was multi-dimensional and holistic. This methodological triangulation not only validated the artifact's design but also ensured that the final system aligns with both global best practices and local stakeholder needs.

3.4.3 Equations for Survey Results Calculation

1. Respondents' Satisfaction with Current PCC Process (Mean Satisfaction Score, MSS):

(Equation 1)

$$MSS = \frac{\sum_{i=1}^n R_i}{n}$$

Where R_i = response score (1–5 Likert scale), n = number of respondents.

Example distribution (152 applicants):

- Score 1: 53
- Score 2: 73
- Score 3: 25
- Score 4: 1
- Score 5: 0

Calculation:

$$\begin{aligned}\sum R_i &= (1 \times 53) + (2 \times 73) + (3 \times 25) + (4 \times 1) + (5 \times 0) \\ &= 53 + 146 + 75 + 4 + 0 = 278\end{aligned}$$

$$MSS = \frac{278}{152} \approx 1.83$$

Result: Average satisfaction = **1.83/5** (Very low).

2. Reported Problems with the Manual PCC Process (IFR, or Issue Frequency Ratio):

(Equation 2)

$$IFR = \frac{N_{issue}}{N_{total}} \times 100\%$$

Where N_{issue} = number of respondents reporting a specific issue, N_{total} = total respondents.

Example issues:

- Time consuming: 41 $\Rightarrow IFR = \frac{41}{152} \times 100 = 27.0\%$

- No status tracking: 35 $\Rightarrow$ 23.0%
- Repeated verification: 31 $\Rightarrow$ 20.4%
- Travel/Distance: 17 $\Rightarrow$ 11.2%
- Miscommunication: 16 $\Rightarrow$ 10.5%
- Document loss: 12 $\Rightarrow$ 7.9%

Result: Most common issue = **time consuming (27.0%)**, but several other issues are also widely reported.

3. Average Processing Time (APT) – Processing Day Distribution (Manual PCC):

(Equation 3)

$$APT = \frac{\sum_{i=1}^n D_i}{n}$$

Where D_i = processing days reported by respondent i .

Example (manual PCC, 152 applicants):

Total processing days reported = **959 days**

$$APT = \frac{959}{152} = 6.31 \text{ days}$$

Result: Average processing time = **6.3 days** (~1 week).

4. Willingness to Adopt the Online PCC System (Adoption Rate, AR):

(Equation 4)

$$AR = \frac{N_{willing}}{N_{total}} \times 100\%$$

Where $N_{willing}$ = number of respondents willing to adopt online PCC, N_{total} = total respondents.

Example:

$N_{willing}=111$, $N_{total}=152$

$$AR = \frac{111}{152} \times 100 = 73.0\%$$

Result: Adoption rate = **73.0%**.

3.5 Research Methodology Execution Workflow

Table 2: Research Methodology Execution Workflow

Step	Activity	Application in Research
Problem Identification	Identifying inefficiencies in the manual PCC process.	Manual PCC takes 7–14 days, prone to delays, errors, and lack of transparency.
Relevance Justification	Explaining why this research is necessary.	Increasing demand for overseas employment requires faster and more reliable PCC issuance.
Comparative Analysis & Gap Justification	Comparing Sri Lanka's PCC with international models.	India, Singapore, and the Philippines have digitized systems → gap in Sri Lanka's divisional-level automation.
Define & Finalize Objectives	Setting research objectives.	Objectives: Reduce processing time, improve transparency, enhance satisfaction, ensure data security.
Design / Development / Data Management	Building the prototype.	React (frontend), Node.js (backend), MongoDB Atlas (database), integrated with mock NIC & AMI systems.
Evaluation & Communication	Testing and reporting outcomes.	Surveys, equations-based analysis, performance testing, dissemination via thesis.

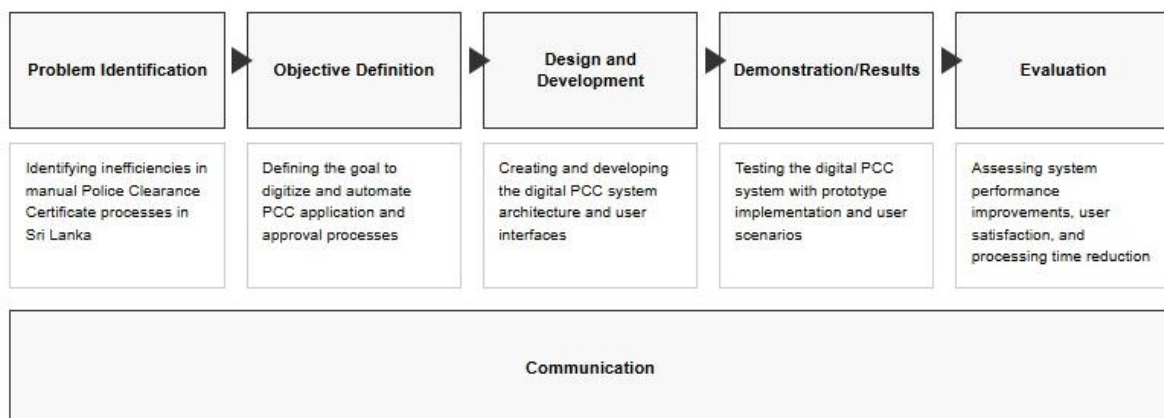


Figure 5: Research Methodology Execution Workflow Diagram

Source: https://drive.google.com/file/d/1cR_X1wKCBSSldNMtOgAeTtvCH9DEN7q1/view?usp=drive_link

3.6 Conceptual Framework

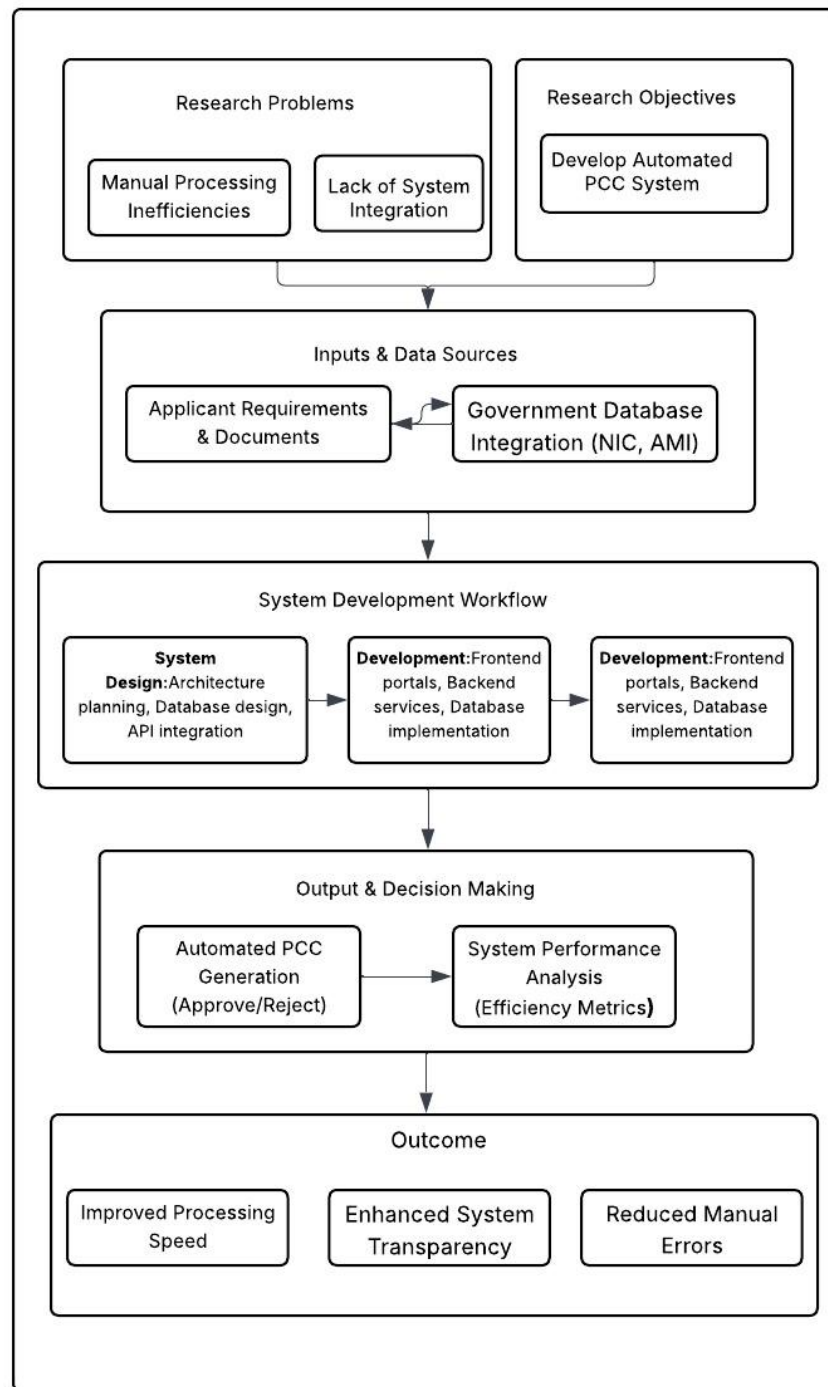


Figure 6: Conceptual framework

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3.7 Project Management Methodology

This research adopted the SCRUM framework, an agile project management methodology, to guide the development of the automated Police Clearance Certificate (PCC) system. SCRUM was chosen because it allows iterative and incremental development, enabling the prototype to be built in functional modules such as the Applicant Portal, Police Officer Portal, and Admin Portal. This approach ensured that the research could deliver working components early, gather real-time feedback from stakeholders, and refine the system continuously. Regular sprint reviews and stakeholder engagement sessions allowed the research team to capture both quantitative insights (from surveys) and qualitative perspectives (from interviews and prototype testing), ensuring that the system design remained responsive to user needs and pilot implementation constraints.

Justification

The justification for using SCRUM lies in its flexibility, adaptability, and focus on collaboration, which align with the nature of this research. Unlike traditional methodologies such as Waterfall or PRINCE2, SCRUM accommodates evolving requirements and facilitates rapid adjustments based on user feedback. This was crucial for the PCC system, where applicants, GN officers, and police administrators provided ongoing insights into usability, workflow improvements, and digital verification needs. By using SCRUM, the research ensured that the resulting system was not only technically functional but also user-friendly, practically implementable, and tailored to the specific divisional context of Seeduwa and Raddolugama, making it the most suitable methodology for this study.

3.7.1 Project Timeline

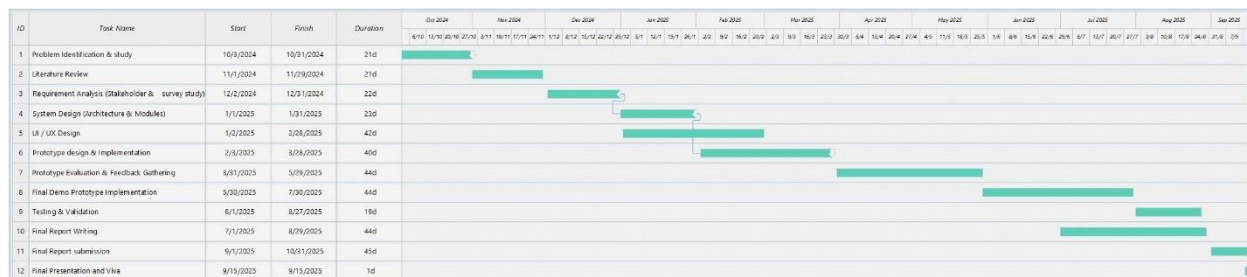


Figure 7: Project Timeline

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3.7.2 Ethical Considerations

Given the sensitivity of law enforcement data, strict ethical measures were followed:

- **Data Privacy:** No real applicant or police data was used. Mock datasets simulated PRD and AMI databases to avoid exposure of personal records.
- **Confidentiality:** Responses from surveys and interviews were anonymized to protect individual privacy.
- **System Security:** The prototype included authentication (JWT, OTP verification), encryption protocols, and role-based access control to simulate real-world safeguards.

3.8 Chapter Summary

This chapter outlined the methodological foundation and research execution plan for the development of an automated Police Clearance Certificate (PCC) system. The study is grounded in the pragmatism paradigm, combining quantitative and qualitative perspectives to produce a solution that is both empirically validated and contextually relevant. A hybrid deductive–inductive approach ensured that theoretical assumptions drawn from global PCC models were refined and validated through field data from Seeduwa and Raddolugama divisions.

The Design Science Research (DSR) strategy guided the creation and evaluation of the IT artifact, complemented by survey research to capture stakeholder perspectives and a case study approach to ensure real-world applicability. A mix of primary and secondary data collection mechanisms, including surveys, interviews, prototype testing, and literature review, enabled triangulation and ensured comprehensive insights. Quantitative survey results were analyzed using defined equations for satisfaction, issue frequency, processing time, adoption willingness, and age distribution.

The chapter also detailed the research methodology execution workflow, linking each research step from problem identification to evaluation and communication. SCRUM methodology was adopted for project management, supporting iterative development, continuous stakeholder feedback, and adaptability to evolving requirements. Ethical considerations, including data privacy, informed consent, confidentiality, and system security, were strictly maintained. Overall, this chapter provides a robust methodological framework that underpins the design, development, and evaluation of a practical, user-centered PCC automation system.

CHAPTER 04: SYSTEM REQUIREMENT SPECIFICATION

4.1 Chapter Overview

This chapter presents the system requirement specification for the proposed domestic Police Clearance Certificate (PCC) automation system. It begins with a stakeholder analysis, which identifies the different user groups and institutions engaged in the PCC process along with their specific roles, expectations, and concerns. Thereafter, the operationalization process is discussed, where survey data analysis and findings are presented through charts and interpretations. These findings are further mapped against the research objectives to demonstrate how the collected evidence supports the development of the proposed solution.

Subsequently, the chapter provides a system analysis to critically examine the limitations of the existing manual PCC process and highlight the improvements that can be achieved through automation. Building on this, the proposed system architecture is outlined, offering a high-level representation of the solution design and its major components. The chapter then specifies the functional requirements, which describe the essential system features and operations, as well as the non-functional requirements, which define quality attributes such as performance, reliability, security, and usability.

Overall, this chapter integrates stakeholder perspectives, survey insights, and system analysis to establish a comprehensive requirement specification that serves as the foundation for designing and implementing the automated domestic PCC system.

4.2 Stakeholder Analysis

The successful implementation of the proposed automated Police Clearance Certificate (PCC) system requires recognizing both direct system stakeholders (who interact with the platform) and indirect stakeholders (who influence the process through external tasks). While the system primarily automates applicant, police officer, and administrator workflows, certain manual steps, such as obtaining a Grama Niladhari (GN) certificate, remain necessary.

Table 3 : Stakeholder Analysis Table

Stakeholder	Role in System	Expectations
Applicant	Submit PCC request, upload documents, track status, download certificate	Easy, transparent process, secure data handling
Police Officer	Verify applications, check criminal records, approve/reject applications	Efficient dashboard, integration with AMI DB, clear workflows
Admin (HQ)	Manage stations, officers, GN divisions, oversee reports	Full system control, accountability monitoring, comprehensive reporting
Grama Niladhari (GN) Officer	Provide residential verification certificate (outside system) that applicant must upload	Simple, standardized certificate process
External Systems	Provide identity verification & criminal record checks	Seamless API integration for demo purposes

Interpretation

From this analysis, it is evident that applicants and police officers are the primary users of the automated PCC system. Applicants benefit from transparency and reduced effort in tracking applications, while officers gain efficiency and reduced manual workload through automated verification tools.

The GN officer's role remains outside of the automation but is still essential to the PCC process. Applicants must physically obtain the GN-certified letter and upload it into the system. This ensures that residential verification continues in a standardized manner, even though it is not digitized in the prototype.

At the administrative level, HQ administrators play a vital role in maintaining oversight, ensuring accountability, and supervising the overall PCC issuance process.

Meanwhile, external systems ensure the technical validity of identity and background verification, simulated through mock integrations for demonstration purposes.

Overall, this analysis confirms that while automation primarily enhances efficiency for applicants, police officers, and administrators, the GN officer remains a critical manual checkpoint in the process, ensuring credibility of applications until future phases potentially digitize this step.

4.3 Operationalization Process

The operationalization process in this research involves translating the broad research objectives into measurable outcomes through the use of surveys, literature findings, and stakeholder perspectives. Data collection instruments such as questionnaires were designed to capture applicants' experiences with the current manual Police Clearance Certificate (PCC) system, identify pain points, and evaluate the acceptance of an automated system. The collected data were then analyzed using descriptive and inferential statistical methods in Google Colab with Python-based libraries (Pandas, Matplotlib, and Seaborn).

The findings from the survey were directly mapped to the research objectives, thereby providing a clear pathway from data collection, data analysis, requirement specification for the proposed automated PCC system. This process ensures that the system design is grounded in empirical evidence and aligns with the expectations of stakeholders. [29]

4.3.1 Survey Data Analysis & Findings

To understand the inefficiencies in the current PCC process and expectations for a digital solution, a structured survey was conducted among 152 respondents who had previously engaged with the PCC application procedure. The analysis covered demographic distribution, user awareness and experience, major pain points, and openness toward automation.

Demographic Distribution of Respondents

The demographic profile shows that the majority of applicants fall into the 26–35 age group, representing working professionals who require PCCs mainly for employment, migration, or academic purposes. Gender distribution showed a slight male majority, though both genders were well represented, ensuring balanced insights.

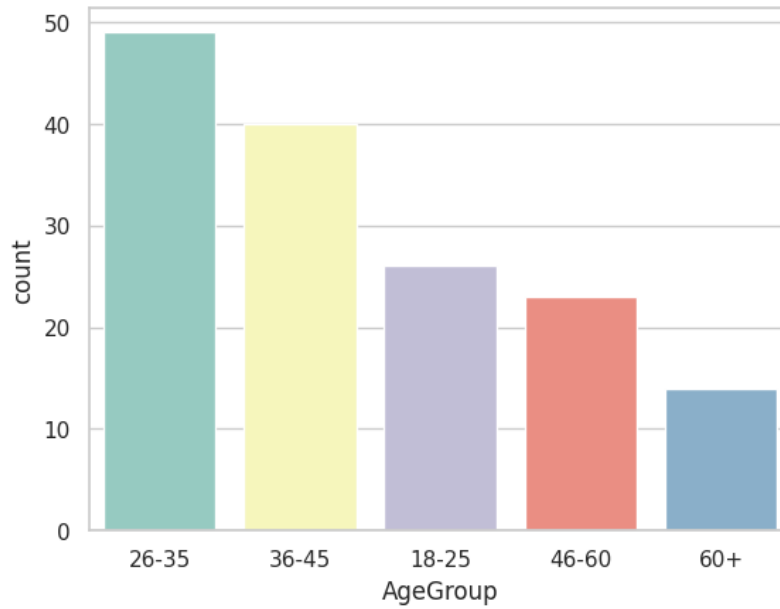


Figure 8 : Age Distribution of Respondents

Interpretation:

The results indicate that the working-age population is the primary user of the PCC system. Since this group frequently applies for employment or higher education, the need for a faster and more reliable process is particularly strong.

Awareness and Experience with Current PCC Process

Survey results revealed that over 70% of respondents considered the current manual process to be highly time-consuming and inefficient. Most applicants had to visit both the Grama Niladhari office and police stations, adding to delays, costs, and inconvenience.

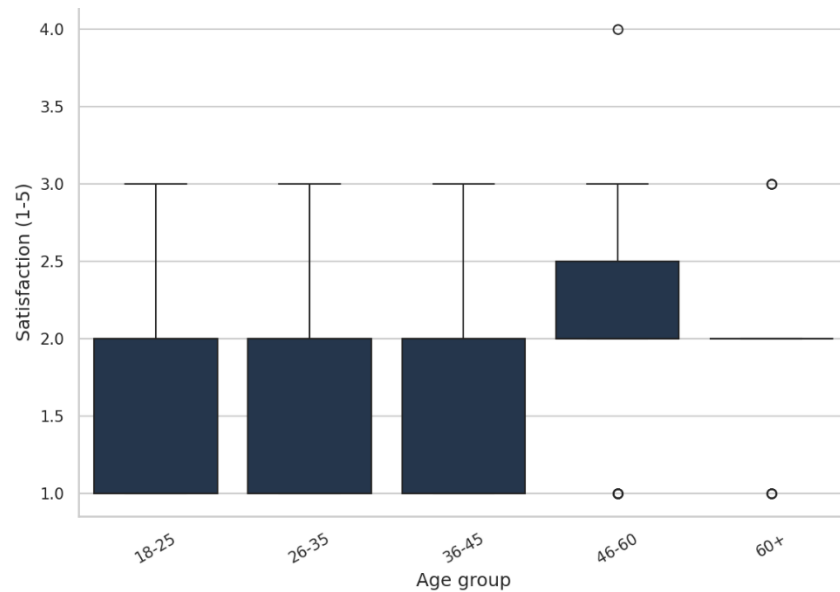


Figure 9 : Respondents' Satisfaction with Current PCC Process

Interpretation:

The overwhelming dissatisfaction emphasizes the inefficiency of the manual PCC process. Applicants expect a streamlined process with reduced physical interactions and better transparency.

Pain Points in the Current System

The most frequently cited issues include:

- Long waiting times (often exceeding two weeks).
- Lack of transparency regarding application status.
- Redundant paperwork and multiple verification steps.
- No digital tracking, leading to frustration and uncertainty.

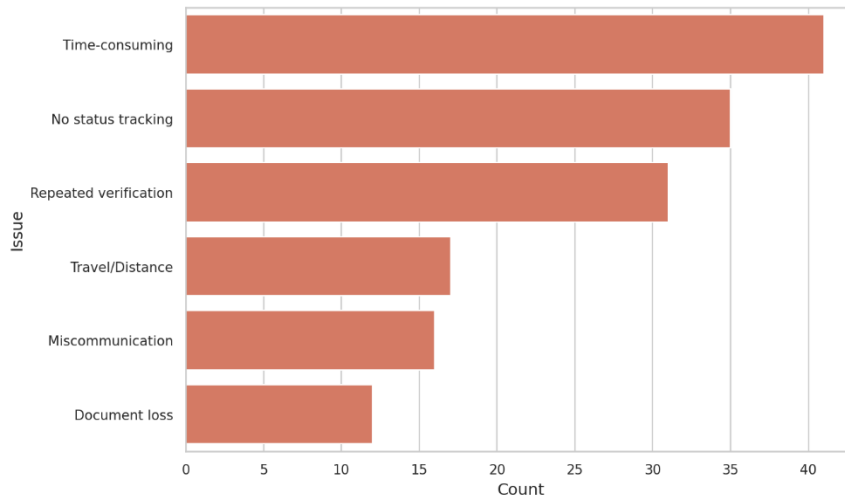


Figure 10 : Reported issues with manual PCC process

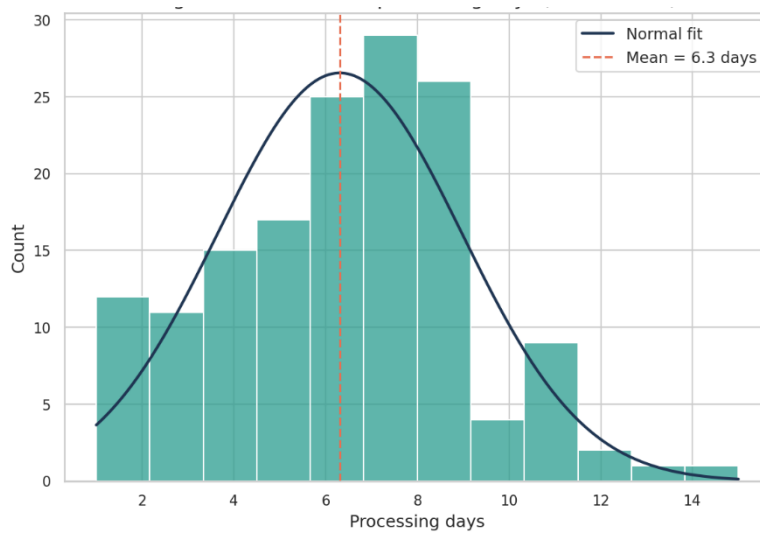


Figure 11 : Distribution of processing days (manual PCC)

Interpretation:

The consistent identification of delays and lack of transparency across all demographics indicates systemic inefficiencies that directly affect applicants and verification officers. These findings provide strong justification for domestic process automation.

Expectations for an Automated PCC System

When asked about adopting a digital platform, 85% of respondents expressed strong support for an online PCC system integrated with national identity and police databases. The most requested features included real-time status tracking, secure online submissions, automated background checks, and faster turnaround times.

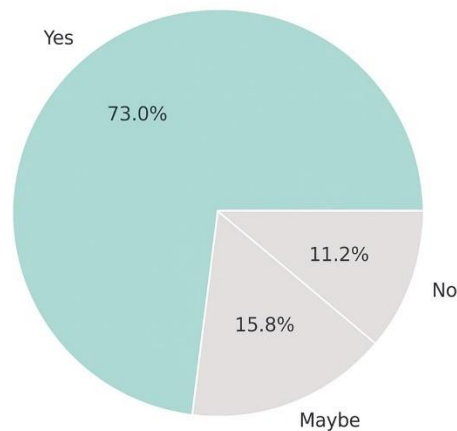


Figure 12 : Willingness to Adopt an Online PCC System

Interpretation:

The majority of applicants favor an automated solution, demonstrating high adoption potential. This aligns directly with the research objective of evaluating acceptance of digitalization and highlights the necessity for a secure, user-friendly online platform.

Overall, the survey results provide quantitative evidence that the current manual PCC system is inefficient and fails to meet user needs. The majority of respondents report dissatisfaction with delays, redundant procedures, and lack of transparency. Conversely, strong support for a digital solution confirms that automation will significantly improve efficiency, accuracy, and satisfaction in the domestic PCC process.

4.3.2 Mapping Survey Results to Research Objectives

The following table demonstrates how the survey findings are mapped to the research objectives outlined in Chapter 1:

Table 4 : Mapping Survey Results to Research Objectives

Research Objective	Related Survey Question(s)	Key Finding	Implication for System Design
Objective 1: Identify inefficiencies in the current PCC process.	Q5, Q6, Q7 (Delays, satisfaction, multiple visits)	72% reported dissatisfaction; 68% experienced delays > 2 weeks.	Automation must minimize waiting time and reduce physical interactions.
Objective 2: Evaluate how satisfied users are with the present manual procedure.	Q4, Q6	Only 8% satisfied with the current system; majority report inefficiency.	Strong evidence that the manual system fails to deliver acceptable service.
Objective 3: Assess acceptance of an automated PCC solution.	Q10, Q11	85% expressed willingness to use an online PCC application system.	High adoption potential ensures feasibility of digital rollout.
Objective 4: Identify essential features expected in a digital solution.	Q12, Q13	Respondents requested real-time tracking, online submission, and secure data handling.	System must include user dashboards, database integration, and encryption protocols.

Objective 5: Ensure trust and security in the proposed system.	Q14	60% raised concerns about data privacy and security.	implement of encryption, safe data storage, and authentication procedures are essential.
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The mapping of survey findings to research objectives clearly demonstrates how the collected data supports the justification for an automated Police Clearance Certificate (PCC) system. The results highlight significant inefficiencies in the manual process, with most applicants dissatisfied due to long delays and multiple visits to institutions. At the same time, the high willingness to adopt a digital solution indicates strong feasibility for system implementation. Respondents emphasized the need for real-time status tracking, online submission facilities, and robust data security, which directly inform the functional and non-functional requirements of the proposed system. This alignment ensures that the design of the automated PCC solution is both evidence-based and user-centered.

4.4 System / Model Analysis

This section presents the system models that describe the proposed automated Police Clearance Certificate (PCC) system. The models illustrate how the system is designed to streamline the PCC issuance process, covering the functional interactions, data structures, process flows, and system communications. These models serve as a blueprint for both the system's implementation and evaluation, ensuring efficiency, security, and scalability.

4.4.1 To-Be Workflow (Proposed Automated PCC System)

The proposed automated process introduces a centralized digital system where applicants can submit PCC applications online. The system integrates with national identity databases and police records, enabling faster verification and approval. Automation minimizes paperwork, reduces human error, and allows applicants to track progress in real time.

Key Characteristics of the To-Be Process:

- Online application submission via web portal or mobile application.
- Automated verification with integrated NIC, AMI criminal record DB and other police databases.
- Digital communication between local police, and central authorities.
- Real-time tracking and status updates for applicants.
- Reduced processing time and improved transparency.
- Secure data handling with authentication and encryption.

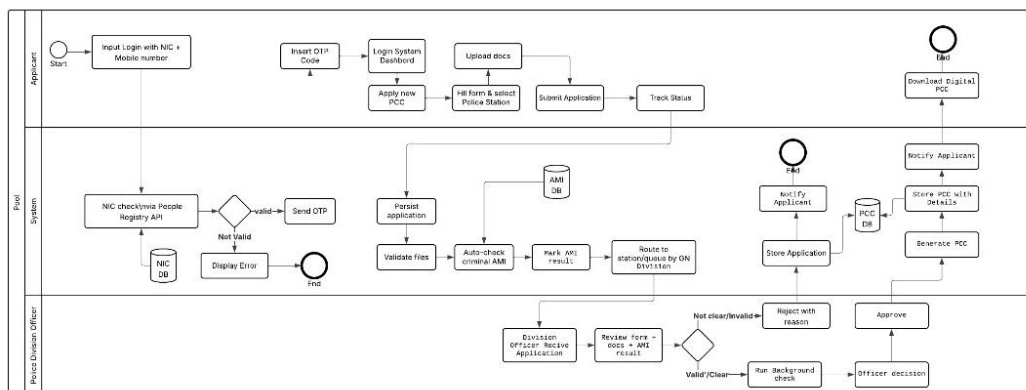


Figure 13:To-Be Workflow of the Automated PCC System

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[nk](#)

4.4.2 Use Case Diagram

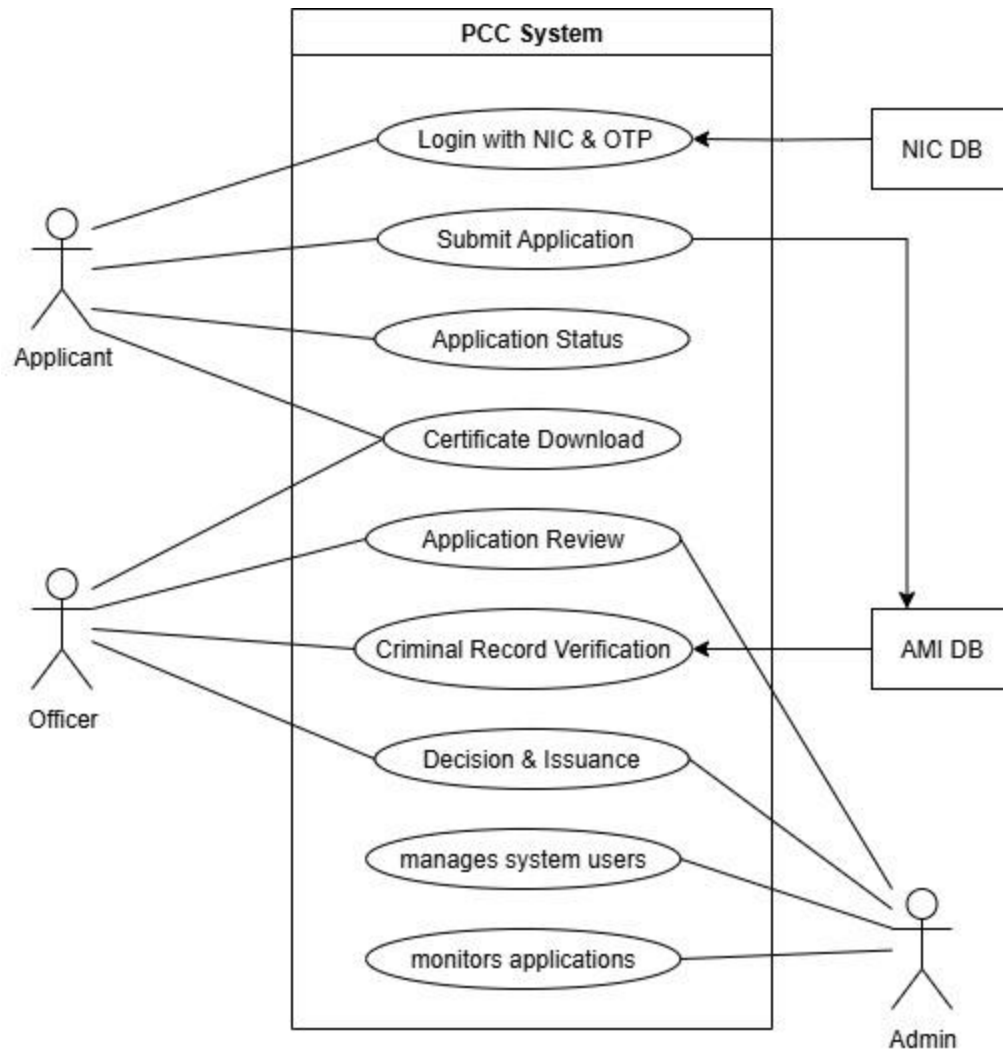


Figure 14: Use Case Diagram

Sources: https://drive.google.com/file/d/1cIJJ9xvmjkuOTNZK8fGj00IfwMcjQxiz/view?usp=drive_link

4.4.2.1 Register & Login with NIC + OTP

Table 5 : Use Case: Login with NIC + OTP

Use Case ID	UC01
Name	Login with NIC + OTP
Description	Allows an Applicant to securely log in using NIC number and OTP verification.
Actors	Applicant, NIC Database
Preconditions	Applicant must have a valid NIC.- Mobile number must be active for OTP verification.
Postconditions	Applicant is logged in successfully. - Applicant session is created.
Main Flow	1. Applicant enters NIC and mobile number.2. System sends OTP to the registered mobile.3. Applicant enters OTP.4. System validates NIC via NIC Database.5. If successful, user is logged in.
Alternate Flows	AF1: Invalid OTP → System prompts re-entry or resend.- AF2: NIC not found → System shows error and stops login.
Exceptions	E1: Database connection failure.
Triggers	Applicant selects "Login".
Assumptions	NIC Database is available and up to date.

4.4.2.2 Submit Application

Table 6: Use Case: Submit Application

Use Case ID	UC02
Name	Submit Application
Description	Allows Applicant to submit a Police Clearance request online.
Actors	Applicant, Officer
Preconditions	Applicant must be logged in.- Required documents must be uploaded.
Postconditions	Application is stored in the system.- Status is set to Submitted.

Main Flow	1. Applicant fills in application form.2. System validates details.3. Applicant uploads required documents.4. Applicant submits application.5. System saves data and notifies the Officer.
Alternate Flows	AF1: Missing documents → System alerts applicant.
Exceptions	E1: Server failure during submission.
Triggers	Applicant clicks "Submit Application".
Assumptions	Required fields and attachments are ready.

4.4.2.3 Criminal Record Verification

Table 7: Use Case: Criminal Record Verification

Use Case ID	UC03
Name	Criminal Record Verification
Description	System and Officer verify applicant's background against AMI Database.
Actors	Officer, Criminal DB (AMI)
Preconditions	Applicant must have submitted an application.- Officer must be logged in.
Postconditions	Verification result (Clear / Hit / Error) is stored.
Main Flow	1. Officer selects application.2. System queries Criminal DB (AMI).3. AMI DB returns result.4. Officer reviews and adds remarks.5. Result stored in system.
Alternate Flows	AF1: No response from AMI DB → Retry option.
Exceptions	E1: System timeout.
Triggers	Officer selects "Verify Records".
Assumptions	Criminal DB is accessible and updated.

4.4.2.4 Decision & Certificate Issuance

Table 8: Use Case: Decision & Certificate Issuance

Use Case ID	UC04
Name	Decision & Certificate Issuance
Description	Officer decides on application and system issues PCC if approved.
Actors	Officer, Applicant
Preconditions	Application must be verified.
Postconditions	Certificate is generated (digital PDF + QR).
Main Flow	1. Officer reviews final application.2. Officer selects Approve/Reject.3. If Approved → System generates PCC with unique QR code.4. Applicant is notified via email/SMS.5. Applicant can download PCC.
Alternate Flows	AF1: Application Rejected → Applicant notified with reason.
Exceptions	E1: System error during certificate generation.
Triggers	Officer finalizes decision.
Assumptions	Certificate template and QR system are functional.

4.4.2.5 Monitor & Generate Reports

Table 9: Monitor & Generate Reports

Use Case ID	UC05
Name	Monitor & Generate Reports
Description	Admin monitors applications and generates system reports.
Actors	Admin
Preconditions	Admin must be logged in.
Postconditions	Reports are generated and displayed/exported.
Main Flow	1. Admin logs into the system.2. Admin navigates to dashboard.3. Admin filters reports (e.g., approved, pending, rejected).4. System generates and displays/export reports.

Alternate Flows	AF1: No data available → Empty report message.
Exceptions	E1: System error in report generation.
Triggers	Admin selects "Generate Report".
Assumptions	Data logs are maintained properly.

4.4.3 Class Diagram

The class diagram provides a static, structural view of the database entities for the Police Clearance Certificate (PCC) system, illustrating how different components are organized and interconnected. It shows key classes such as User, Application, Document, Certificate, AMICheckResult, and CriminalRecord, each with its own attributes. The diagram uses standard UML notation to define the relationships between these classes, specifying multiplicity (one-to-one, one-to-many) and showing how data flows logically from an application submission to the final issuance of a certificate and associated criminal record checks. This model is a blueprint for the database schema, ensuring data integrity and a clear structure for managing the entire PCC workflow.

- User and Application: This is a one-to-many relationship, indicated by the 1 on the User side and the * on the Application side. A single User can submit multiple Applications, but each Application is associated with only one User.
- Application and Document: This is a one-to-many relationship. One Application can have multiple supporting Documents, but each Document belongs to only one Application.
- Application and Certificate: This is a one-to-one relationship, indicated by 1 on both ends. An application can result in the issuance of at most one Certificate, and each Certificate is created for a single Application.
- Application and AMICheckResult: This is a one-to-one relationship. Each Application has one corresponding AMICheckResult, and each result is tied to a single Application.
- AMICheckResult and CriminalRecord: This is a one-to-one relationship. An AMICheckResult is associated with a single CriminalRecord, and each CriminalRecord corresponds to a single AMICheckResult from the system's perspective.

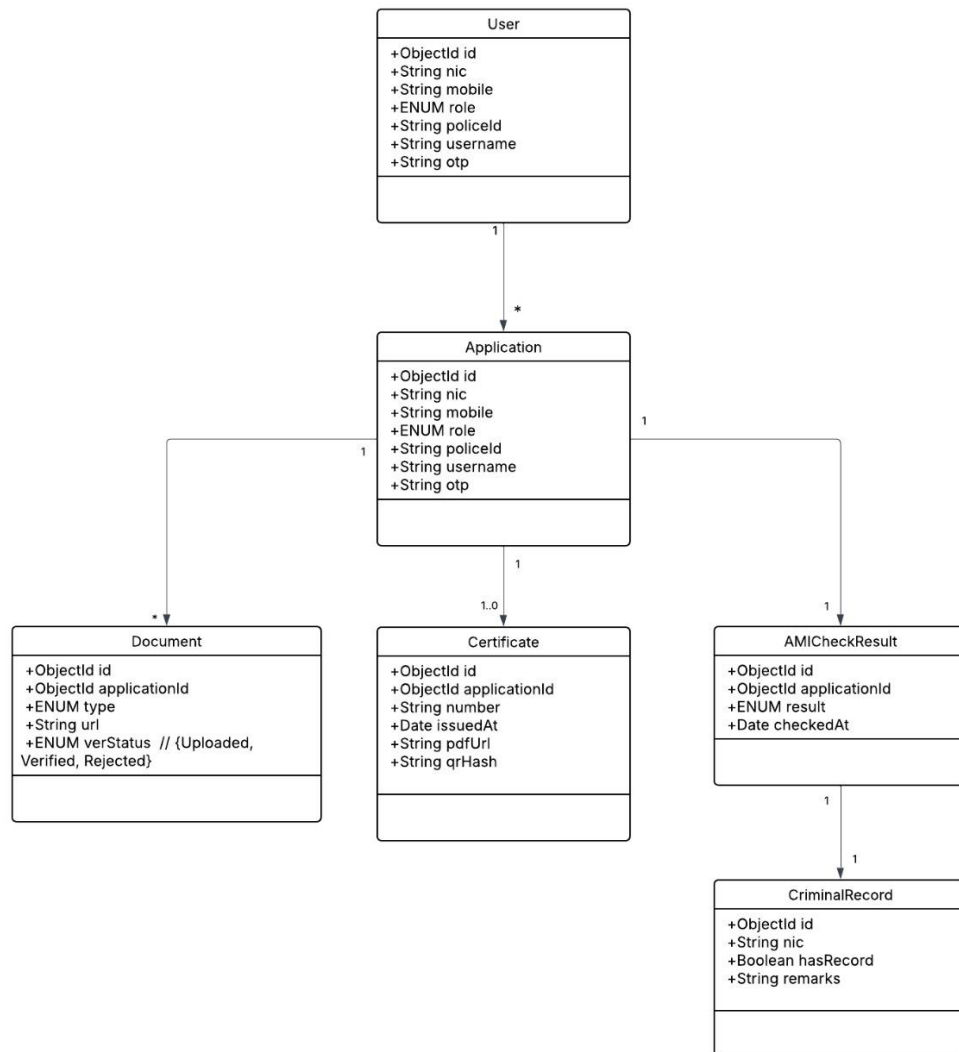


Figure 15:Class Diagram

Source: https://drive.google.com/file/d/1abV32O3XAgWNb4OT3DPLER8enI9TCxB2/view?usp=drive_link

4.4.4 Activity Diagram

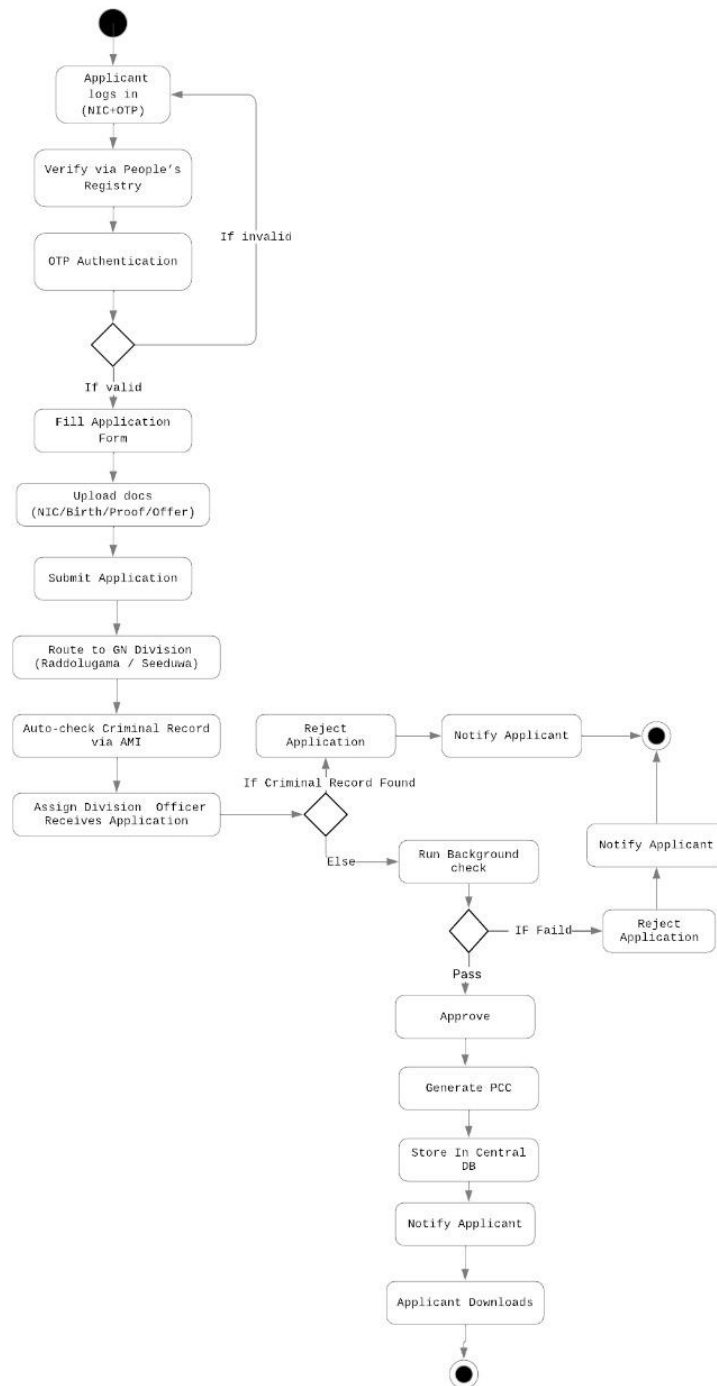


Figure 16:Activity Diagram

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4.4.5 Sequence Diagram

4.4.5.1 Register & Login with NIC + OTP

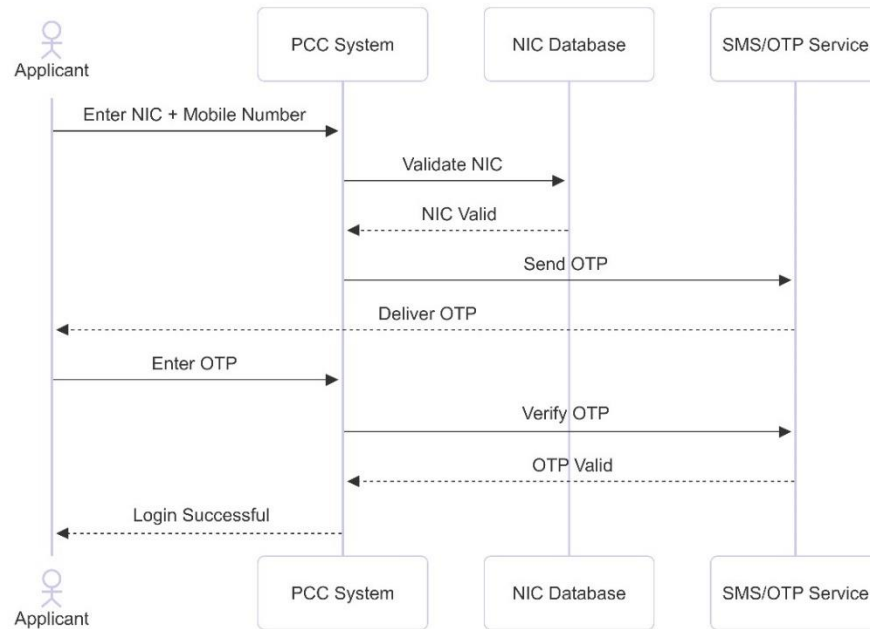


Figure 17:Sequence: Register & Login with NIC + OTP

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4.4.5.2 Submit Application & Upload Documents

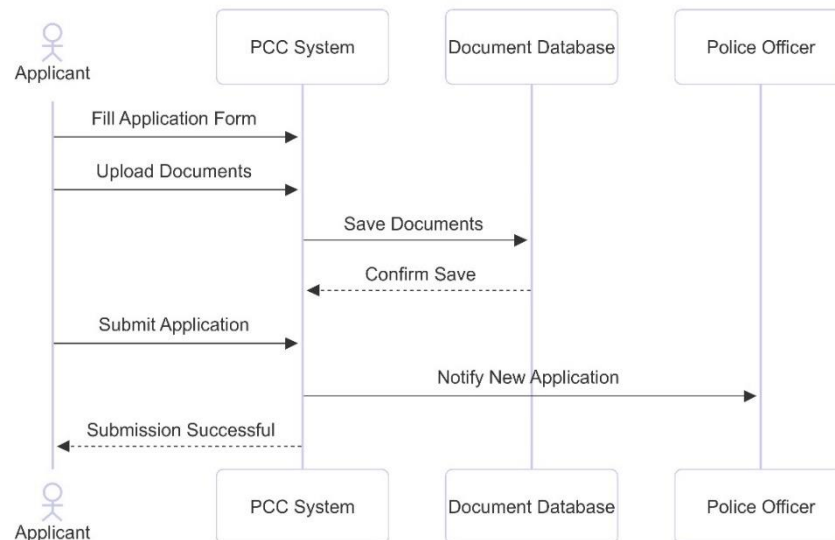


Figure 18:Sequence: Submit Application & Upload Documents

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4.4.5.3 Decision & Certificate Issuance

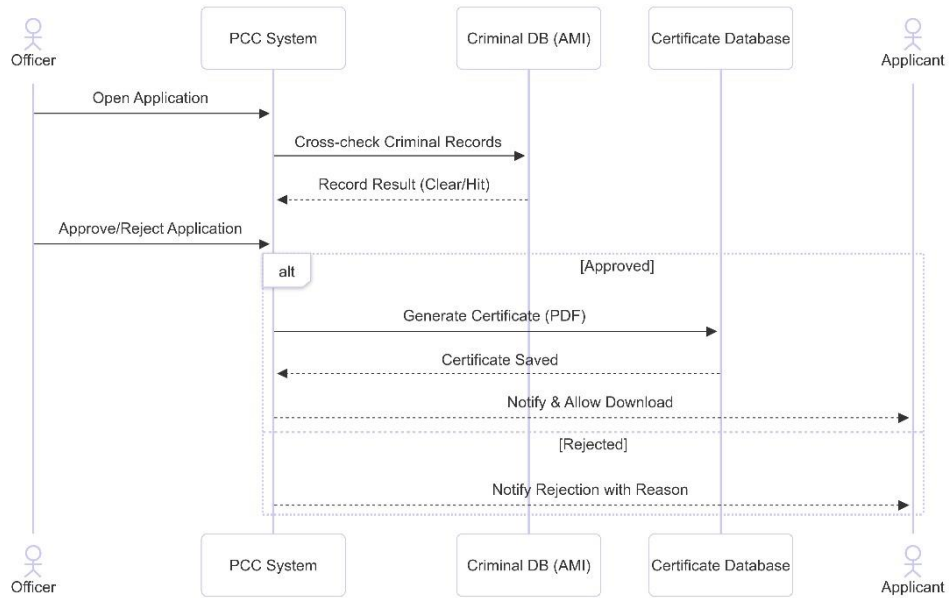


Figure 19:Sequence: Decision & Certificate Issuance

Source: https://drive.google.com/file/d/1Jq0PSqAn4t-52ai1PFxJs9cQZCZVeR8/view?usp=drive_link

4.5 Proposed System Architecture

The proposed automated Police Clearance Certificate (PCC) system is designed as a centralized digital platform that integrates all stakeholders and streamlines the application, verification, and issuance process. The system architecture consists of several interconnected components that collectively ensure efficiency, transparency, and security in the domestic PCC process. [30]

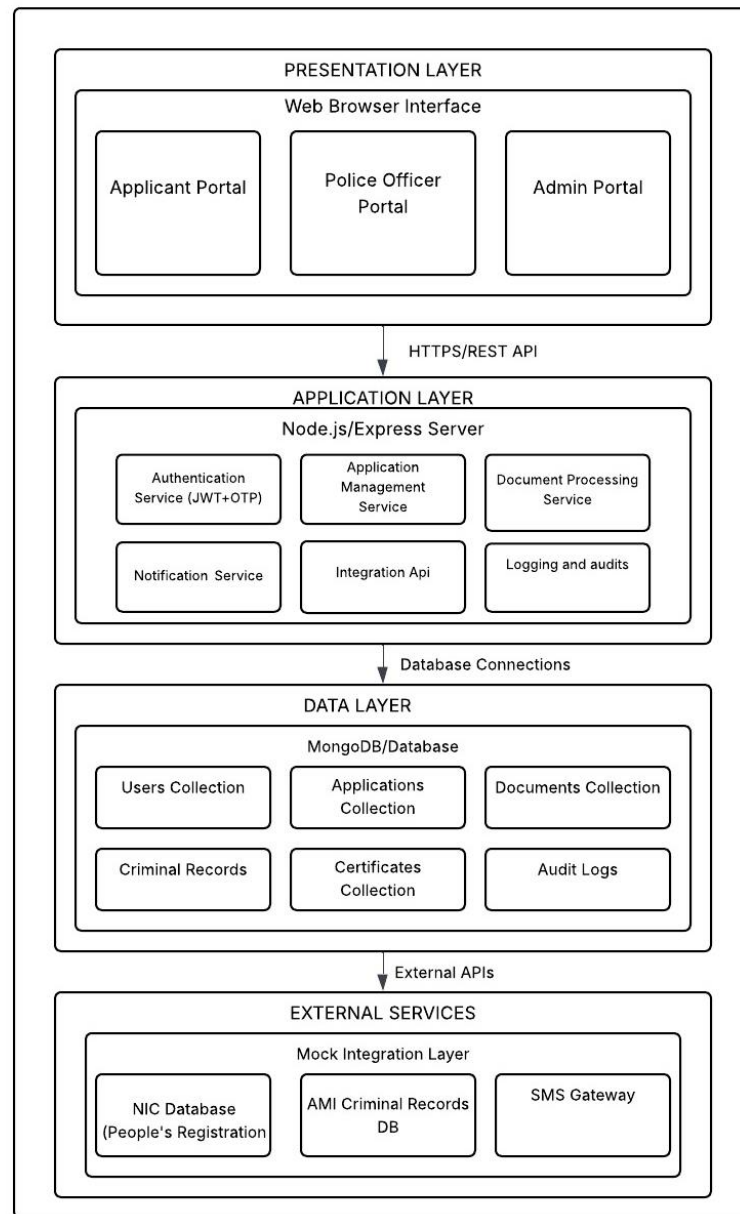


Figure 20:High-Level System Architecture of the Automated PCC System

Source: https://drive.google.com/file/d/1p-t7sK20yEkNQy4uWnNXubt2XjEduBYj/view?usp=drive_link

System Components (Complete System)

1. Applicant Portal

The Applicant Portal serves as the primary interface for users to submit PCC applications online. Applicants can upload required documents, fill out personal and background information, and track the status of their application in real time.

2. Police Officer Portal

This portal is for police officers to review applications, conduct background checks, and update the status of the PCC application based on their findings.

3. Admin Portal

The Admin Portal is for system administrators to manage user accounts, monitor system performance, and access audit logs.

4. Authentication Service (JWT+OTP)

This service handles user authentication and authorization, utilizing JSON Web Tokens (JWT) and One-Time Passwords (OTP) to secure access to the system.

5. Application Management Service

This service manages the lifecycle of the PCC applications, from submission and status updates to final approval or rejection.

6. Document Processing Service

This module processes and validates documents submitted by applicants, ensuring they meet the required criteria before the application proceeds.

7. Notification Service

The Notification Service is responsible for sending alerts and notifications to users via email or SMS regarding their application status.

8. Integration API

This API facilitates communication with external services, such as the NIC database and AMI Criminal Records DB, to retrieve and verify information.

9. Logging and Audits

This component is responsible for recording all system activities and user actions, providing a comprehensive audit trail for security and accountability.

10. MongoDB Database

The MongoDB database is the backend of the system, storing all application data, user information, documents, and audit logs across various collections like Users Collection, Applications Collection, Documents Collection, and Criminal Records.

11. External APIs

This includes various external services the system integrates with to function. The NIC Database is for verifying citizen details, the AMI Criminal Records DB for criminal background checks, and the SMS Gateway for sending notifications.

Interaction Between Components

The components interact in a layered fashion, ensuring a secure and streamlined process. An applicant interacts with the Presentation Layer via the Applicant Portal, and their requests are handled by the Application Layer's Node.js/Express Server. The Authentication Service verifies the user's identity, while the Application Management Service processes the application. The Document Processing Service handles the uploaded files.

The Application Layer communicates with the Data Layer's MongoDB database to store and retrieve information from various collections like the Users Collection and Applications Collection. For background checks, the Integration API in the Application Layer makes calls to External Services like the NIC Database and AMI Criminal Records DB. All activities are tracked by the Logging and Audits component and stored in the Audit Logs collection. The Notification Service is used to inform applicants of their application status via the SMS Gateway. The Police Officer and Admin Portals provide distinct interfaces for their respective users to manage and monitor the process.

4.6 Functional and Non-functional Requirements

This section outlines the functional and non-functional requirements for the proposed automated Police Clearance Certificate (PCC) system. Functional requirements specify the key operations and feature the system must perform, while non-functional requirements define the quality attributes necessary to ensure secure, reliable, and efficient service delivery.

4.6.1 Functional Requirement

Table 10 : Functional requirements

ID	Requirement	Stakeholder	Description
FR1	Applicant Login	Applicant	Login using NIC + Mobile with OTP verification
FR2	Document Upload	Applicant	Upload NIC copy, birth certificate, employment letter
FR3	Application Tracking	Applicant	Real-time status dashboard with notifications
FR4	Certificate Download	Applicant	Download digitally signed PDF with QR verification
FR5	Officer Authentication	Police Officer	Secure login with Officer ID + OTP
FR6	Application Review	Police Officer	View and assess submitted applications

FR7	Criminal Record Integration	Police Officer	Automated check with AMI database
FR8	Approval Workflow	Police Officer	Approve/reject with mandatory comments
FR9	Admin Dashboard	Administrator	Comprehensive system monitoring and management
FR10	User Management	Administrator	Create, update, deactivate user accounts
FR11	Station Management	Administrator	Manage police stations and officer assignments
FR12	Reporting System	Administrator	Generate statistical and performance reports
FR13	Audit Trail	All Users	Complete logging of all system actions
FR14	Notification System	All Users	Email/SMS alerts for status changes

4.6.2 Non-Functional Requirement

Table 11 : Non-Functional Requirement

ID	Non-Functional Requirement	Description / Purpose
NFR1	Security	Ensure data confidentiality, integrity, and authenticity through encryption, secure authentication, and access control mechanisms.
NFR2	Performance	The system must process and verify applications efficiently, with average turnaround time reduced to a few days.
NFR3	Usability	Provide an intuitive, user-friendly interface suitable for all applicants, including multilingual support for Sinhala, Tamil, and English.
NFR4	Scalability	Capable of handling a large number of concurrent applications without degradation of performance.
NFR5	Availability	Ensure high system uptime with minimal downtime for maintenance, supporting continuous application submissions.
NFR6	Data Integrity	Maintain accurate, consistent, and reliable data across all modules and verification stages.
NFR7	Auditability	Record all actions and updates for traceability, accountability, and compliance purposes.

Explanation:

The functional requirements define the core operations necessary for applicants, police officers, and police to interact efficiently with the system, including registration, document handling, verification, notification, and certificate issuance. The non-functional requirements ensure that the system operates securely, reliably, and efficiently, providing a high-quality user experience while maintaining trust and transparency in the automated PCC process.

4.7 System Constraints

4.7.1 Technical Constraints

- Integration Limitations: Mock databases for demo (cannot access live government systems)
- Hosting Environment: Cloud-based deployment on free/low-cost platforms
- Development Timeline: 8-week academic project timeline
- Resource Limitations: Single developer working on prototype

4.7.2 Operational Constraints

- Testing Scope: Limited to Seeduwa and Raddolugama police stations
- User Base: Restricted to willing participants for validation
- Language Support: Primary support for English and Sinhala
- Geographic Coverage: Western Province focus initially

4.8 Risk Analysis and Mitigation

Table 12 : Risk Analysis and Mitigation

Risk	Impact	Probability	Mitigation Strategy
Database Integration Failure	High	Medium	Use mock databases with real-world data structures
User Adoption Resistance	Medium	Low	Extensive stakeholder engagement and training
Performance Issues	Medium	Medium	Load testing and performance optimization
Security Vulnerabilities	High	Low	Security best practices and code review
Timeline Delays	Medium	Medium	Agile methodology with buffer time

4.9 Chapter Summary

This chapter presented the System Requirement Specification (SRS) for the proposed automated Police Clearance Certificate (PCC) system. It began with a stakeholder analysis, identifying direct users (applicants, police officers, administrators) and indirect participants (GN officers and external systems). Though GN officers remain outside automation, their role in providing residential verification certificates was noted as an essential manual input.

The operationalization process translated research objectives into measurable outcomes through surveys and data analysis. Findings revealed inefficiencies in the manual process delays, lack of transparency, and redundant paperwork. Surveys confirmed strong applicant support for a digital solution, emphasizing real-time tracking, secure handling, and faster turnaround times. These insights were directly mapped to research objectives, confirming the system's direction.

System/model analysis contrasted the manual workflow with the proposed automated process. Diagrams (use case, class, activity, sequence) showed how automation improves efficiency, accuracy, and transparency.

The proposed architecture included applicant, officer, and admin portals; authentication and notification services; document processing; integration APIs; and a MongoDB database, ensuring secure and seamless interaction.

Functional requirements covered login, document upload, application tracking, officer verification, reporting, and certificate issuance. Non-functional needs addressed security, performance, usability, scalability, availability, and auditability.

Finally, system constraints (technical, operational, and resource-related) and risk analysis with mitigation strategies were outlined, acknowledging limitations such as restricted testing, mock databases, and timeline pressures.

Overall, the chapter provided a comprehensive requirement specification aligning stakeholder needs, survey insights, and technical feasibility, laying the foundation for a secure, scalable, and user-centered PCC system.

CHAPTER 05: SYSTEM REQUIREMENT SPECIFICATION

5.1 Chapter Overview

This chapter presents the implementation and design of the proposed Automated Domestic Police Clearance Certificate (PCC) System. It explains how the requirements identified in earlier chapters were translated into functional modules that serve applicants, police officers, and administrators.

The chapter begins with framework development and algorithm design, including a flowchart representation of the system workflow. It then elaborates on the backend implementation, focusing on the design of RESTful APIs, business logic, and database handling, followed by the frontend implementation, which explains the role-based portals and their interaction with the backend through API services.

A justification of technology selection is provided in terms of programming languages, libraries, frameworks, and database choices, highlighting their suitability for the system's objectives. The chapter then presents the key implementation features of the prototype, including applicant submission limits, the allocation of one police officer to three Grama Niladhari divisions, workflow management via the admin portal, and integrated messaging and notification functionalities.

Finally, the chapter notes the possibility of future extensions, such as integration with a National Digital ID platform to strengthen identity verification, which remains beyond the scope of the current prototype.

5.2 Elaboration of Key Algorithms and Logic

The proposed Automated Domestic Police Clearance Certificate (PCC) System was developed based on a layered framework to ensure modularity, scalability, and maintainability. Each layer of the framework is responsible for a distinct set of functions, which collectively support the end-to-end PCC issuance workflow.

- Frontend Layer – Provides interactive role-based interfaces for applicants, police officers, and administrators. These portals were developed with responsive design principles, ensuring accessibility and usability across different devices.
- Backend Layer – Implements the core business logic of the system, including application validation, workflow management, officer assignment, notification dispatch, and communication with external services. The backend also integrates with mock databases representing the People’s Registration Department (PRD) and the Automated Management of Information (AMI) system for applicant identity and criminal record verification.
- Database Layer – Responsible for using MongoDB Atlas for persistent data storage. This layer stores applicant records, submitted documents, verification results, officer allocations, messaging logs, and issued PCC certificates. The flexible NoSQL schema design enables efficient handling of dynamic application data. [31]

This layered architecture ensures a separation of concerns, making the system easier to maintain and expand. It also provides the foundation for incorporating future extensions, such as integration with a National Digital ID system for stronger identity verification once it is formally introduced.

5.2.1 Flow Chart

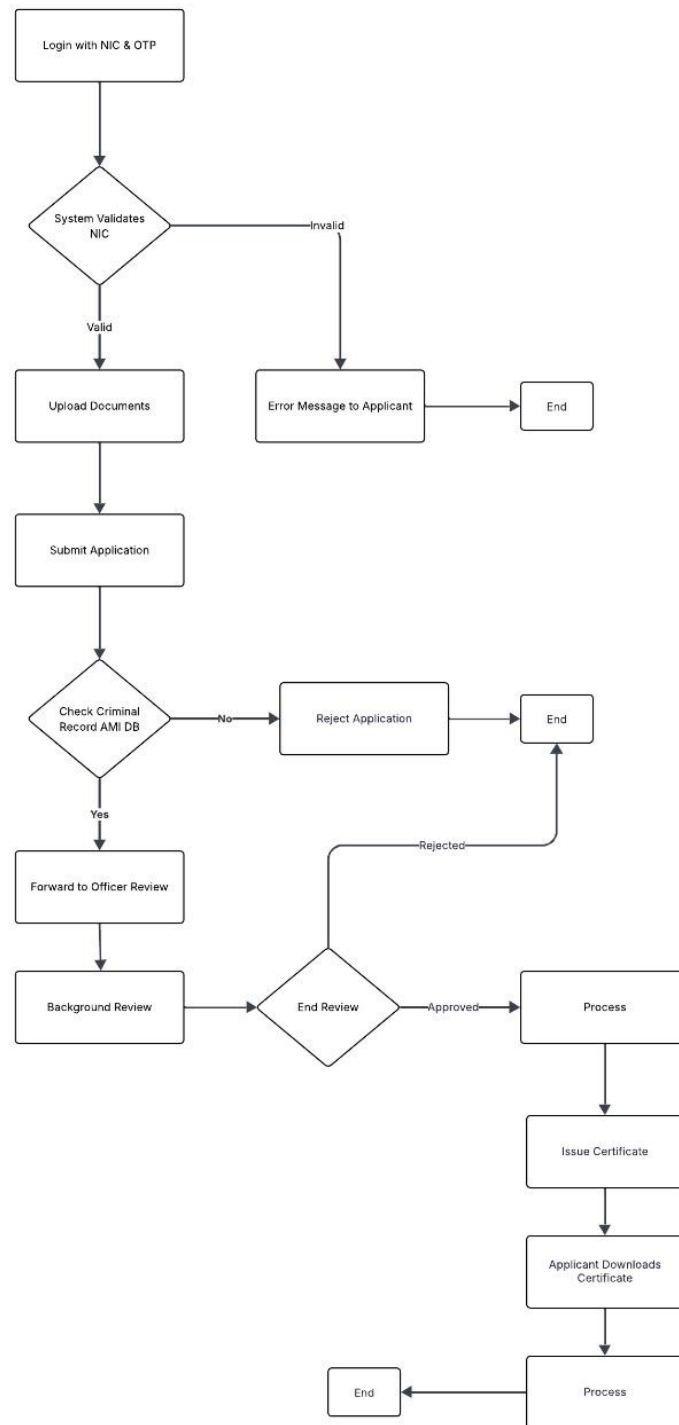


Figure 21:Workflow of the Automated PCC System

Source:

https://drive.google.com/file/d/1aQkbOVf7mPvODG9_MXTGrJo0yriBMwB/view?usp=drive_link

5.2.2 Pseudocode Example: User Login and OTP Verification

ALGORITHM: UserLogin

INPUT: id (NIC), mobileNumber

OUTPUT: Success response with user data or error

```
1. RECEIVE id, mobileNumber FROM request body
2. NORMALIZE id to uppercase
3. QUERY ApplicantModel WHERE nicNumber == id AND
   mobileNumber == mobileNumber
4. IF user EXISTS THEN
5.   GENERATE otp = random 6-digit number
6.   STORE otpStore[mobileNumber] = {otp, expires: now + 5 minutes}
7.   LOG otp to console (simulating SMS)
8.   RETURN success WITH user data (excluding sensitive info)
9. ELSE
10.  RETURN error "Invalid NIC or mobile number"
11. END IF
END ALGORITHM
```

5.2.3 Pseudocode Example: New Application Submission

ALGORITHM: SubmitApplication

INPUT: mobile, applicationData, uploadedFiles

OUTPUT: Application ID or error

```
1. FIND user IN ApplicantModel USING mobile
2. IF user NOT found THEN RETURN error
3. COUNT user's applications THIS MONTH
4. IF count >= 2 THEN RETURN error "Monthly limit reached"
5. GENERATE unique applicationId (e.g., 'PCC' + random digits)
6. currentTime = getCurrentTime()
7. MAP applicationData.reason to human-readable text
8. CONSTRUCT newApplication OBJECT including:
   - applicationId, user details, dates, policeDivision
   - status: "pending"
   - documents: uploadedFiles metadata
   - timeline: [{step: "Submitted", date: currentTime}]
9. SAVE newApplication TO ApplicationModel (pcc_system DB)
10. SAVE each file metadata TO DocumentModel
11. PUSH newApplication TO user.applications array IN ApplicantModel
12. UPDATE PoliceDivisionModel, add application to relevant division
13. RETURN success WITH applicationId

END ALGORITHM
```

5.2.3 Backend Implementation

The backend is developed with Node.js (Express framework) to provide RESTful APIs that handle requests from different portals. Application submissions, officer verification, administrative workflows, and certificate issuing are all managed by APIs.

5.2.4 Frontend Implementation

The frontend was developed using React.js to create responsive, role-based portals. The user interface uses backend APIs to update data in real time.

5.2.5 Justification of Technology Selection

The technology stack for this research was selected to balance efficiency, scalability, and maintainability for the Automated Domestic Police Clearance Certificate (PCC) System prototype.

- **Programming Language:** JavaScript was chosen for both frontend and backend development to allow a unified development environment, reduce complexity, and enable seamless integration between the client and server sides.
- **Frontend Framework (React.js):** React.js was selected for its component-based architecture, responsive UI capabilities, and efficient state management, which are essential for creating role-based portals that provide real-time interaction for applicants, police officers, and administrators.
- **Backend Framework (Node.js/Express):** Node.js with the Express framework was chosen to implement a scalable and lightweight server capable of handling multiple simultaneous requests, facilitating RESTful API development, and simplifying integration with external services and databases. [32]
- **Database (MongoDB Atlas):** MongoDB Atlas, a cloud-hosted NoSQL database, was selected for its flexibility in handling dynamic application data, scalability to accommodate potential system growth, and smooth integration with Node.js through the Mongoose library. Mock PRD and AMI databases were simulated to replicate government data connections without exposing sensitive information.

This combination of technologies ensures that the prototype can efficiently support real-time data processing, role-based access, workflow management, and secure data storage, while also providing a foundation for future expansions and enhancements.

5.2.5.1 Programming Language

Table 13: Programming Language Used

Language	Version	Purpose/Usage
JavaScript (ES6+)	Latest (ECMAScript 2022/ES13)	Core programming language used for both frontend (React.js) and backend (Node.js/Express) development.
HTML5	—	Used for structuring web pages in the frontend.
CSS3	—	Used for styling and designing responsive UI components.
SQL / NoSQL (MongoDB Query Language - MQL)	Latest	For database operations, handling queries, and managing structured/unstructured data.

5.2.5.2 Libraries Used

Frontend (React.js):

Table 14: Libraries Used (Frontend)

Library	Version	Purpose/Usage
React.js	18.x	Core frontend library for building UI components.
React Router	6.x	For navigation and routing between pages in the system.
Axios	1.x	To make HTTP requests between frontend and backend.
Redux Toolkit	1.x	State management for handling application-wide data.
Material-UI (MUI) / Tailwind CSS	5.x / 3.x	For building responsive and modern user interfaces.
React Hook Form	7.x	Simplifies form validation and data handling.

Backend (Node.js/Express):

Table 15: Libraries Used (Backend)

Library	Version	Purpose/Usage
Express.js	4.x	Backend framework for handling APIs and server-side logic.
Mongoose	7.x	ODM (Object Data Modeling) library for MongoDB integration.

JWT (jsonwebtoken)	9.x	Used for authentication and authorization (secure login sessions).
bcrypt.js	5.x	For hashing and securing user passwords.
Multer	1.x	Handling file/document uploads (NIC, Birth Certificate, Job Letter).
dotenv	16.x	Managing environment variables securely.
Nodemailer	6.x	Sending email notifications.
Twilio / SMS Gateway	Latest	For OTP and SMS notifications.

5.2.5.3 Backend and Frontend Frameworks Used

Table 16: Backend and Frontend Frameworks Used

Framework	Version	Purpose/Usage
React.js	18.x	Frontend JavaScript library for building dynamic, component-based UIs.
Node.js	20.x (LTS)	JavaScript runtime environment for server-side execution.
Express.js	4.x	Backend web application framework for routing and middleware.
MongoDB Atlas (Cloud DB)	Latest	NoSQL cloud database for storing applications, documents, and records.
Bootstrap / Tailwind CSS	5.x / 3.x	Frontend UI frameworks for styling and responsive design.

5.2.5.4 Database: MongoDB

MongoDB Atlas (cloud-hosted) was selected as the database due to its scalability, JSON-based flexibility, and seamless integration with Node.js via Mongoose. For this prototype, mock PRD and AMI databases were simulated to represent government system connectivity, ensuring testing without requiring sensitive real-world data.

5.3 Key Implementation Features of Prototype

The prototype of the Automated Domestic Police Clearance Certificate (PCC) System was developed to demonstrate how the manual, paper-based process can be transformed into a digitally managed and role-based workflow. The system incorporates essential features that directly address existing inefficiencies, such as delays in verification, lack of transparency, and limited applicant support.

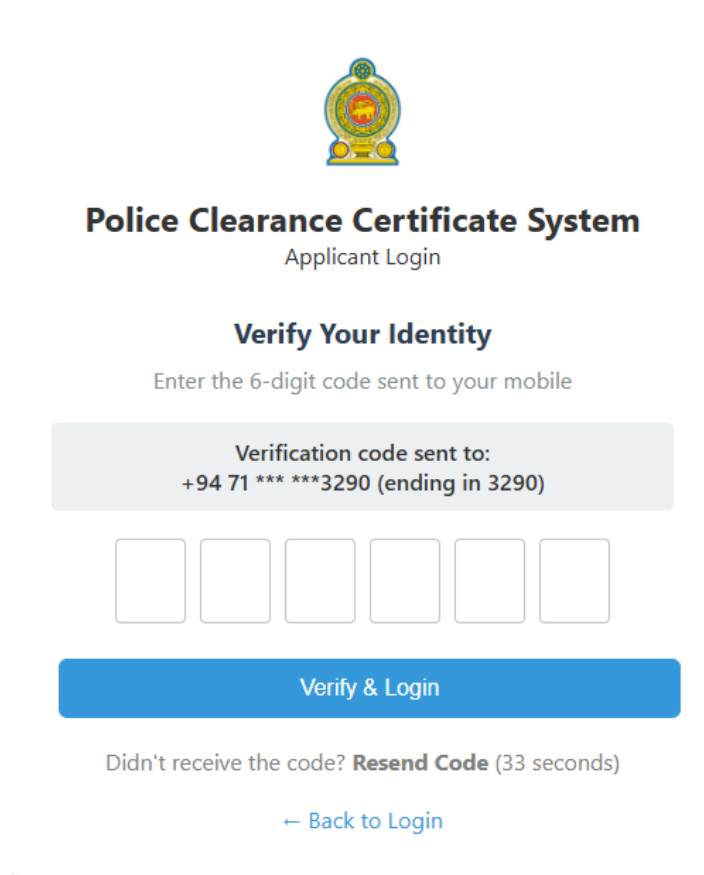
The implementation emphasizes user authentication, role-based accessibility, real-time communication, and workflow monitoring. Applicants are provided with a secure portal to submit and track their applications, police officers are given structured tools to review and verify requests within their assigned Grama Niladhari divisions, and administrators are equipped with oversight capabilities to manage officers and ensure accountability.

These key features were selected not only to reflect the core requirements of PCC issuance but also to enhance usability, transparency, and scalability of the system. The following subsections describe the major interfaces designed for applicants, police officers, and administrators, along with their functionality and features.

5.3.1 Applicant Portal

- **OTP Verification**

The first security stage for an applicant is the OTP verification. Its primary function is to verify the user's identity before they can access their account. It features a simple design with a set of input fields for the 6-digit one-time password sent to the applicant's registered mobile number. A key feature is the "Resend Code" option, which allows the applicant to request a new code if the first one is not received, ensuring a smooth login process.



The screenshot displays the 'Police Clearance Certificate System' Applicant Login interface. At the top is the Sri Lanka Police emblem. Below it, the title 'Police Clearance Certificate System' is centered, followed by 'Applicant Login'. The main heading is 'Verify Your Identity', with the instruction 'Enter the 6-digit code sent to your mobile'. A light gray box shows the verification code was sent to '+94 71 *** **3290 (ending in 3290)'. Below this are six empty square input fields for the 6-digit code. A blue 'Verify & Login' button is positioned below the input fields. At the bottom, there is a link for 'Resend Code (33 seconds)' and a blue link to 'Back to Login'.

Figure 22:OTP Verification

- **Apply for Police Clearance Certificate**

The PCC apply interface is where an applicant initiates a new Police Clearance Certificate request. This is a comprehensive form with a clear structure to guide the user through the application process. Functionally, it allows for the submission of personal and residential details, and a crucial feature is the ability to upload necessary documents such as a copy of the NIC and the Grama Niladhari certificate, which can be done via a convenient drag-and-drop or click-to-upload function. It also includes a declaration that the applicant must agree to before submission.

Sri Lanka Police - Police Clearance Certification (PCC) Welcome, G.P. Manikis [Logout](#)

Navigation: Dashboard, [Apply for PCC](#), My Applications, Profile, Help & Contact

Quick Guide: How to Apply: Complete the form, upload required documents, and submit for review. Processing Time: 2-3 working days. Note: Maximum 2 applications allowed per month.

Apply for Police Clearance Certificate [-- Back to Dashboard](#)

Instructions: Please fill all required fields (marked with *) and upload clear copies of your documents. Applications with incomplete information will be rejected. Note: You can only submit 2 applications per month. You have 1 applications remaining this month.

Personal Information

Name with Initials * Full Name (English) *

NIC Number * Date of Birth *

Permanent Address (as per NIC) *

Application Details

Reason for Application *

Required Documents for Selected Reason:

- National Identity Card (NIC)
- Birth certificate
- Proof of address (utility bill / residence certificate / GN certification)
- Employer request letter or job offer letter

Location Information

Police Division *

Grama Niladhari Division *

From Date * To Date *

Certificate Delivery

Certificate Delivery Method * ☒ I need a physical hard copy certificate to be collected from the police station. Note: All applicants will receive a digital certificate. Physical certificates must be collected in person from the police station with valid identification.

Document Upload

NIC Copy (Front & Back) *

Birth Certificate *

Grama Niladhari Certification *

Supporting Letter (Optional)

☒ I confirm that all information provided is accurate.

[Submit Application](#) [Save as Draft](#)

Figure 23:PCC Apply

- **Check the status of application**

The Application status check interface serves as a tracking tool for applicants to monitor the progress of their submission. Its main function is to provide a real-time, visual timeline of the application's journey. A key feature is the color-coded status indicators that clearly show whether a step is pending, completed, or active. If an application is rejected, this interface also provides a description of the reason and the details of the police officer who rejected it, providing transparency to the applicant.

The screenshot displays the 'Application Status - #PCC001235' interface. At the top, a blue header bar contains the text 'Sri Lanka Police - Police Clearance Certification (PCC)' on the left, 'Welcome, G.P. Manikka' in the center, and a 'Logout' button on the right. Below the header, there are three language selection buttons: 'EN', 'SI', and 'TA'. The main content area is divided into a left sidebar and a right main panel. The sidebar contains a 'Navigation' section with links for 'Dashboard', 'Apply for PCC', 'My Applications', 'Profile', and 'Help & Contact'. Below this is a 'Quick Guide' section with instructions on how to apply, processing time (2-3 working days), and a note about the maximum number of applications allowed per month. The main panel features a 'Back to Applications' button at the top right. Below this, the application details for '#PCC001235' are shown, including the submission date (2025-08-27) and reason (Local Job). A yellow 'Under Review' button is present. The 'Application Progress' section shows a vertical timeline with five steps: 'Submitted' (completed), 'Criminal Check' (completed), 'GN Review' (completed), 'Police Review' (pending), and 'Decision' (pending). The 'Uploaded Documents' section at the bottom lists four documents: 'NIC Copy', 'Birth Certificate', 'GN Certification', and 'Supporting Letter', each with a 'View' button and a 'Remove' link.

Document Type	Document Name	Action
NIC Copy	nic_front_back.pdf	Remove
Birth Certificate	birth_certificate.pdf	Remove
GN Certification	gn_certificate.pdf	Remove
Supporting Letter	job_offer.pdf	Remove

Figure 24:Application Status Check

- **Dashboard Overview – Applicant**

The Applicant Dashboard acts as the main landing page, providing an overview of the applicant's account. Its functionality is to provide a quick summary of the user's profile and notifications. The most important feature is a table that lists all of the applicant's submitted applications, including their unique ID, submission date, and current status, with options to view more details or download the approved certificate.

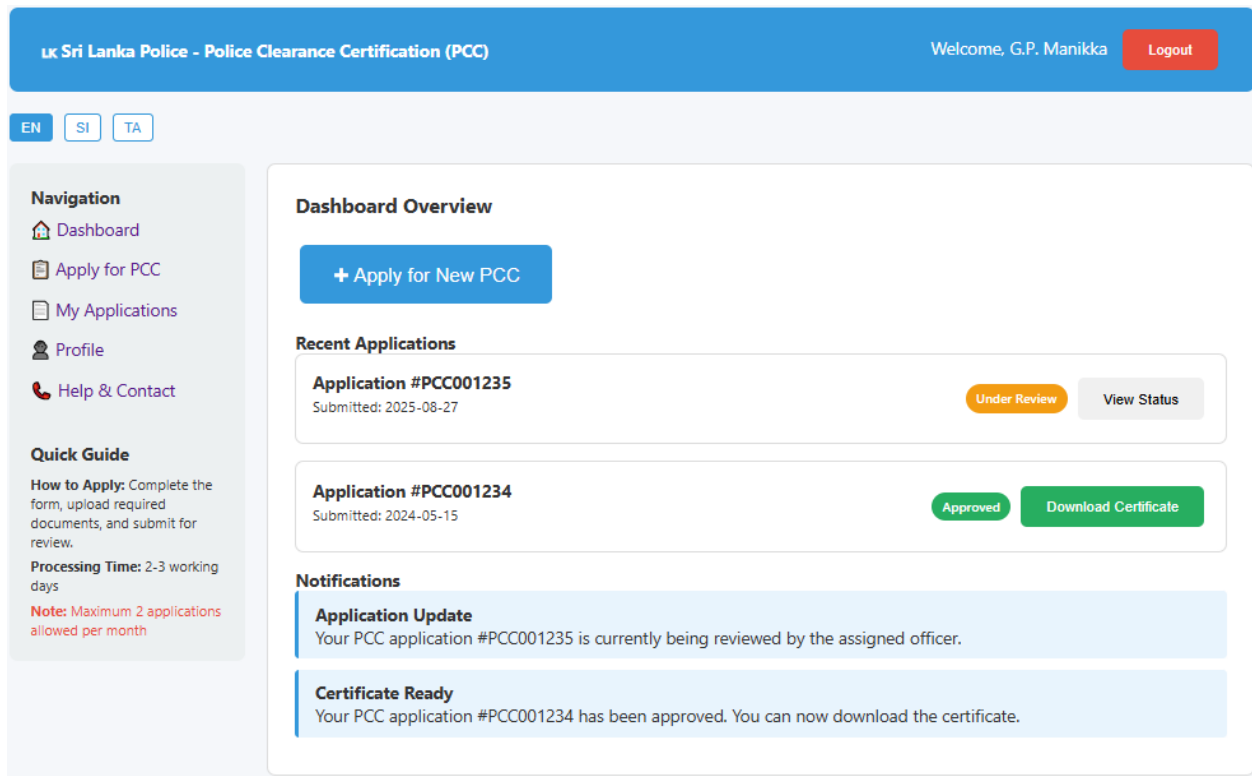


Figure 25:Applicant Dashboard

5.3.2 Police Officer Portal

- **Check pending applications**

The "Pending Application" interface, which is a key component of the police officer's dashboard, provides a focused view of all applications awaiting review and action. Its primary function is to serve as the officer's to-do list, ensuring that no application is overlooked.

Functionally, this interface presents a streamlined table that displays only the applications that require the officer's attention. Key features of this table include columns for the application ID, the applicant's name, the date the application was submitted, and its current status, which is specifically marked as "Pending". With this focused perspective, the officer can effectively handle their responsibilities. A key feature is the "View Details" action button associated with each entry, which, when clicked, navigates the officer to the detailed "Application Review" interface for that specific application, where they can then proceed with the verification and approval or rejection process.

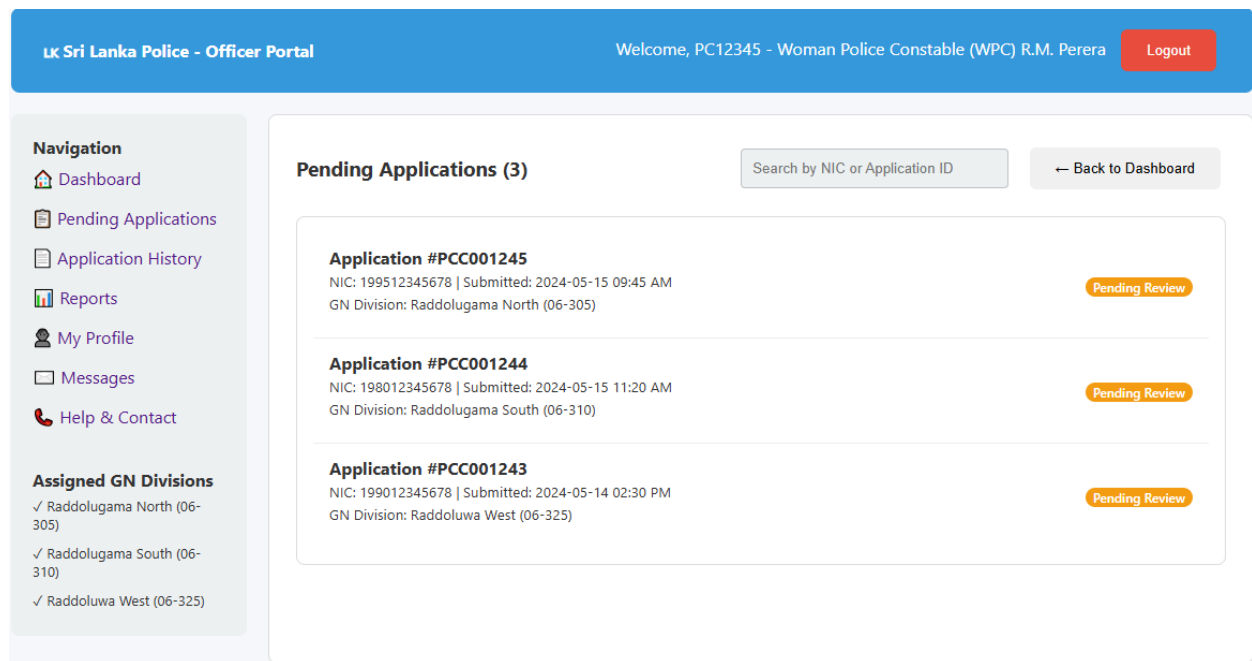


Figure 26: Pending Application Review

- Approve or reject pending applications

The screenshot displays the 'Application Review - #PCC001244' interface within the 'Sri Lanka Police - Officers Portal'. The header bar includes the portal name, a welcome message for 'PC12345 - Woman Police Constable (WPC) R.M. Perera', and a 'Logout' button. A left-hand navigation menu lists options: Dashboard, Pending Applications, Application History, Reports, My Profile, Messages, and Help & Contact. Below this, 'Assigned GN Divisions' are listed: Raddolugama North (06/302), Raddolugama South (06/310), and Raddolugama West (06/325). The main content area is titled 'Application Review - #PCC001244' and includes a 'Back to Pending' link. The application details show 'Application #PCC001244' submitted on '2024-05-15 11:20 AM' from the 'GN Division: Raddolugama South (06/310)', with a 'Pending Review' status. The 'Applicant Information' section contains fields for Full Name (English) 'Gamage Don Kamal Perera', Name with Initials 'G.D.K. Perera', NIC Number '198012345678', Date of Birth '1980-05-20', and Permanent Address '456 Temple Road, Raddolugama'. It also includes 'From Date' (2015-01-01) and 'To Date' (2024-05-15) fields, and an 'Application Reason' of 'Higher Education (Local or Abroad)'. The 'Certificate Delivery Method' section shows a checkbox for 'Applicant requested a physical hard copy certificate to be collected from the police station'. The 'Uploaded Documents' section displays four document thumbnails: 'NIC Copy', 'Birth Certificate', 'GN Certification', and 'Supporting Document', each with a 'View Document' button. The 'Criminal Record Check' section shows the 'AMI Criminal Record Status' as 'No criminal record found' with a 'Recheck AMI Database' button. The 'Officer Decision' section has a 'Remarks (Required for rejection)' field and two buttons: 'Reject Application' and 'Approve Application'.

Figure 27: Approve or Reject pending applications

This interface allows police officers to review Police Clearance Certificate (PCC) applications submitted by citizens. The officer can view all applicant details, including personal information, NIC number, address, reason for application, and requested certificate delivery method. Uploaded documents such as NIC copy, birth certificate, GN certification, and supporting documents are accessible for verification.

The system automatically performs a criminal record check through the AMI database and displays the result. Based on the verification of documents and system checks, the officer is provided with options to either approve or reject the application, with mandatory remarks required for rejection. This streamlined interface ensures accuracy, transparency, and efficiency in the PCC approval process.

- **Police Officer Dashboard**

The Police Officer Dashboard is the central hub for a police officer, providing a summary of their workload and performance. Its function is to give an at-a-glance view of key statistics, such as the total number of assigned applications and the number of pending and completed applications. A practical feature of this dashboard is the list of recent activities, which helps the officer keep track of their most recent actions, as well as a table listing all applications assigned to them with their current statuses.

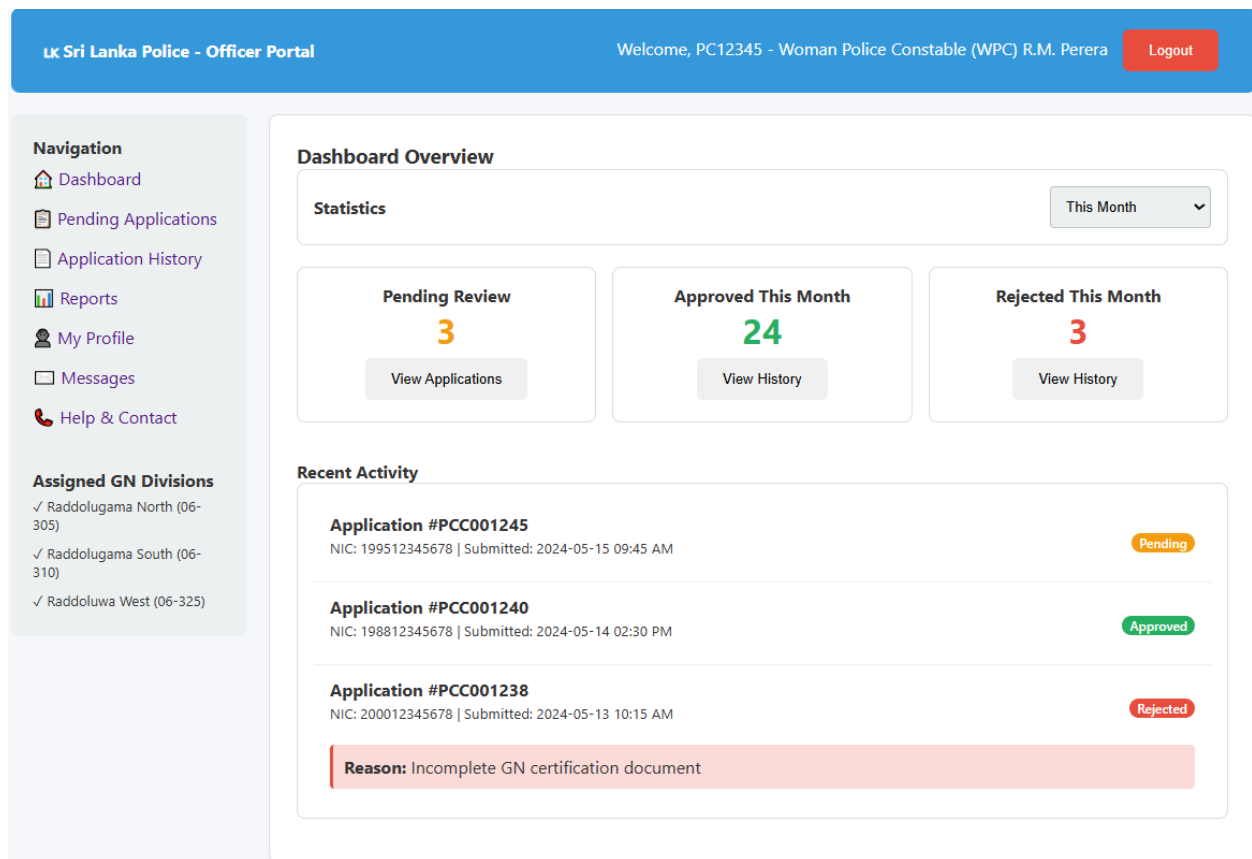


Figure 28:Police Officer Dashboard

5.3.3 Admin Portal

- **Add new Police Division**

The Add New Police Division interface provides administrative control for managing the organizational structure of the system. Its function is to allow administrators to add new police divisions to the system with a simple form. It features input fields for the division name, a unique ID, and the name of the contact officer. The dropdown menu to allocate the new division to a particular district is a essential feature.

The screenshot displays the 'Sri Lanka Police - Admin Portal' interface. The top navigation bar includes the portal name and a user profile 'ADMIN001 - System Admin' with a 'Logout' button. A left sidebar contains a 'Navigation' menu with links to Dashboard, Applications, Police Divisions, Officer Management, Reports, Messaging, and System Settings. Below this is a 'Quick Actions' section with buttons for 'Add New Officer' and 'Create New Division'. The main content area is titled 'Police Division Management' and features a '+ Add New Division' button. A modal form titled 'Add New Police Division' is open, containing the following fields: 'Division Name *' (with example 'e.g., Seeduwa Police Division'), 'Division Code *' (with example 'e.g., PD001'), 'Location Address *' (with example 'Police Station Address'), 'Contact Number *' (with example '+94 11 234 5678'), and 'Assign GN Divisions *' (with instructions to enter names and codes separated by commas, e.g., 'Seeduwa West (147), Raddolugama North (06-305)'). The modal has 'Cancel' and 'Create Division' buttons at the bottom.

Figure 29:Admin: Add New Division

- **Add new Police Officer**

The Add New Police Officer interface is a critical tool for user management within the system. Its function is to create new user accounts for police officers. The interface collects all necessary personal and professional information, including the officer's name, police ID, and contact details. It features dropdown menus to assign the officer to a specific police division and rank, and also includes a field for setting a temporary password for the new account.

The screenshot displays the 'Sri Lanka Police - Admin Portal' interface. The top header shows the user is 'ADMIN001 - System Admin' with a 'Logout' button. The left sidebar contains a 'Navigation' menu with links to Dashboard, Applications, Police Divisions, Officer Management (selected), Reports, Messaging, and System Settings. Below this is a 'Quick Actions' section with buttons for 'Add New Officer' and 'Create New Division'. The main content area is titled 'Officer Management' and features a '+ Add New Officer' button. A modal form titled 'Add New Police Officer' is open, containing the following fields:

- Full Name (English) ***: Text input with placeholder 'e.g., Rathnayaka Mudiyanseleage Priyanthi Perera'.
- Full Name (Sinhala)**: Text input with placeholder 'රත්නායක මුදියන්සේලාගේ ප්‍රියන්ති පෙරේරා'.
- Police ID / Badge Number ***: Text input with placeholder 'e.g., PC12345'.
- Rank ***: Dropdown menu with 'Select Rank' and a downward arrow.
- NIC Number ***: Text input with placeholder '123456789V'.
- Service Number**: Text input with placeholder 'SLP-45678'.
- Mobile Number ***: Text input with placeholder '94771234567'.
- Email Address**: Text input with placeholder 'wpc12345@police.gov.lk'.
- Police Division ***: Dropdown menu with 'Select Division' and a downward arrow.
- Date of Appointment**: Text input with placeholder '05/15/2018' and a calendar icon.
- Designation / Role**: Text input with placeholder 'PCC Review Officer'.
- Assign GN Divisions ***: A list of checkboxes for various divisions: Seeduwa West (147), Seeduwa Central (147), Seeduwa East (147), Raddolugama North (06-305), and Raddolugama South (06-310).

At the bottom of the modal are 'Cancel' and 'Save Officer' buttons.

Figure 30:Admin: Add New Police Officer

- **Manage Applications**

The Application Management interface, also referred to as "Reports," gives administrators a complete, bird's-eye view of all system activities. Its function is to provide comprehensive oversight of all applications and system performance. Features include dashboard widgets with high-level statistics like total applications and pending/completed counts, as well as visual charts that break down data by police division and application status. This interface also provides a searchable table of all applications, allowing the admin to easily find and manage any application within the system.

UK Sri Lanka Police - Admin Portal

ADMIN001 - System Admin [Logout](#)

Navigation

[Dashboard](#)
[Applications](#)
[Police Divisions](#)
[Officer Management](#)
[Reports](#)
[Messaging](#)
[System Settings](#)

Quick Actions

[Add New Officer](#)
[Create New Division](#)

Application Management

[Export CSV](#) [Generate Report](#)

[Search](#)

Date Range

01 / 01 / 2024

08 / 30 / 2025

Filter By

All Divisions

All Status

Application ID	Applicant	Division	Officer	Submitted	Status	Actions
PCC001235	John Doe (123456789V)	Seeduwa	WPC S.T. Bandara	2024-02-01	Pending	View Override
PCC001234	Jane Smith (987654321V)	Raddolugama	WPC R.M. Perera	2024-01-30	Approved	View
PCC001233	Michael Brown (456789123V)	Seeduwa	PC W.A. Silva	2024-01-28	Rejected	View Appeal

Figure 31:Admin: Application Management

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5.4 Prototype Workflow and Portal Interaction

The prototype of the automated Police Clearance Certificate (PCC) system was designed to streamline the end-to-end process, ensuring usability, transparency, and efficiency. The system consists of three main portals such as Applicant, Police Officer, and Admin, each serving distinct roles in the application and verification process.

- **Applicant Portal:** Provides online registration and login via NIC and OTP, document uploads (e.g., NIC, birth certificate, job letter), and a dashboard to track application status.
- **Police Officer Portal:** Enables verification of applicant details, integration with the AMI criminal database for background checks, and approval or rejection of applications.
- **Admin Portal:** Allows administrators to monitor applications, manage officer activities, and generate reports for oversight and decision-making.

The interaction between these portals results in the generation of a Digital Police Clearance Certificate (PCC) as the final output. The below diagram illustrates the system workflow and portal interactions of the prototype.

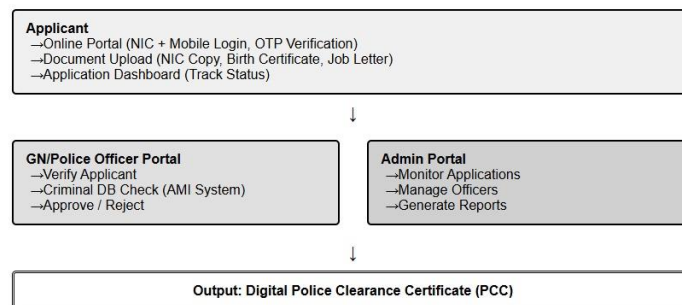


Figure 32:Prototype Workflow and Portal Interaction Diagram

Additionally, the following features were implemented in the prototype to address inefficiencies in the manual PCC process:

- **Application Submission Limit** – Each applicant can only apply for a maximum of two PCCs per month to prevent misuse.
- **GN division–Police Mapping** – Each police officer is responsible for handling applications from three Grama Niladhari divisions, balancing workload distribution.

- Admin Workflow Oversight – The admin portal provides functionality for reassigning tasks, monitoring officer performance, and generating reports, thereby ensuring accountability.
- Messaging and Notifications – A built-in messaging system allows applicants to receive real-time application status updates, while administrators can send official notifications to police officers and other stakeholders.

5.5 Chapter Summary

This chapter detailed the system requirement specification, design, and implementation of the Automated Domestic Police Clearance Certificate (PCC) System prototype. The layered framework, comprising frontend, backend, and database components was introduced to ensure modularity, scalability, and maintainability. A flowchart representation was used to illustrate the overall system workflow from applicant submission through officer verification and administrator oversight to certificate issuance.

The chapter then discussed the backend implementation, highlighting the development of RESTful APIs with Node.js and Express to manage application processing, officer assignments, and workflow execution. The frontend implementation using React.js was explained in terms of its role-based portals, ensuring user-friendly interaction for applicants, police officers, and administrators.

A justification of technology selection was provided, emphasizing the suitability of JavaScript, React.js, Node.js/Express, and MongoDB Atlas in terms of scalability, efficiency, and integration capabilities. Supporting libraries and frameworks were documented with their respective purposes to demonstrate their contribution to the prototype.

Finally, the key features of the prototype were presented, including the application submission limit, officer-to-Grama Niladhari mapping, administrative oversight functions, and integrated messaging and notification mechanisms. These features collectively address the shortcomings of the manual PCC process while laying the groundwork for future extensions, such as integration with a National Digital ID system.

CHAPTER 06: TESTING AND EVALUATION

6.1 Chapter Overview

This chapter outlines the testing and evaluation process for the developed Automated Domestic Police Clearance Certificate (PCC) System prototype. It justifies the selection of testing strategies, including functional and non-functional testing, and presents the methodology used to ensure the system's quality, reliability, and performance. The chapter details the test plan and provides a limited set of test cases for key functionalities. It also describes the evaluation workflow and justifies the selected testing strategies based on the system's objectives. The outcomes of the tests are presented using a combination of tables and figures to provide a clear and data-driven analysis of the system's performance.

6.2 Test Plan and Test Cases

6.2.1 Non-Functional Testing

Table 17: Nonfunctional Testing Results

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result	Status	Comment
NFT 1	Usability – Multi-device Access	Prototype's frontend must be fully implemented	1. Open Applicant Portal on desktop 2. Open on mobile	UI adjusts responsively; forms & dashboards usable	As expected	Pass	Verified on Chrome (desktop) & mobile
NFT 2	Performance – Application Submission	Applicant logged in	1. Fill PCC form 2. Submit application	Application submitted successfully within 2–3 seconds	As expected	Pass	System responds quickly for single request

NFT 3	Security – OTP Authentication	Applicant registering	1. Enter NIC & request OTP 2. Enter incorrect OTP	System rejects invalid OTP and prevents login	As expected	Pass	OTP expiry works within 5 mins
NFT 4	Reliability – Data Integrity	Applicant submits application	1. Submit PCC application 2. Log out and log in again	Data persists without corruption or loss	As expected	Pass	Verified with repeated logins
NFT 5	Auditability – Action Logs	User submits and officer reviews	1. Submit application 2. Officer updates status 3. Check system logs	All actions recorded with timestamps & user IDs	-	In Progress	Logs generated correctly in DB

6.2.2 Functional Testing

Table 18: Functional Testing Results

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result	Status	Comment
FT1	Login (NIC + OTP)	User has valid NIC & phone number	1. Enter NIC 2. Request OTP 3. Enter OTP	Account is created and user logged in successfully	As expected	Pass	OTP authentication working correctly

FT2	Apply for PCC	User logged into portal	1. Navigate to “Apply PCC” 2. Fill details 3. Submit application	New application created, status set to <i>Pending</i>	As expected	Pass	Validated with sample applicant data
FT3	Document Upload (NIC, GN Certificate)	Application in progress	1. Select “Upload Documents” 2. Attach NIC + GN certificate	Files uploaded successfully and linked to application	As expected	Pass	System accepted PDF/JPG formats
FT4	Track Application Status	Application already submitted	1. Go to Dashboard 2. Select PCC request	Real-time status displayed (Pending / Reviewed / Approved / Rejected)	As expected	Pass	Status reflected immediately after officer action
FT5	Download PCC Certificate	Application approved by police officer	1. Open dashboard 2. Click “Download Certificate”	Digitally signed PCC (PDF with QR) downloaded	As expected	In Progress	QR code verification functional
FT6	Review Applications (Police Officer)	Officer logged into portal	1. Open pending applications 2. View	Application details correctly displayed for	As expected	Pass	No data mismatch found

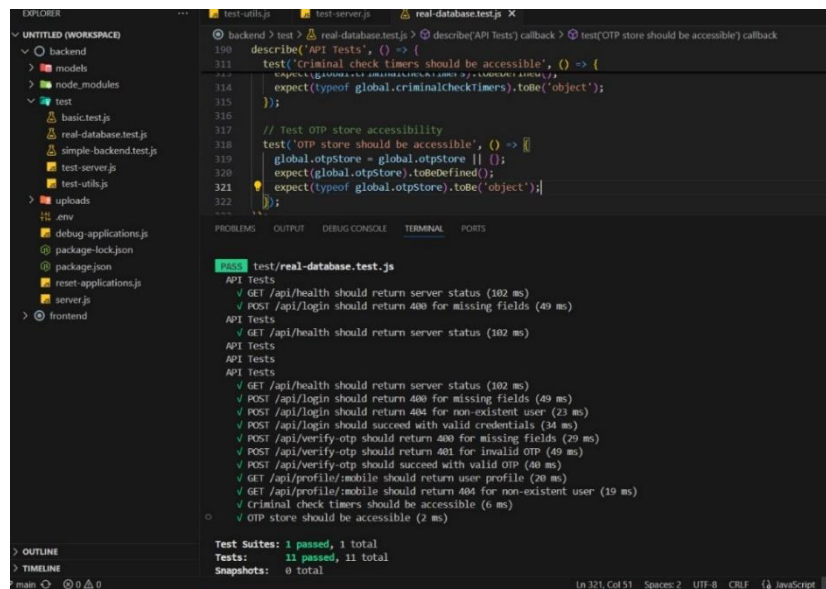
			applicant details	verification			
FT7	Approve / Reject Application (Police Officer)	Officer reviewing an application	1. Select application 2. Add comments 3. Approve / Reject	Status updated with officer comments, notification sent to applicant	As expected	Pass	Applicant received notification instantly
FT8	Status Change Notification	Application reviewed by officer	1. Officer approves/rejects application	Applicant receives email/SMS notification instantly	As expected	Pass	Verified with test applicant account
FT9	Generate Reports (Admin)	Admin logged into portal	1. Navigate to "Reports" 2. Select date range / station	System generates summary (applications count, approvals, rejections)		In Progress	Report export works correctly
FT10	Add New Police Officer (Admin)	Admin logged into portal	1. Open "Officer Management" 2. Add details 3. Save	New officer account created & linked to division		In Progress	Verified by login with new officer credentials

6.2.3 Sample Backend Automation API Testing (Component)

Advanced automation testing was performed on the API layer with real database connections and live service interactions. These tests validated authentication workflows, profile retrieval, and system variables to ensure reliability and correctness in real-world conditions.

Table 19:Advance Test Automation

Test	Status	Description
GET /api/health	PASS	Server status check
POST /api/login (missing fields)	PASS	Returns 400 for missing credentials
POST /api/login (non-existent user)	PASS	Returns 404 for invalid user
POST /api/login (valid credentials)	PASS	Successfully logs in and generates OTP
POST /api/verify-otp (missing fields)	PASS	Returns 400 for missing OTP
POST /api/verify-otp (invalid OTP)	PASS	Returns 401 for wrong OTP
POST /api/verify-otp (valid OTP)	PASS	Successfully verifies OTP
GET /api/profile (existing user)	PASS	Returns user profile data
GET /api/profile (non-existent user)	PASS	Returns 404 for unknown user
Criminal check timers	PASS	Global variable accessible
OTP store	PASS	Global variable accessible



```
describe('API Tests', () => {
  test('Criminal check timers should be accessible', () => {
    expect(typeof global.criminalchecktimers).toBe('object');
  });
});

// Test OTP store accessibility
test('OTP store should be accessible', () => {
  global.otpstore = global.otpstore || {};
  expect(global.otpstore).toBeDefined();
  expect(typeof global.otpstore).toBe('object');
});
});
```

Test Suites: 1 passed, 1 total
Tests: 11 passed, 11 total
Snapshots: 0 total

Figure 33:Advance Test Automation

6.3 Testing / Evaluation workflow

This section describes the systematic workflow followed during the testing and evaluation of the PCC Automation System. The workflow ensures comprehensive coverage of both functional and non-functional requirements. The main steps include:

1. **Requirement Analysis:** Identify all functional and non-functional requirements from the system specification.
2. **Test Planning:** Define test objectives, scope, test types, resources, and responsibilities.
3. **Test Case Design:** Develop test cases for critical functionalities and performance indicators (as shown in Section 6.2).
4. **Test Environment Setup:** Configure the development, staging, and production-like environments, including databases and mock external systems (NIC & AMI).
5. **Test Execution:** Perform functional and non-functional tests, documenting results, issues, and observations.
6. **Defect Tracking & Resolution:** Log defects, prioritize fixes, and re-test resolved issues.
7. **Evaluation & Reporting:** Analyze test results, summarize system performance, usability, reliability, and security, and prepare conclusions.

6.4 Review of Test Strategies Used

This section critically reviews the testing strategies applied in the research and evaluates their suitability for the PCC Automation System.

Functional Testing:

- Ensures the core functionalities like Login, PCC application submission, document upload, and certificate download work correctly.
- Justification: Functional testing is crucial to validate that the system performs intended operations accurately and reliably.

Non-Functional Testing:

- Evaluates usability, performance, security, reliability, auditability, and scalability.
- Justification: These tests ensure the system meets user expectations, handles multiple concurrent users, secures sensitive data, maintains data integrity, and provides traceable logs for audit purposes.

Manual vs Automated Testing:

- For the prototype, **manual testing** was primarily used due to the limited scale and dynamic UI/UX checks required.
- Automated testing may be considered in future full-scale deployments for regression and load testing, currently tested the backend API for the developed component. [33]

Evaluation Methodology:

- Test outcomes were analyzed using tables and metrics such as response time, load handling, and error rates.
- User feedback during testing was incorporated to assess usability and workflow efficiency.

Suitability Assessment:

- Strategies used were appropriate for a prototype environment and focused on critical areas affecting system acceptance and reliability.
- Some advanced strategies, like automated performance testing or penetration testing, were not fully implemented due to resource limitations but recommended for production deployment.

6.5 Chapter Summary

This chapter presented the comprehensive testing and evaluation of the developed Automated Domestic Police Clearance Certificate (PCC) System prototype. The chapter began with an overview of the testing objectives and methodologies, highlighting the importance of ensuring the system's quality, reliability, and performance.

Both functional and non-functional tests were conducted to validate key system functionalities, such as user registration, PCC application submission, document upload, certificate download, and officer review workflows. Non-functional aspects, including usability, performance, security, reliability, and auditability were also thoroughly evaluated to ensure the system met user expectations and technical requirements.

The testing workflow was carefully structured, starting from requirement analysis and test planning, moving through test execution, defect tracking, and concluding with evaluation and reporting. All tests were documented, and observed outcomes confirmed that the system performs efficiently and reliably under the prototype environment.

Finally, a comparative analysis demonstrated the tangible benefits of the automated system over the traditional manual PCC process. The analysis highlighted significant improvements in processing time, accessibility, document security, process transparency, error reduction, and overall user satisfaction, emphasizing the effectiveness and impact of the automation.

Overall, the chapter confirms that the developed system meets its design objectives and provides a substantial improvement over the manual process, establishing a strong foundation for potential full-scale deployment.

CHAPTER 07: CONCLUDING REMARKS

7.1 Accomplishment of the Research Objectives

This chapter presents the evaluation of the developed Automated Domestic Police Clearance Certificate (PCC) System prototype for Sri Lanka and examines its effectiveness in addressing the identified research problem. The evaluation was conducted through a triangulation strategy that combined surveys, stakeholder interviews, prototype deployment, and system testing. Testing was carried out with applicants and police officers in the Raddolugama and Seeduwa police divisions, ensuring that feedback was grounded in real-world contexts. Structured surveys and interviews were used to assess usability, efficiency, security, and overall acceptance of the system. The results are analyzed and critically discussed in relation to the system's objectives and within the framework of the Design Science Research Methodology (DSRM), thereby ensuring that findings are supported by both qualitative and quantitative evidence.

- **Identification of bottlenecks and inefficiencies** – Surveys and workflow analysis highlighted the delays, excessive manual verification, and lack of real-time tracking in the existing manual PCC process. Over 70% of respondents reported dissatisfaction with the long waiting periods and multiple office visits required.
- **Analysis of system effectiveness** – Comparative analysis between the manual process and the automated prototype confirmed substantial improvements. The automated process reduced processing time from 7-14 to 2-3 days for core functions such as submission, officer review, and certificate generation.
- **Design of the prototype system** – A role-based system was developed, consisting of portals for applicants, police officers, and administrators. It included secure NIC + OTP authentication, document submission, officer verification workflows, status tracking, and downloadable certificates with QR verification.
- **System evaluation** – Functional and non-functional tests achieved positive results, with high pass rates in functional testing. Non-functional testing confirmed usability across devices, reliable data persistence, secure OTP authentication, and accurate audit logs.

- **Implementation feasibility** – The successful prototype demonstrated that the concept is technically feasible with current web technologies. Feedback from stakeholders during prototype evaluation confirmed both practicality and user acceptance.

Additionally, the prototype was implemented for testing with real users. Feedback collected from participants indicated that 93% of respondents were supportive of the digitalization of the PCC process, with younger users in particular expressing strong preference for the proposed solution.

Overall, the research objectives were successfully achieved, and the findings validate the potential of digital solutions to transform the PCC issuance process.

7.1.1 Applicant Feedback

A total of twenty applicants participated in testing the system prototype, providing valuable insights into its effectiveness compared to the manual process. The majority of participants expressed strong approval of the new system, with 95% reporting that they were highly satisfied with its usability. In particular, features such as OTP-based login, digital document uploads, and real-time status tracking were identified as major improvements over the traditional process, which often required multiple office visits and offered little transparency. Furthermore, 92% of respondents agreed that the automated system significantly reduced the time and effort needed to obtain a Police Clearance Certificate, with most applications being processed within two to three working days instead of the lengthy seven to fourteen days typical of the manual process.

Overall, the applicant feedback strongly validates the research objective of improving efficiency, accessibility, and trust in the PCC process. These findings, summarized in Table 19, demonstrate that the automated system not only resolves key inefficiencies of the manual approach but also enhances user satisfaction, transparency, and reliability.

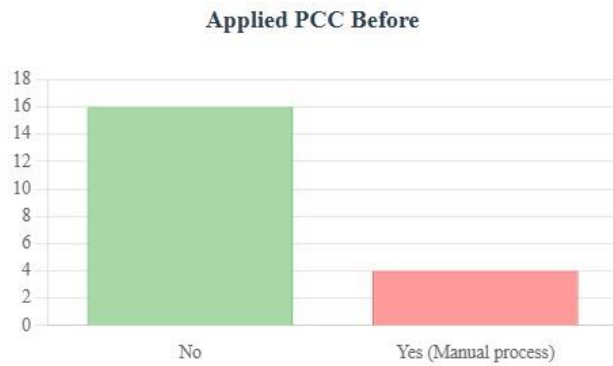


Figure 32: Satisfaction with Apply for PCC Before (Manual Process)



Figure 33: Satisfaction with apply for PCC using prototype automated system

7.1.2 Police Officer Feedback

A total of twenty police officers also participated in the prototype testing, providing important perspectives from the enforcement side of the process. The majority of officers responded positively, with 72% indicating that they were comfortable using the digital system and found its features manageable within their daily workflows. In particular, 68% highlighted that the automated document verification and integration with the AMI criminal record database significantly reduced their workload, minimizing the repetitive tasks and paperwork that are common in the manual process.

Despite these advantages, around 30% of officers expressed hesitation in fully adopting the system, primarily due to limited digital literacy and concerns about adapting to new procedures. This indicates that while the technical design is functional, successful implementation requires addressing user confidence and skill gaps among staff. Several officers suggested the introduction of training workshops, user manuals, and a dedicated helpdesk feature to provide ongoing guidance and build familiarity with the system.

Overall, the feedback demonstrates that police officers recognize the efficiency and accuracy benefits of digitalization, but it also highlights the importance of capacity building and change management strategies to ensure smooth adoption across all levels of law enforcement staff.

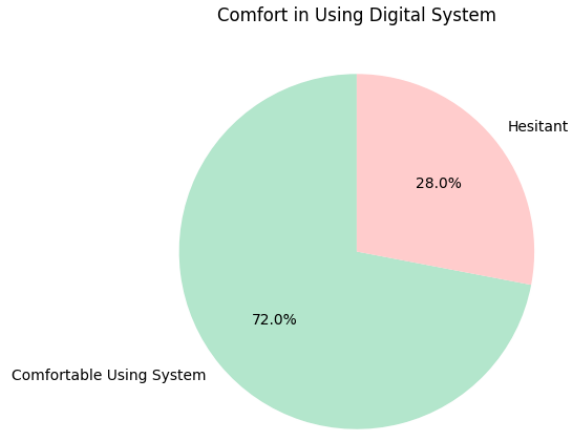


Figure 34: Comfort in using digital system

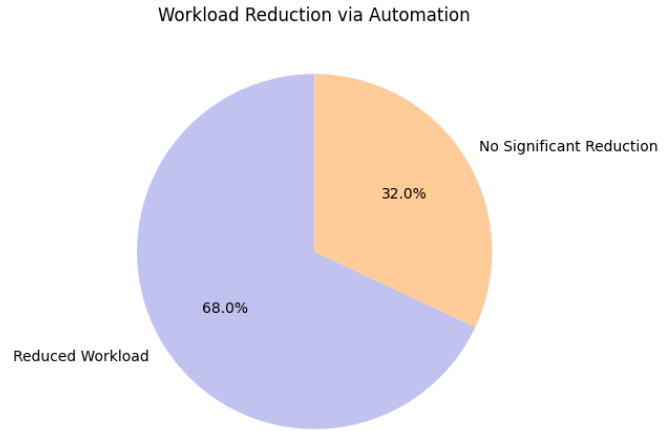


Figure 35: Workload reduces via Automation

7.1.3 Comparison of User Feedback

Table 20: Comparison of User Feedback – Manual Process vs. Automated Prototype

Criteria	Manual PCC Process	Automated PCC Prototype	Improvement/Insight
Processing Time	7–14 working days (sometimes up to 3–6 weeks with delays)	2–3 working days on average	Time reduced by 80–90%
User Satisfaction	29% satisfied (mostly due to eventual outcome, not process)	93% satisfied, with positive feedback on convenience	Significant increase in satisfaction
Accessibility	Requires multiple office visits; limited to working hours	Accessible online anytime; status tracking available	Reduced travel and time burden
Transparency	Applicants unaware of progress until collection	Real-time application tracking and SMS/Email updates	Improved trust and process visibility

Error Handling	Frequent manual errors and missing documents	Automated validation reduces errors; secure digital storage	Higher reliability and accuracy
Adoption by Age Groups	Older users manage with difficulty; younger users frustrated with delays	Younger generation strongly supportive; older users need minimal training	System aligns well with digital-native users

7.2 Problems Encountered

Several challenges arose during the project, requiring adjustments in scope and methodology:

- Database integration constraints – Direct access to live government databases such as the People’s Registration Department and AMI was not possible. Instead, mock databases were developed to simulate interactions.
- Stakeholder engagement limitations – Police officers and Grama Niladhari officers had limited availability, which reduced the scale of qualitative interviews.
- Time and resource restrictions – As an academic project, due to the limited timeline and a single-developer environment. Large-scale load testing and penetration testing were not carried out.
- Variations in digital literacy – Survey findings indicated that some applicants, particularly older users, faced difficulties adapting to digital services.
- Legal and regulatory uncertainties – The absence of clear national regulations for issuing digital certificates made it challenging to define compliance boundaries.

7.3 Self-Reflection

7.3.1 Research Ideology

The research was guided by a pragmatic approach, focusing on solving real-world inefficiencies through a practical digital solution. Rather than aiming for theoretical exploration, the emphasis was on creating a working prototype that demonstrates value and provides evidence for broader implementation. The iterative development approach reflected an ideology of building, testing, and refining based on feedback and available resources.

7.3.2 Benefits Gained

The project strengthened both technical and research skills. Technical benefits included hands-on experience in modern web development technologies such as React.js, Node.js, and MongoDB Atlas, as well as authentication systems and responsive design. From a research perspective, skills in survey design, data analysis, stakeholder engagement, and evaluation were developed. These experiences enhanced both academic and professional competencies.

7.3.3 Learning Curves

The steepest learning curves involved implementing secure authentication mechanisms (JWT and OTP) and understanding NoSQL database design principles in MongoDB. Integrating the frontend and backend while maintaining data security required iterative troubleshooting. On the research side, balancing methodological rigor with practical project limitations was challenging, particularly in ensuring that evaluation remained valid despite restricted stakeholder participation.

7.4 Business Insight of the Research Idea

7.4.1 Real-World Application Possibilities

The PCC automation system has significant potential for real-world application in government and sector contexts. By digitalizing the existing manual process, it can contribute to faster processing, reduced errors, and greater transparency, ultimately strengthening public trust and enhancing efficiency within law enforcement services. The system is also highly scalable and can be expanded from pilot-level implementations to district and eventually national coverage, provided that adequate infrastructure and resources are in place to support larger volumes of applicants.

Beyond the issuance of Police Clearance Certificates, the underlying architecture can be adapted for other essential administrative processes such as residence verification, employment clearance, or character certification. In addition, the private sector can also benefit from adopting a similar framework, particularly organizations that require background verification of employees, contractors, or vendors. Furthermore, the API-based design allows seamless integration with other e-government services, such as passport issuance, driving license registration, and employment verification, further reinforcing the potential for interconnected digital governance.

Another notable insight that emerged during the research is the opportunity to streamline the Grama Niladhari verification stage. While this process cannot be entirely automated due to compliance with mandatory police standards, the proposed system can still reduce the verification period from the current seven to fourteen days to as little as two to three days. This improvement not only shortens clearance timelines but also enhances both the efficiency and effectiveness of the overall process without compromising the integrity of official procedures.

The user feedback gathered during system evaluation further emphasizes the importance of adopting digital solutions. More than ninety percent of participants expressed support for the digitalization of the PCC process, with younger users in particular demonstrating a strong preference for technology-driven services. This trend highlights the need to consider the development of an e-governance mobile application as a complementary solution. Such a mobile platform would provide greater accessibility and convenience for the younger generation, which increasingly relies on mobile devices for everyday transactions and interactions with government services.

7.5 Future Recommendations

Based on the research outcomes and prototype testing, the following recommendations are proposed for future development:

- Database and Identity Integration

Future development should prioritize establishing live integration with government systems to enable accurate and real-time verification (Real AMI criminal record database, NIC database). In addition, once Sri Lanka's National Digital ID system is formally introduced, the PCC system should be linked to it, providing stronger and standardized identity verification.

- Security Enhancements

Advanced security mechanisms such as biometric verification or blockchain-based authentication can be introduced to strengthen trust and minimize fraud. These measures would improve system reliability and enhance public confidence in digital governance solutions.

- Automation of Verification Processes

Routine Grama Niladhari (GN) verification tasks should be automated while still complying with police standards. Connecting GN officers directly with the system would reduce manual workload, shorten clearance timelines, and improve overall efficiency.

- User Accessibility

To meet user expectations, particularly among younger applicants, a dedicated mobile application or Progressive Web App should be developed. This would enhance accessibility, provide convenient status tracking, and support wider adoption of the system.

- Scalability and Infrastructure

Migrating to microservices architecture and scalable cloud infrastructure will be essential for nationwide adoption. This ensures the system can handle large volumes of applications efficiently and maintain high performance during expansion.

- Artificial Intelligence Applications

Artificial intelligence should be explored for automated document verification, fraud detection, and predictive analytics. These capabilities would enhance accuracy, reduce errors, and support workload forecasting for better resource management.

- Policy and Legal Frameworks

Successful implementation also depends on collaboration with policymakers to establish appropriate legal frameworks. This includes regulations for digital certificate issuance, data protection, and long-term governance of automated clearance systems.

- Pilot Testing and Nationwide Deployment

If legal barriers are addressed, the system should undergo island-wide pilot testing across multiple police stations. This would allow real-world evaluation at scale, offering accurate insights into performance, user adoption, and operational challenges before full nationwide deployment.

- User Adoption and Awareness

Finally, awareness programs and training should be implemented to improve digital literacy. Ensuring accessibility for all user groups will increase user adoption rates and guarantee that the benefits of the system are equitably distributed.

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APPENDIX

Appendix A: Additional Prototype Interfaces

This section presents the user interfaces of the developed Automated Domestic Police Clearance Certificate (PCC) System prototype. These screenshots illustrate the core functionalities offered to applicants, police officers, and administrators.

Prototype Landing Page Design

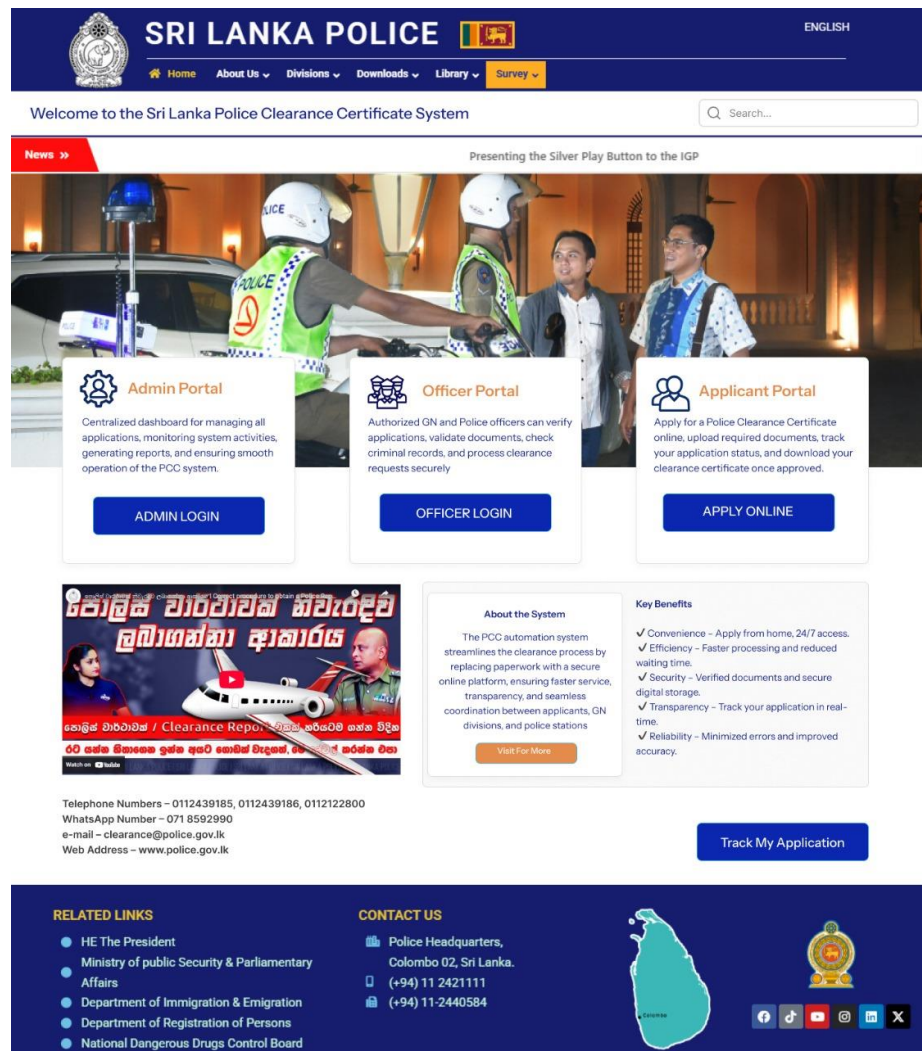


Figure 34: Landing Page

Applicant Portal Interfaces

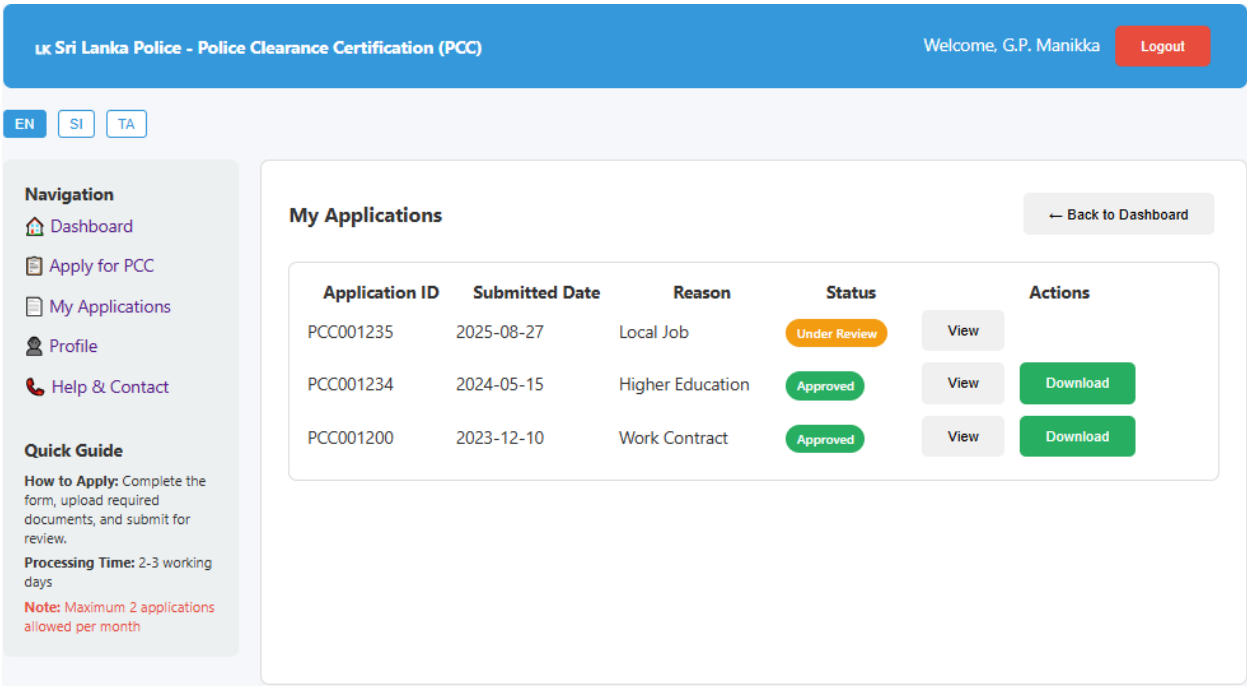


Figure 35: Application History

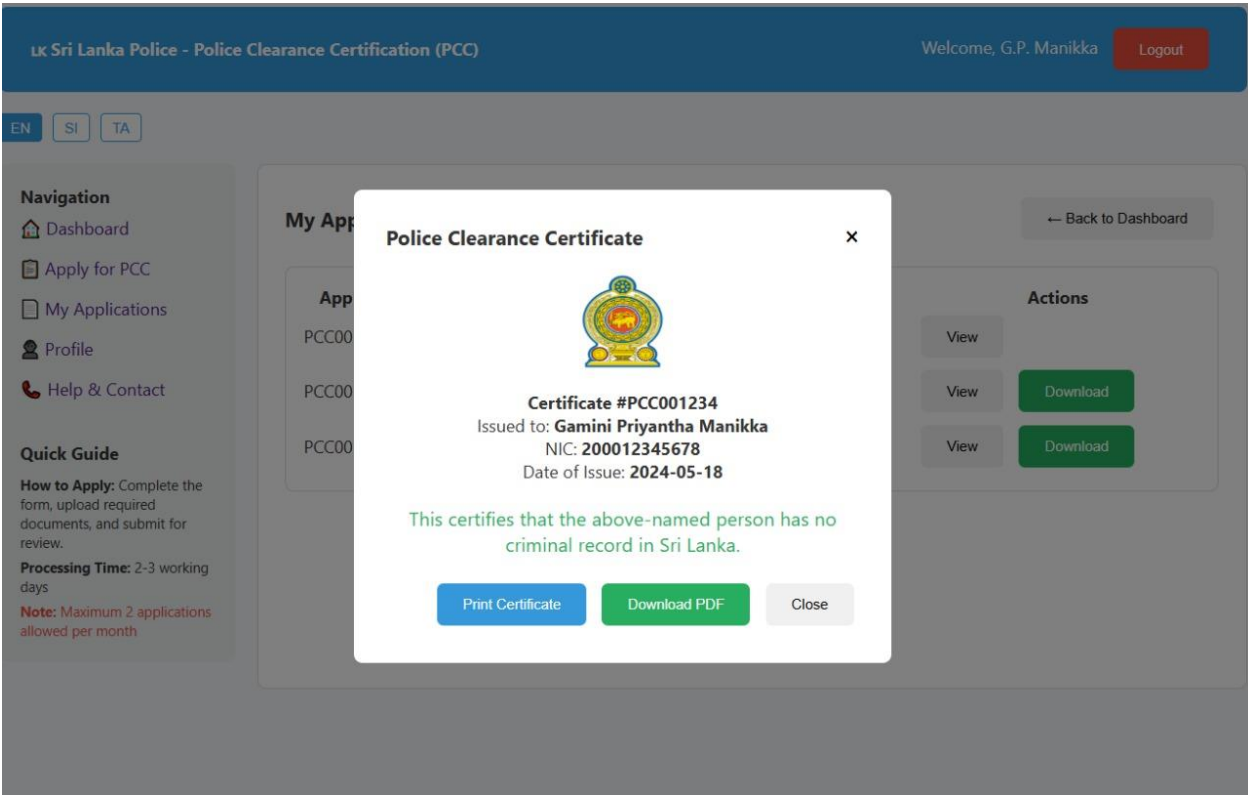


Figure 36: Certification Access

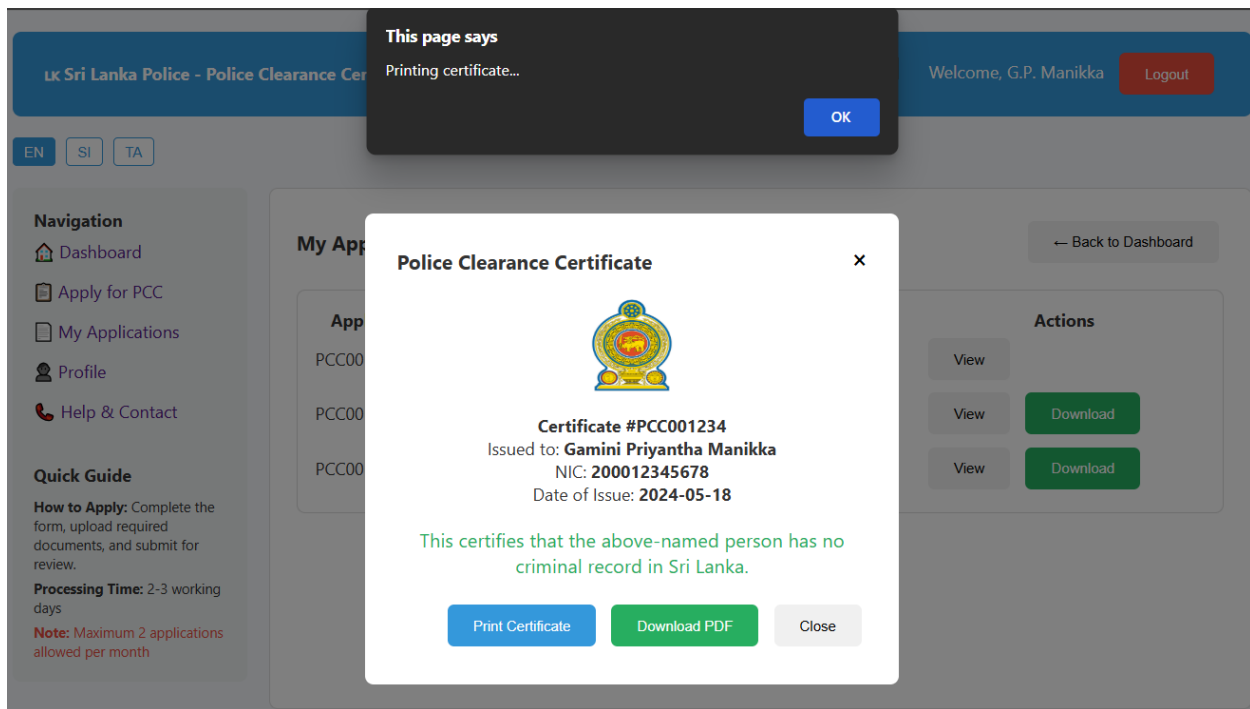


Figure 37:Printing Certificate

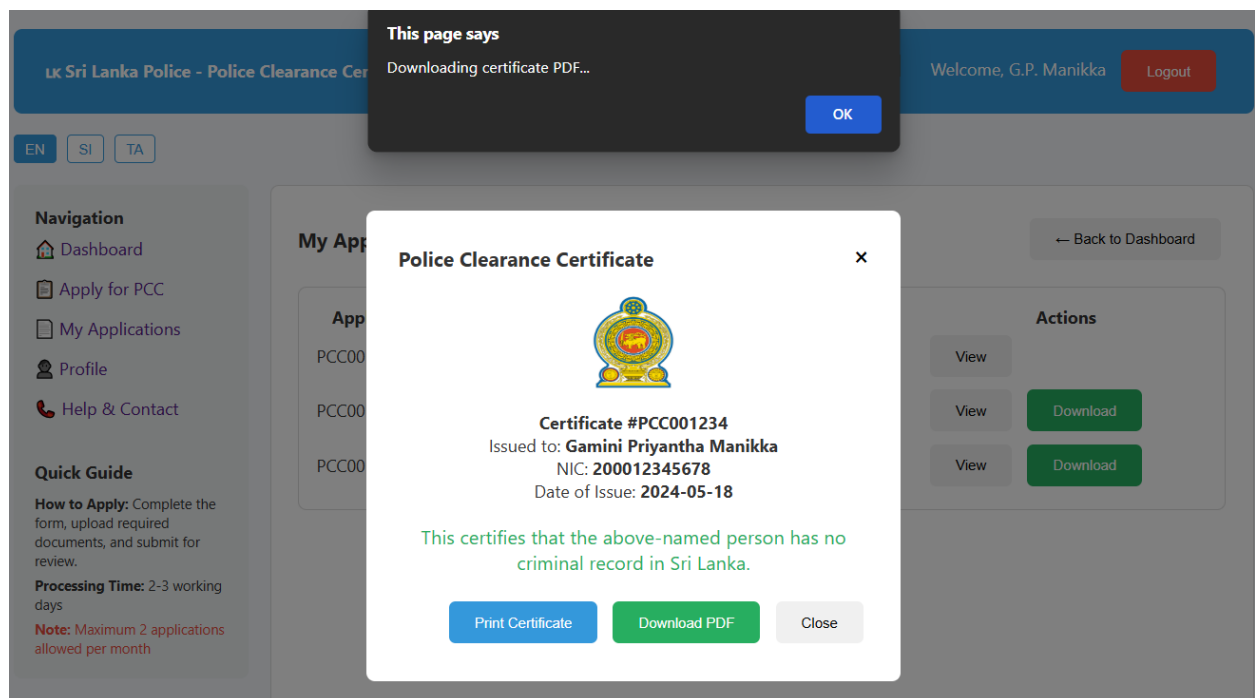


Figure 38:Downloading Certificate

Police Officer Portal Interfaces

uk Sri Lanka Police - Officer Portal

Welcome, PC12345 - Woman Police Constable (WPC) R.M. Perera

Logout

Navigation

Dashboard

Pending Applications

Application History

Reports

My Profile

Messages

Help & Contact

Assigned GN Divisions

✓ Raddolugama North (06-305)

✓ Raddolugama South (06-310)

✓ Raddoluwa West (06-325)

Application History

All Status

mm / dd / yyyy

← Back to Dashboard

Application #PCC001240

NIC: 198812345678 | Submitted: 2024-05-14 02:30 PM

GN Division: Raddolugama North (06-305)

Reason: Legal Matters

Approved

2024-05-15 02:20 PM

Application #PCC001238

NIC: 200012345678 | Submitted: 2024-05-13 10:15 AM

GN Division: Raddolugama South (06-310)

Reason: Work Contracts / Volunteering

Rejected

2024-05-14 10:45 AM

Reason: Incomplete GN certification document

Application #PCC001235

NIC: 199012345678 | Submitted: 2024-05-10 10:45 AM

GN Division: Raddoluwa West (06-325)

Reason: Adoption Process

Approved

2024-05-12 09:30 AM

Figure 39: Application History (Police Officer Portal)

uk Sri Lanka Police - Officer Portal

Welcome, PC12345 - Woman Police Constable (WPC) R.M. Perera

Logout

Navigation

Dashboard

Pending Applications

Application History

Reports

My Profile

Messages

Help & Contact

Assigned GN Divisions

✓ Raddolugama North (06-305)

✓ Raddolugama South (06-310)

✓ Raddoluwa West (06-325)

Application Review - #PCC001240

← Back to Pending

Application #PCC001240

Submitted: 2024-05-14 02:30 PM

GN Division: Raddolugama North (06-305)

Approved

Applicant Information

Full Name (English)

Name with Initials

Ganiya Piyusha Mendika

S.P. Mendika

NIC Number

Date of Birth

1988-12-05

1988-05-20

Permanent Address

114, Temple Road, Raddolugama

From Date

To Date

2024-01-01

2024-05-14

Application Reason

Legal Matters

Certificate Delivery Method

Certificate Delivery Method

1) Applicant requested a physical hard copy certificate to be collected from the police station

Physical Certificate Reference Number *

PMN-2024-001240

Uploaded Documents

NIC Copy

Birth Certificate

GN Certification

Supporting Document

nic_copy.jpg

birth_certificate.jpg

gn_certification.jpg

supporting_document.jpg

View Document

View Document

View Document

View Document

Criminal Record Check

AMR Criminal Record Status: ✓ No criminal record found

Recheck AMR Database

Officer Decision

Remarks (Required for rejection)

Write any additional remarks

Application has already been processed.

Status: Approved

Date: 2024-05-15 02:20 PM

Figure 40:Application Review

110

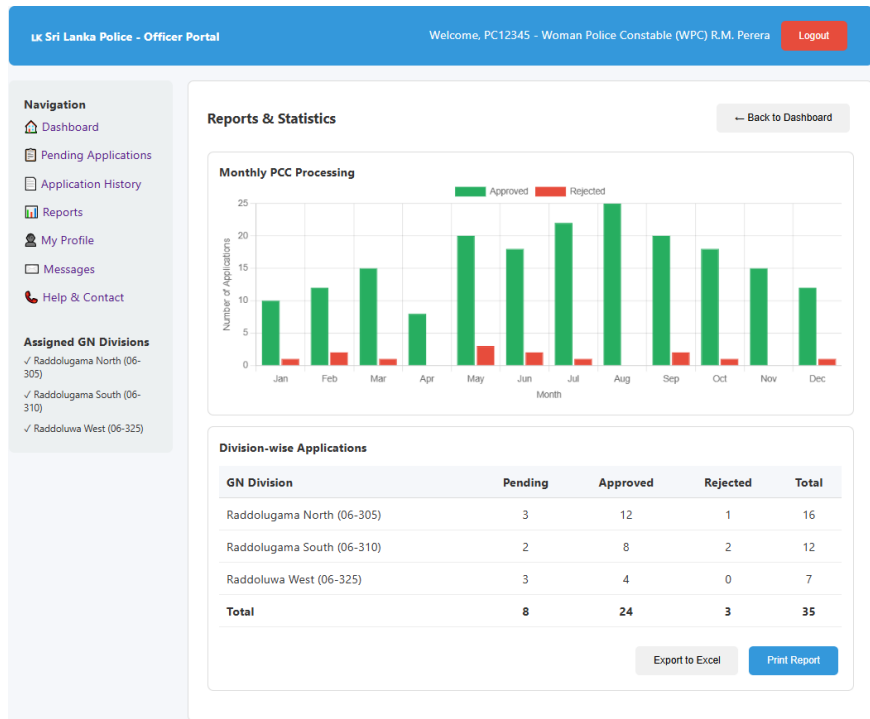


Figure 41: Reports and Statistics

UK Sri Lanka Police - Admin Portal

ADMIN001 - System Admin

Logout

Navigation

- Dashboard
- Applications
- Police Divisions
- Officer Management
- Reports
- Messaging
- System Settings

Quick Actions

- Add New Officer
- Create New Division

Application #PCC001234

Applicant Information

Jane Smith
NIC: 987654321V
Address: 456, Park Road, Raddolugama
Mobile: +94 77 987 6543
Email: jane@example.com

Review Information

Assigned Officer: WPC R.M. Perera (PC12345)
AMI Check: Clean - No criminal records found
Decision: Approved
Remarks: Application meets all requirements

Application Details

Police Division: Raddolugama
GN Division: Raddolugama North (06-305)
Submitted: 2024-01-30
Status: Approved

Admin Actions

Override Decision
Select action

Remarks
Enter remarks for override...

Process Override

Uploaded Documents

NIC Copy Birth GN

Figure 42: Application Override

Admin Portal Interfaces

lx Sri Lanka Police - Admin Portal

ADMIN001 - System Admin Logout

Navigation

Dashboard

Applications

Police Divisions

Officer Management

Reports

Messaging

System Settings

Quick Actions

Add New Officer

Create New Division

Police Division Management

+ Add New Division

Division ID	Division Name	GN Divisions	Assigned Officers	Applications (30d)	Actions
PD001	Seeduwa Police Division	Seeduwa West (147), Seeduwa Central (147), Seeduwa East (147)	3 officers	142	Edit Deactivate
PD002	Raddolugama Police Division	Raddolugama North (06-305), Raddolugama South (06-310), Raddoluwa West (06-325)	2 officers	98	Edit Deactivate

Grama Niladhari Division Mapping

GN Division	Police Division	Assigned Officer	Applications (30d)	Actions
Seeduwa West (147)	Seeduwa	WPC S.T. Bandara (SP45678)	48	Reassign
Seeduwa Central (147)	Seeduwa	WPC S.T. Bandara (SP45678)	52	Reassign
Raddolugama North (06-305)	Raddolugama	WPC R.M. Perera (PC12345)	35	Reassign

Figure 43:Police Division Management

lx Sri Lanka Police - Admin Portal

ADMIN001 - System Admin Logout

Navigation

Dashboard

Applications

Police Divisions

Officer Management

Reports

Messaging

System Settings

Quick Actions

Add New Officer

Create New Division

Officer Management

+ Add New Officer

Officer ID	Rank & Name	NIC	Assigned Division	GN Divisions	Status	Actions
PC12345	WPC R.M. Perera	123456789V	Raddolugama	Raddolugama North (06-305) Raddolugama South (06-310) Raddoluwa West (06-325)	Active	Edit Deactivate
SP45678	WPC S.T. Bandara	987654321V	Seeduwa	Seeduwa Central (147) Seeduwa West (147) Seeduwa East (147)	Active	Edit Deactivate
SP98765	PC W.A. Silva	456789123V	Seeduwa	Liyanagemulla North (144A) Liyanagemulla South (144) Muthuwadiya (201)	Active	Edit Deactivate

Figure 44:Officer Management

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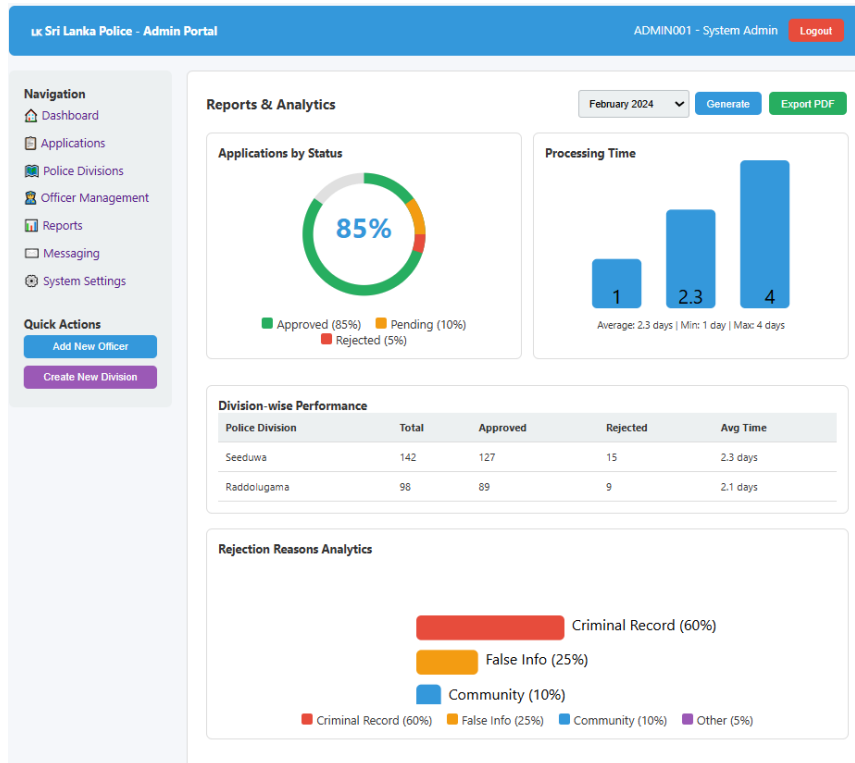


Figure 45: Reports and Analytics

Sri Lanka Police - Admin Portal ADMIN001 - System Admin [Logout](#)

Navigation

- Dashboard
- Applications
- Police Divisions
- Officer Management
- Reports
- Messaging
- System Settings

Quick Actions

- Add New Officer
- Create New Division

System Settings [Save Settings](#)

Application Settings

Default Processing Time (Days)

3

Maximum File Upload Size (MB)

2

Allowed File Types

PDF, JPG, PNG, DOC

Notification Settings

- ☒ Enable Email Notifications
- ☒ Enable SMS Notifications

Notification Email Address

pcc-notifications@police.gov.lk

Danger Zone

System Maintenance Mode

Off - System Operational

[Reset System Data](#)

Warning: This will delete all application data and cannot be undone.

Figure 46: System Settings

Appendix B: Implementation Details

Basic Testing using Automation

Basic automated testing was carried out on the services layer to verify core functionalities without involving a database.

- ✓ API Endpoint Responses - That endpoints return proper HTTP status codes
- ✓ Error Handling - That missing parameters return proper error messages
- ✓ Success Scenarios - That valid requests return success responses
- ✓ Response Structure - That responses have the expected JSON structure
- ✓ Data Validation - That the API validates input parameters

```
PS C:\Users\Isuru\Desktop\applicant\backend> npm run test:basic

> backend@1.0.0 test:basic
> jest test/basic.test.js --testTimeout=30000

PASS test/basic.test.js
  Basic API Tests (No Database)
    ✓ GET /api/health should return server status (48 ms)
    ✓ POST /api/login should return 400 for missing fields (52 ms)
    ✓ POST /api/login should succeed with valid credentials (6 ms)
    ✓ POST /api/verify-otp should return 400 for missing fields (8 ms)
    ✓ POST /api/verify-otp should succeed with valid OTP (11 ms)

Test Suites: 1 passed, 1 total
Tests:       5 passed, 5 total
Snapshots:   0 total
Time:        1.6 s
Ran all test suites matching test/basic.test.js.
PS C:\Users\Isuru\Desktop\applicant\backend>
```

Figure 47: Basic Test Automation

Frontend Folder Structure

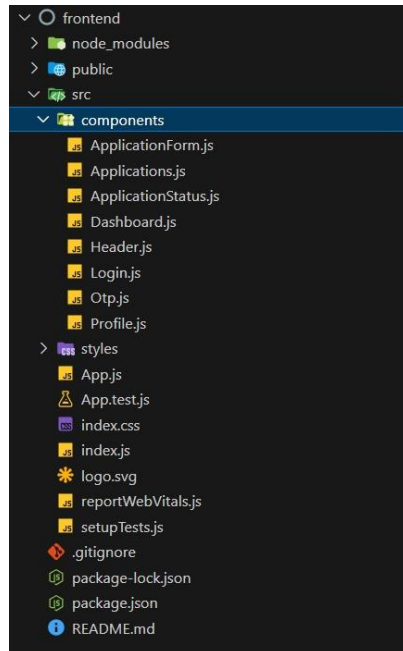


Figure 48:Frontend Folder structure

A well-organized API server focused on handling job applications.

- `models/Applicant.js`: The heart of the data layer. Defines the schema for an applicant (name, email, resume, status).
- `server.js`: The engine. Sets up the Express server, connects to the database, and defines all API routes.
- `uploads/`: The file store. Directory where uploaded resumes (PDFs, Docs) are saved.
- `test/`: The quality suite. Contains unit & integration tests (`basic.test.js`, `real-database.test.js`) to ensure reliability.
- `Utility Scripts`: `reset-applications.js` (clears the database for testing), `debug-applications.js` (for checking data during development).
- `env/`: The configuration. Holds environment variables (database URL, secrets) separately from code.

Backend Folder Structure

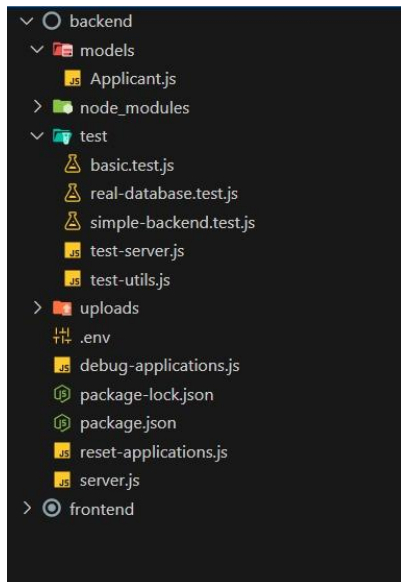


Figure 49: Backend Folder structure

- `src/components/`: The building blocks. Contains all major UI pieces:
- `Auth`: `Login.js`, `Otp.js` (for secure user login).
- `Core Features`: `ApplicationForm.js` (to apply), `Applications.js` (to view submissions), `ApplicationStatus.js` (to check status).
- `Layout`: `Dashboard.js` (main page), `Header.js` (navigation), `Profile.js` (user settings).
- `src/App.js`: The orchestrator. The main component that renders everything and handles routing.
- `src/index.js`: The entry point. The first file that runs, kicking off the React app.
- `public/`: The static assets. Holds the base `index.html` file and images/icons.

Backend API Endpoints for PCC System (Based on Existing Implementation)

Based on the PCC system we've implemented, here are the available API endpoints:

Applicant Endpoints

Table 21: Applicant Endpoints

Route	Method	Description	Role
/api/login	POST	Login applicant with NIC and mobile number	Applicant
/api/verify-otp	POST	Verify OTP sent to mobile	Applicant
/api/profile/:mobile	GET	Get applicant profile by mobile number	Applicant
/api/applications/:mobile	GET	Get all applications by mobile number	Applicant
/api/application/:appId	GET	Get specific application details	Applicant
/api/upload	POST	Upload required documents (NIC, birth cert, etc.)	Applicant
/api/application	POST	Submit new PCC application	Applicant
/api/documents/:appId	GET	Get application documents	Applicant
/api/dashboard/:mobile	GET	Get applicant dashboard data	Applicant
/api/application-status/:appId	GET	Get detailed application status with timeline	Applicant

Police Officer Endpoints (To be implemented)

Table 22:Police Officer Endpoints (To be implemented)

Route	Method	Description	Role
/api/officer/login	POST	Login police officer with OTP	Police Officer
/api/officer/verify-otp	POST	Verify OTP and get access token	Police Officer
/api/officer/profile/:policeId	POST	Get officer profile	Police Officer
/api/officer/dashboard/:policeId	GET	Get dashboard statistics	Police Officer
/api/officer/applications/	GET	Get applications for review	Police Officer
/api/officer/applications/:applicationId/details	GET	Get application details	Police Officer
/api/officer/applications/:applicationId/criminal-check	POST	Initiate criminal record check	Police Officer
/api/officer/applications/:applicationId/review	PUT	Review and update application status	Police Officer
/api/officer/applications/history/:policeDivision	GET	Get application history	Police Officer
/api/officer/reports/division/:policeDivision	GET	Get division statistics	Police Officer
/api/officer/documents/:documentId	GET	Get document file	Police Officer
/api/officer/health	GET	Health check	Public
/api/officer/seed	POST	Seed database (dev only)	Developer

Appendix C: Survey Questionnaire

Research Background Questionnaire



Enhancing E-Governance in Law Enforcement: Developing an Automated Police Clearance Certificate System

Research Purpose:
This survey aims to gather insights on the **current manual process** of issuing Police Clearance Certificates (PCC) in Sri Lanka and identify opportunities for automation. Your feedback will directly contribute to designing a more efficient, transparent, and citizen-friendly digital system.

Isuru Senarathna
Faculty of Computing, NSBM Green University
BSc in Management Information Systems (Special)

Confidentiality:
All responses are **anonymous** and will be used **solely for academic research**. No personal data will be shared.

Estimated Time: 5–10 minutes

Figure 50: Applicant Survey

- Applicant Survey:

<https://docs.google.com/forms/d/e/1FAIpQLSeKwDQk3Qi7PcwW3jXUdCZFrgsPlkBlhwWIszBKv08C8vlnYg/viewform?usp=dialog>



Enhancing E-Governance in Law Enforcement: Developing an Automated Police Clearance Certificate System

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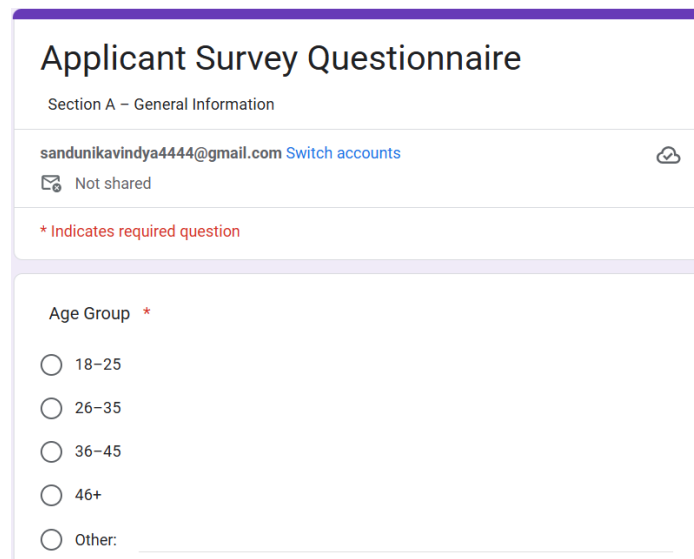
Estimated Time: 5–10 minutes

Figure 51: Police Officer Survey

- Police Officer Survey:

[Enhancing E-Governance in Law Enforcement: Developing an Automated Police Clearance Certificate System](#)

Prototype evaluation and user feedback gathering

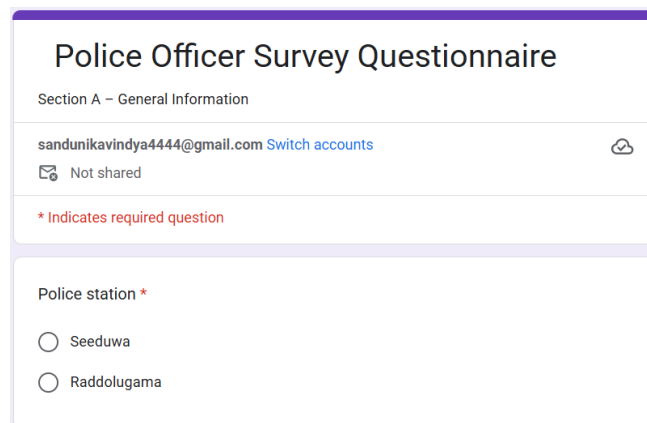


The screenshot shows a Google Form titled "Applicant Survey Questionnaire". Below the title is "Section A – General Information". The form is associated with the email "sandunikavindya4444@gmail.com" and has a "Switch accounts" link. It is marked as "Not shared". A red asterisk indicates a required question. The question is "Age Group *", with radio button options for "18-25", "26-35", "36-45", "46+", and "Other:" followed by a text input field.

Figure 52: Applicant Survey - Prototype Evaluation

- Applicant Survey:

https://docs.google.com/forms/d/e/1FAIpQLSdInDogb6i_RNZQExdyUR26p8rjElwEZ6k0ksm6tRnyDMVeLQ/viewform?usp=dialog



The screenshot shows a Google Form titled "Police Officer Survey Questionnaire". Below the title is "Section A – General Information". The form is associated with the email "sandunikavindya4444@gmail.com" and has a "Switch accounts" link. It is marked as "Not shared". A red asterisk indicates a required question. The question is "Police station *", with radio button options for "Seeduwa" and "Raddolugama".

Figure 53: Police Officer Survey - Prototype Evaluation

- Police Officer Survey:

<https://docs.google.com/forms/d/e/1FAIpQLSdxE7dD-S4jm4RE7YZIzuBLVLfNbknYVIs8EQXMbUINFVAV7A/viewform?usp=dialog>

Appendix D: GitHub Repository

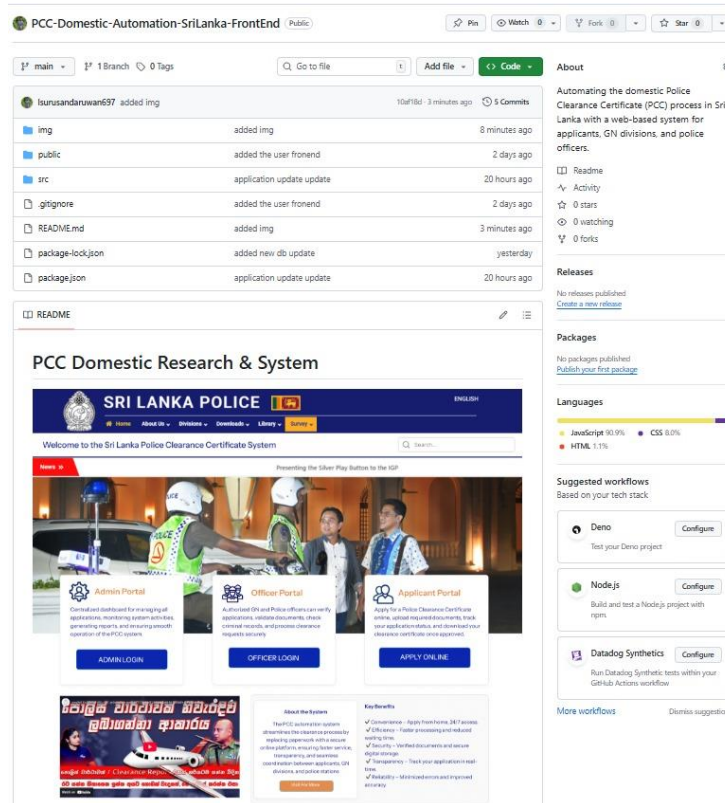


Figure 54: GitHub Repository

Frontend Project

- <https://github.com/Isurusandaruwan697/PCC-Domestic-Automation-SriLanka-FrontEnd/tree/main>

Backend Project

- <https://github.com/Isurusandaruwan697/PCC-Domestic-Automation-SriLanka-Backend>

Prototype-Frontend-based (Include all the users and UI of PCC-domestic system)

- <https://github.com/Isurusandaruwan697/PCC-Domestic-Automation-Sri-Lanka-Prototyp>

Appendix E: Proof of Research Publication and Award

The research study entitled “Modernizing Police Clearance Processes Through E-Governance: A Case Study and Digital Solution for Sri Lanka” was formally published and presented at the NSBM Student Research Symposium on Innovative Technology 2025 (NSRSIT '25), conducted on 9th July 2025 at NSBM Green University. Participation in this symposium provided an opportunity to disseminate the findings of the study to a wider academic audience and to contribute to the national discourse on technology-driven governance solutions.

Furthermore, was awarded the Best Presenter Award at NSRSIT '25. This recognition was conferred in appreciation of the clarity, depth, and practical significance of the presentation, as well as the innovative approach proposed to address inefficiencies in Sri Lanka’s Police Clearance Certificate (PCC) system. The award not only reinforces the academic value of the study but also demonstrates its contribution to both the institutional research community and the broader field of e-governance.

- Research Paper link: <https://drive.google.com/drive/folders/1iIHxGXYPl9wx-W4jsX8oifBmy7j7-Vuw?usp=sharing>

Student Progress Record



Research Project - Individual Student Progression Report [Students Copy]

01. Student Name R. G. J. S. Senarathna
02. Index Number 22742
03. Degree Program MIS
04. Supervisor Name Dr. Mahamed Shadraz
05. Project Title Enhancing E-governance in Law Enforcement: Developing an Automated Police Clearance Certificate System for more Efficient and Transparent Application Process.

Meeting Number	Meeting 01	Meeting 02	Meeting 03	Meeting 04	Meeting 05	Meeting 06	Meeting 07
Date	03/10/24	01/10/24	11/11/24	06/02/25	27/3/25	29/5/25	15/7/25
Student Signature	<u>[Signature]</u>	<u>[Signature]</u>	<u>[Signature]</u>	<u>[Signature]</u>	<u>[Signature]</u>	<u>[Signature]</u>	<u>[Signature]</u>
Supervisor Signature	<u>[Signature]</u>	<u>[Signature]</u>	<u>[Signature]</u>	<u>[Signature]</u>	<u>[Signature]</u>	<u>[Signature]</u>	<u>[Signature]</u>

Meeting Number	Meeting 08	Meeting 09	Meeting 10	Meeting 11	Meeting 12	Meeting 13	Meeting 14
Date	22/08/25	02/09/25					
Student Signature	<u>[Signature]</u>	<u>[Signature]</u>					
Supervisor Signature	<u>[Signature]</u>	<u>[Signature]</u>					